



University of Glasgow | School of
Computing Science

Honours Individual Project Dissertation

TRUST AWARE STOCK TREND PREDICTION AND INVESTMENT STRATEGY OPTIMISATION USING DEEP NEURAL NETWORKS

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Abstract

This dissertation addresses the gap in stock market prediction between accuracy-based metrics and economic value by integrating multi-modal predictions, probability calibration and principled capital allocation into a single end-to-end framework, evaluating whether diverse predictive signals fed into multiple deep-learning architectures (LSTM & GRU variants) trained to predict stock direction and magnitude can yield robust and trustworthy forecasts implemented with risk-aware capital allocation strategies.

The framework incorporates prior model predictions and their historical errors as meta-features, enabling the system to learn when its outputs should be trusted. In addition, a variable-horizon forecasting approach is employed to move beyond traditional fixed-horizon models, allowing the system to capture both short-term fluctuations and long-term trends.

Results show that this approach improves decision stability and reduces excessive turnover compared to uncalibrated baselines, furthermore, risk-aware capital allocation strategies consistently outperform naïve staking strategies, with the hybrid Kelly-Modern Portfolio Theory achieving the strongest risk-adjusted return of **TODO: XX%** over 100 trading days, **TODO: X** times higher than the previous baseline set over the same time-frame. These findings lay the groundwork for future research into trust-aware financial prediction systems that prioritises both accuracy and economic value.

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1 | Introduction

1.1 Motivation

Financial markets are inherently noisy, non-stationary systems in which stock prices are influenced by complex interactions between historical price dynamics and rapidly evolving external information. Despite decades of research, consistently forecasting market behaviour remains challenging, not only due to stochastic price movements but also due to the relationship between predictive signals and future returns varying across time and market regimes. As a result, even small modelling assumptions can significantly impact downstream decision-making.

Recent advances in deep learning have led to measurable improvements in stock movement prediction, particularly through the use of sequence models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, as well as the incorporation of unstructured textual data such as news and social media. However, much of the existing literature evaluates predictive success primarily through accuracy-based metrics, often without sufficient consideration of how model outputs are subsequently used in financial decision-making. In practice, predictive accuracy alone is insufficient, investment decisions depend critically on the reliability of predicted probabilities and the ability to translate forecasts into appropriately sized positions under risk.

In practical investment settings, model outputs are rarely consumed as hard class labels, instead, they are interpreted as probabilities that inform position sizing and risk management. This places importance on both the directional accuracy and the reliability of the prediction. A well-calibrated model produces probabilities estimates whose empirical frequencies align with observed outcomes in practice, whilst poorly calibrated models generate unreliable probabilities that are treated as meaningful inputs to allocation rules leading to over- or under-sized positions, unstable trading behaviour and amplified drawdowns, even when headline accuracy metrics appear strong.

Beyond calibration, a further limitation of many existing approaches is the implicit assumption that a model's predictive reliability is static. In reality, model performance fluctuates over time due to regime shifts, changes in market microstructure and evolving information flows. Models rarely account for their own historical errors when making new predictions and thus may continue to express high confidence even during periods of deteriorating performance. Incorporating information about prior predictions and their realised errors offers a principled way for a system to assess its own reliability and adapt its behaviour accordingly.

Finally, prediction and capital allocation are often treated as separate problems, with models evaluated independently of the investment strategies they enable. This separation obscures the true economic value of predictive improvements and limits the interpretability of reported results. Investment performance depends jointly on the quality of predictive signals, the reliability of associated probabilities and the manner in which capital is allocated under uncertainty.

This project is motivated by the need for an integrated framework that unifies multimodal prediction, model self-assessment, probability calibration and principled capital allocation within a single coherent system. By combining heterogeneous data sources with information derived from prior predictions and their realised errors, the framework seeks to improve both the reliability of forecasts and their practical utility under non-stationary market conditions. Through the joint

evaluation of statistical performance and economic outcomes, this work aims to bridge the gap between predictive modelling and real-world investment decision-making, demonstrating how certainty-aware learning and risk-sensitive allocation can support more stable and economically meaningful trading strategies in dynamic environments.

1.2 Purpose

The purpose of this project is to design and evaluate an end-to-end, multi-horizon trading system that integrates multimodal predictive signals, probability calibration and risk-aware capital allocation within a single coherent pipeline. Rather than optimising predictive accuracy in isolation, the system is designed to assess the reliability and economic viability of model-driven investment decisions under realistic market conditions.

This work builds upon *Stock Movement Prediction from Tweets and Historical Prices* by Xu and Cohen (2018), which is used as a baseline for comparison. However, the focus of this project extends beyond next-day prediction and single-model evaluation by exploring variable forecasting horizons, multimodal feature integration and multiple predictive model formulations. Crucially, the value of these predictive signals is assessed through their interaction with capital allocation strategies in a simulated trading environment, rather than through benchmarking metrics alone. To account for non-stationarity and regime shifts, the proposed system incorporates prior predictions and their realised errors as meta-features, enabling the model to capture time-varying predictive reliability.

To this end, the dissertation evaluates multiple deep-learning architectures and systematically examines their integration with several capital allocation strategies, including full Kelly, fractional Kelly, mean-variance portfolio theory (MPT) and a hybrid fractional Kelly-MPT approach. By jointly analysing predictive performance, calibration quality and financial outcomes, this project aims to bridge the gap between statistical forecasting and practical investment decision-making.

1.2.1 Contributions

The following key contributions are made:

- **Stock direction and magnitude prediction formulations**, including binary classification for directional movement, regression for return magnitude and an ordinal classification approach that discretises returns into buckets to estimate expected return.
- **An end-to-end multimodal prediction and investment framework** that integrates historical price data, textual signals and auxiliary features within a single automated pipeline, designed for realistic walk-forward evaluation.
- **A variable-horizon trading system with an early-exit mechanism**, enabling positions to be held for multiple days while allowing premature exit when predicted downside risk increases, thereby improving flexibility relative to fixed-horizon strategies.
- **The explicit incorporation of model self-assessment through meta-features**, using prior predictions and their realised errors to capture time-varying predictive reliability in non-stationary financial markets.
- **A systematic application of probability calibration to deep-learning models**, enabling model confidence scores to be interpreted as meaningful probabilities suitable for downstream risk-sensitive decision-making.
- **A comparative evaluation of capital allocation strategies**, including full Kelly, fractional Kelly, mean-variance portfolio theory (MPT) and a hybrid fractional Kelly-MPT approach, assessed using economic performance measures.

- **An evaluation methodology grounded in economic outcomes**, assessing return, draw-down and risk-adjusted performance alongside traditional accuracy-based predictive metrics.

1.2.2 Novelty

The novelty of this work lies not in proposing new model architectures, but in the integrated treatment of multimodal prediction, uncertainty estimation, model reliability and capital allocation. By embedding these components within a unified framework and evaluating them through economic outcomes, this dissertation advances existing stock prediction research towards more trustworthy and practically deployable system.

1.3 Report Structure

This dissertation is organised into eight chapters.

Chapter 2, *Background and Related Work*, reviews prior research in stock movement prediction and outlines the key theoretical foundations of the project, including LSTM and GRU models, bidirectional variants, ordinal regression, BERT, probability calibration methods (isotonic regression and temperature scaling), the Kelly Criterion, mean-variance portfolio theory (MPT) and the use of meta-features.

Chapter 3, *Analysis and Research Questions*, formalises the problem setting, defines the prediction tasks, discusses constraints such as temporal integrity and non-stationarity as well as presenting the research questions guiding the evaluation.

Chapter 4, *Design*, describes the architecture of the end-to-end trading framework, including multimodal data integration, model formulations for direction and magnitude prediction, meta-feature incorporation, calibration strategy and the investment simulation design.

Chapter 5, *Implementation*, outlines the practical realisation of the system, covering data pre-processing, feature engineering, model training, calibration, and capital allocation within a walk-forward simulation environment.

Chapter 6, *Results*, reports predictive performance, calibration quality, and economic outcomes such as return, drawdown, and risk-adjusted measures, alongside comparative analyses of model and allocation strategies.

Chapter 7, *Discussion*, interprets the findings in relation to the research questions and reflects on the interaction between multimodal prediction, model reliability, calibration, and capital allocation.

Chapter 8, *Conclusion and Future Work*, summarises the main contributions, discusses limitations, and outlines directions for future research.

Additional material, including extended results and implementation details, is provided in the Appendices.

2 | Background & Related Work

2.1 Stock Movement Prediction

Stock movement prediction (SMP) aims to forecast future price direction or return magnitude using historical market data and auxiliary information. A fundamental challenge in this domain is the time-varying and regime-dependent nature of financial markets. Early empirical evidence by Schwert (1989) demonstrated that stock market volatility is persistent, clustered and strongly associated with macroeconomic uncertainty, recessions, and financial crises. Crucially, volatility dynamics are not constant over time, and no single macroeconomic factor fully explains their variation. This work established that financial markets are inherently non-stationary, with structural shifts in risk conditions occurring over time.

The implication for prediction models is clear in that the assumptions of stable statistical relationships are often violated in practice. Forecasting systems must therefore operate under changing volatility regimes and evolving market conditions, motivating adaptive and sequence-aware modelling approaches.

2.1.1 From Statistical Models to Deep Learning

Early approaches to SMP relied on classical statistical models such as ARIMA, which assume linear relationships and weak stationarity. While these models can capture short-term autocorrelation patterns, their performance is limited in the presence of non-linearity and structural breaks (Ho et al. 2021).

With the emergence of machine learning and deep learning, recurrent architectures such as LSTM and GRU networks became widely adopted for financial time-series modelling. These models are capable of capturing long-term dependencies and non-linear interactions within sequential data. Large-scale empirical evidence by Fischer and Krauss (2018) shows that LSTM-based models trained on S&P 500 constituents (1992–2015) outperform classical baselines such as ARIMA, Random Forests, and shallow neural networks when evaluation respects temporal ordering. However, performance was shown to decay post-2010, highlighting regime sensitivity and the fragility of learned patterns.

These findings suggest that while deep sequence models can extract predictive structure, their effectiveness depends heavily on market conditions and proper evaluation protocols.

2.1.2 Multimodal Extensions

Beyond price-only models, researchers have incorporated textual information such as news headlines and social media posts to capture sentiment and narrative signals not immediately reflected in historical prices. Xu and Cohen (2018) introduced *StockNet*, modelling market uncertainty using a recurrent variational autoencoder combined with an auxiliary temporal prediction head. Evaluated on Twitter and price data across 88 U.S. stocks (2014–2016), the model outperformed ARIMA, Random Forest, and text-only baselines. However, the use of dead-zone filtering, which excludes small-movement days, may have inflated reported accuracy by simplifying the classification task.

Building upon this direction, He and Gu (2022) proposed a Multimodal Attention Network (MMAN) that integrates credibility-aware textual features with price encoders through intra- and inter-modal attention mechanisms. Ablation studies demonstrate that both credibility modelling and cross-modal attention contribute positively to performance, suggesting that structured multimodal fusion can enhance predictive capability.

In parallel, robustness to market noise has been explored through adversarial training. Feng et al. (2019) introduced an adversarially trained LSTM (Adv-LSTM), incorporating worst-case perturbations during training to improve generalisation under stochastic conditions. Reported improvements over prior baselines indicate enhanced resilience to noisy inputs.

More recent work has investigated multi-scale modelling and extended temporal windows, demonstrating that incorporating longer historical context can further improve statistical performance. Nevertheless, across these multimodal approaches, evaluation remains predominantly focused on classification metrics such as accuracy and MCC. Label filtering and the absence of cost-aware economic assessment are recurring limitations, leaving open the question of whether statistical improvements translate into meaningful financial performance.

2.2 Deep Learning for Time-Series Modelling

Early recurrent neural networks (RNNs) were theoretically well-suited for sequence modelling but struggled in practice due to the vanishing and exploding gradient problem. As demonstrated by Hochreiter and Schmidhuber (1997), gradients propagated through time either decay or grow exponentially as sequence length increases, preventing effective learning of long-term dependencies. This limitation makes standard RNNs unreliable for tasks requiring memory over extended time horizons.

To address this, Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory (LSTM) architecture. LSTMs incorporate gated memory cells designed to preserve stable gradient flow across long temporal spans. By regulating information flow through input, forget and output gates, LSTMs enable controlled memory retention and selective forgetting. This architecture allows the network to learn dependencies spanning hundreds or even thousands of time steps, overcoming the instability inherent in traditional RNN training.

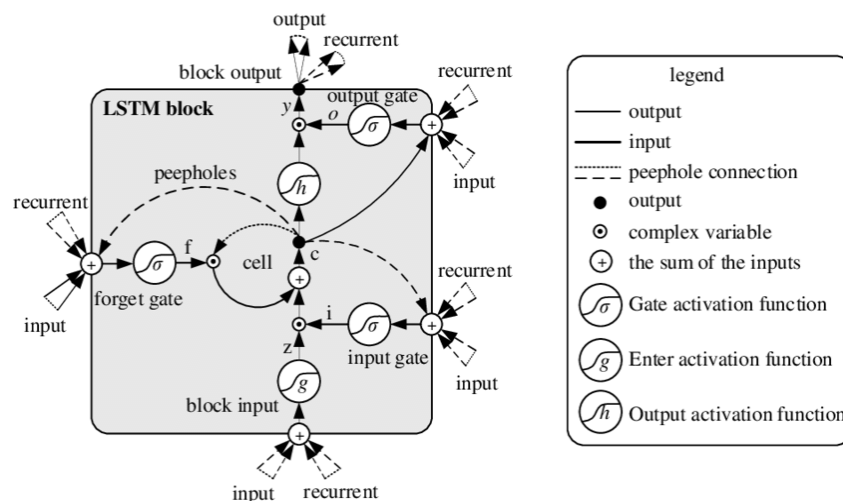


Figure 2.1: Schematic diagram of LSTM model (Shuangao et al. 2020).

2.2.1 Long Short-Term Memory (LSTM)

Subsequent analyses, including Staudemeyer and Morris (2019), clarify that the strength of LSTMs lies in their gated structure and explicit memory cell. The separation between hidden state and cell state permits stable error propagation while protecting memory from irrelevant noise. This design enables LSTMs to model long-term temporal dependencies more reliably than earlier recurrent architectures.

In time-series domains such as finance, where regime persistence and volatility clustering occur, the ability to retain information over extended periods is particularly relevant. However, LSTMs are not without limitations. Model capacity is finite, architectural design choices significantly influence performance, and increasing network size does not automatically guarantee improved long-term reasoning.

2.2.2 Gated Recurrent Units (GRU)

The Gated Recurrent Unit (GRU) was introduced as a simplified variant of the LSTM, merging the input and forget gates into a single update gate and removing the explicit memory cell. This reduces architectural complexity and parameter count while retaining gating mechanisms that regulate temporal information flow.

Empirical comparisons, such as those conducted by Cahuantzi et al. (2023), suggest that GRUs often achieve performance comparable to LSTMs on moderately complex sequence tasks while offering improved computational efficiency. However, for highly complex or long-range dependency structures, LSTMs may demonstrate superior stability and representational capacity.

In financial time-series modelling, this trade-off implies that GRUs may suffice for short-term pattern extraction, whereas LSTMs may better capture longer-term regime persistence and volatility dynamics. Consequently, both architectures remain widely adopted in stock movement prediction research.

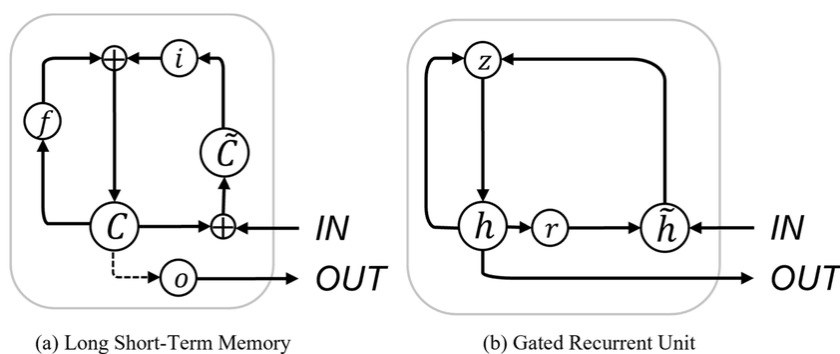


Figure 2.2: Diagram of LSTM and GRU structure. (a) The structure diagram of a vanilla LSTM architecture. “ C ” represents the memory cell, “ i ” denotes the input gate, “ f ” represents the forget gate, and “ o ” denotes the output gate; (b) The simple architecture of GRU. “ h ” is the activation function, “ z ” refers to the update gate, and “ r ” stands for the reset gate. (Liu et al. 2022).

2.2.3 Long Short-Term Memory (LSTM)

2.2.4 Bidirectional Variants

Bidirectional recurrent neural networks (BRNNs) extend standard RNN architectures by processing sequences in both forward and backward directions (Schuster and Paliwal 1997). By

combining hidden representations from past-to-future and future-to-past passes, bidirectional models leverage full-sequence context for each time step.

Bidirectional architectures have demonstrated improved performance in tasks such as speech recognition and sequence classification (Graves and Schmidhuber (2005), Huang et al. (2015)), and they form the basis of widely used models such as BiLSTM and BiGRU. However, in financial time-series applications, strict temporal ordering must be preserved to avoid information leakage. While bidirectional models may be appropriate in offline feature extraction or representation learning contexts, their use in predictive trading systems requires careful consideration to ensure causal integrity.

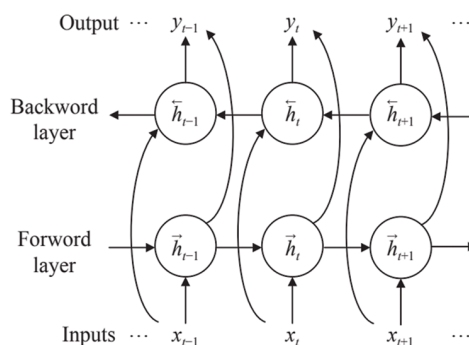


Figure 2.3: Bidirectional RNN model diagram (Wenqiu et al. 2023).

2.2.5 Implications for Stock Movement Prediction

The progression from classical RNNs to gated recurrent architectures and their bidirectional extensions reflects a broader shift toward models capable of handling long-term dependencies, non-linearity, and temporal complexity. In financial markets, characterised by regime shifts, volatility clustering, and evolving dynamics, such properties are particularly valuable.

Nevertheless, while deep sequence models have demonstrated strong statistical performance relative to classical baselines, their effectiveness remains sensitive to market conditions and evaluation methodology. This reinforces the importance of coupling architectural advances with robust evaluation, calibration, and economic assessment.

2.2.6 Ordinal Regression for Financial Returns

Financial returns are inherently ordered quantities rather than purely categorical variables. Binary classification (up/down) ignores magnitude information, while regression directly models continuous returns but may be sensitive to noise. Ordinal regression provides an intermediate formulation by discretising returns into ordered buckets, preserving directional and magnitude structure while improving robustness. This formulation enables the estimation of expected return ranges rather than simple direction, offering more informative signals for downstream capital allocation.

$$P(Y \leq c \mid x) = F(\theta_c - \eta(x)), \quad c = 1, \dots, K - 1,$$

where θ_c are ordered thresholds and $\eta(x)$ is the model score induced by the input features. This allows the model to exploit the natural ordering of return ranges and, in turn, supports the estimation of expected return ranges rather than simple direction alone (Kook et al. 2021).

2.3 BERT for Natural Language Processing

The integration of textual information into stock prediction models has attracted increasing attention, as market prices often reflect narratives and sentiment not immediately captured by historical returns alone. Xu and Cohen (2018) demonstrated that incorporating Twitter data alongside price time-series improves predictive performance relative to price-only baselines. Early approaches typically relied on static word embeddings (e.g., Word2Vec or GloVe) combined with recurrent or convolutional encoders to extract textual features. For example, He and Gu (2022) encoded textual posts using neural attention mechanisms alongside price features, while Li et al. (2020) model textual relationships through graph-based structures to capture inter-stock dependencies.

These methods, however, depend on context-independent embeddings, where each word is represented by a fixed vector regardless of surrounding context. Transformer-based language models, particularly BERT (Devlin et al. 2019), address this limitation by generating contextualised embeddings in which a word's representation depends on its sentence-level context. BERT employs a bidirectional self-attention mechanism, enabling it to capture long-range linguistic dependencies and nuanced semantic relationships more effectively than earlier recurrent encoders.

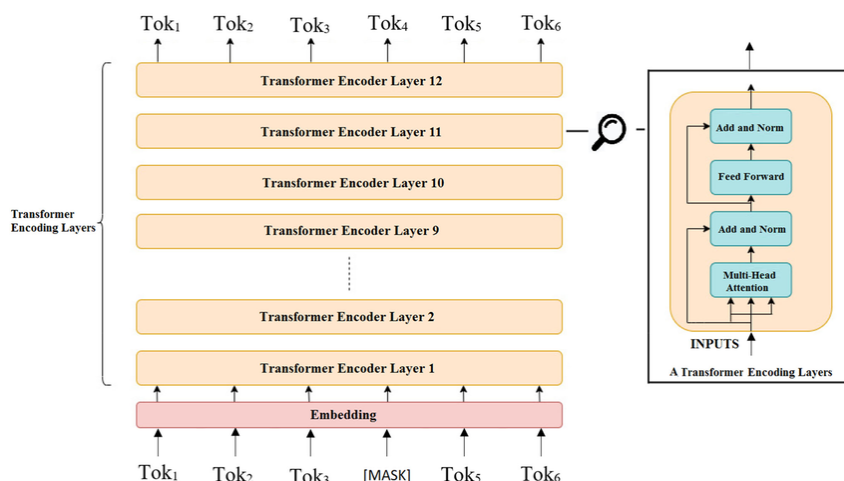


Figure 2.4: BERT model architecture (Noorian et al. 2023).

Domain-specific adaptations such as FinBERT by Araci (2019), further improve performance in financial applications fine-tuning the model on financial corpora, enhancing its ability to interpret domain-specific terminology, sentiment, and stance. Empirical studies report consistent improvements across sentiment classification and financial text analysis tasks when using such pre-trained transformer models.

Despite these advances, textual signals in financial markets remain noisy, sparse, and temporally irregular. Careful alignment between text timestamps and market data is therefore essential to prevent information leakage. Furthermore, improvements in textual representation quality do not automatically translate into economic gains, reinforcing the need for integrated evaluation within realistic trading frameworks.

2.4 Probability Calibration

In classification models, outputs are often constrained to lie between zero and one, typically through sigmoid or softmax activations, and are therefore interpreted as probabilities. However,

these values represent model confidence rather than true probabilistic quantities in the formal sense of Kolmogorov’s axioms (von Plato 2005), as demonstrated by Guo et al. (2017). A neural network does not estimate the objective probability of a stock moving up or down, rather, it produces a score reflecting patterns learned from the training data and encoded in its parameters.

Even under ideal conditions—clean data, large sample sizes, and well-optimised architectures—it is unrealistic to expect model outputs to coincide exactly with empirical outcome frequencies. Deep neural networks are known to exhibit systematic overconfidence, particularly when trained to minimise cross-entropy loss. As a result, a predicted probability of 0.7 does not necessarily imply that the event occurs approximately 70% of the time.

This distinction is particularly consequential in financial applications. Capital allocation strategies such as the Kelly Criterion and MPT explicitly require probability estimates to determine optimal position sizing. If predicted probabilities are miscalibrated, allocation decisions can systematically overestimate or underestimate risk, leading to excessive leverage, unstable trading behaviour, and increased drawdowns.

A model is therefore said to be well-calibrated when its predicted probabilities align with observed empirical frequencies. Probability calibration methods aim to correct systematic distortions in model confidence, enabling outputs to be interpreted more reliably for downstream risk-sensitive decision-making.

2.4.1 Isotonic Regression

Isotonic regression is a non-parametric post-hoc calibration technique that fits a monotonic transformation to model outputs in order to align predicted confidence scores with empirical outcome frequencies. Originally proposed for probability estimation in supervised learning by Zadrozny and Elkan (2002), isotonic regression relaxes parametric assumptions and instead learns a piecewise-constant monotonic mapping using the pool-adjacent-violators (PAV) algorithm.

This flexibility allows isotonic regression to correct systematic distortions in model confidence while preserving ranking. Empirical studies demonstrate that isotonic calibration can significantly improve probability estimates for binary and multiclass classification tasks Zadrozny and Elkan (2001); Niculescu-Mizil and Caruana (2005). However, because it is non-parametric and highly flexible, isotonic regression may overfit when calibration datasets are small, leading to unstable probability estimates in limited-data regimes Guo et al. (2017).

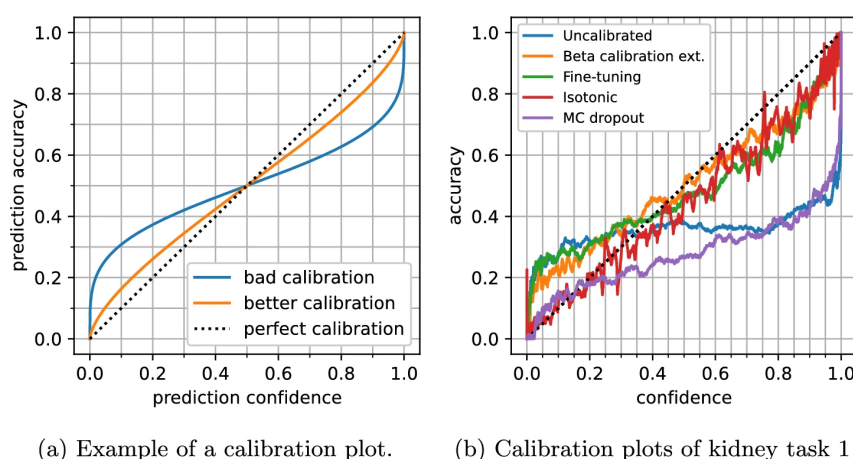


Figure 2.5: Calibration plots of an example (a) and of the results on the kidney task (b). As the plot moves closer to the diagonal, the predicted confidences become a better reflection of the accuracy, and the model is better calibrated (Rousseau et al. 2025).

2.4.2 Temperature Scaling

Temperature scaling is a parametric post-hoc calibration method introduced for modern deep neural networks by Guo et al. (2017). The method rescales the logits of a trained classifier using a single scalar temperature parameter $T > 0$, learned on a held-out validation set. The calibrated probabilities are obtained by applying the softmax function to the rescaled logits (Figure 2.6).

Unlike non-parametric methods such as isotonic regression, temperature scaling preserves the relative ranking of predictions while correcting systematic overconfidence. Empirical evidence demonstrates that deep neural networks trained with cross-entropy loss are often miscalibrated, typically exhibiting overconfident predictions, and that temperature scaling substantially reduces calibration error without affecting classification accuracy Guo et al. (2017).

Because it introduces only a single additional parameter, temperature scaling is less prone to overfitting than flexible non-parametric approaches and performs reliably even with limited calibration data. As a result, it has become a standard baseline for post-hoc probability calibration in deep learning applications.

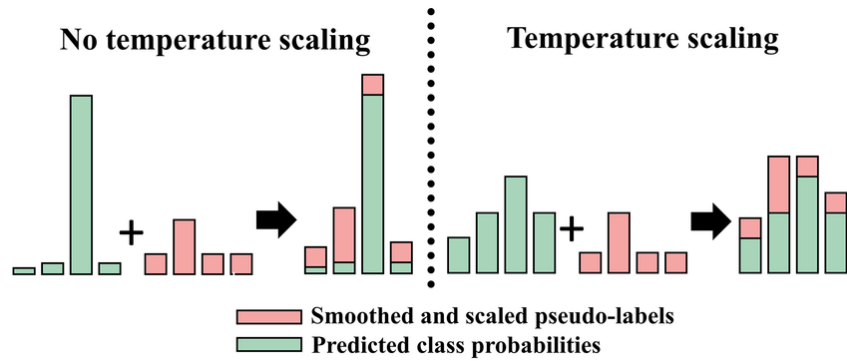


Figure 2.6: Effect of temperature scaling on ordinal pseudo-label smoothing. Without temperature scaling, predicted class probabilities remain sharp and poorly aligned with the smoothed pseudo-label distribution. Applying temperature scaling softens the predicted probabilities, improving their alignment with the target distribution and yielding better-calibrated outputs (Maracani et al. 2024).

2.5 Meta-Features and Model Self-Assessment

Financial markets are inherently non-stationary, with volatility, return dynamics, and cross-asset relationships evolving across economic cycles and structural shifts. As surveyed by Pesaran and Timmermann (2002), financial time-series exhibit high volatility, nonlinear dependencies, and regime changes associated with macroeconomic shocks, crises, and policy transitions. These characteristics imply that predictive relationships are unstable over time and that model performance may deteriorate as market conditions change.

Despite this, many predictive systems implicitly assume static reliability, treating model outputs as equally trustworthy across all market regimes. Prior forecasting errors are rarely incorporated into subsequent predictions, despite containing information about recent model performance and potential regime shifts.

Pesaran and Timmermann (2002) discuss the value of hybrid approaches in financial forecasting, combining complementary techniques such as evolutionary algorithms and neural networks to enhance predictive robustness in complex and potentially non-stationary market environments. Similarly, Varshney and Sharma (2024) show that stacked deep learning architectures can achieve

consistently lower out-of-sample forecasting errors, reporting reductions in MSE and MAE across heterogeneous time-series datasets, suggesting a potential pathway towards improved predictive stability under evolving data conditions.

Incorporating prior predictions and realised errors as meta-features extends this adaptive perspective by modelling time-varying predictive reliability directly within the forecasting pipeline. Rather than assuming constant model confidence, such an approach allows the system to adjust implicitly to recent performance degradation. This idea is conceptually related to meta-learning and adaptive model weighting, yet its integration within deep learning-based financial trading systems remains relatively limited.

2.6 Capital Allocation Strategies

Prediction alone does not determine profitability. Position sizing plays a crucial role in investment performance

2.6.1 The Kelly Criterion

Introduced by Kelly (1956), the Kelly Criterion derives the optimal fraction of capital to allocate to a favourable bet in order to maximise long-term logarithmic wealth growth (Kelly Jr. 1956). The original formulation arose from information theory and the analysis of signal transmission over noisy communication channels. It was later adapted to gambling and financial markets, where it provides a principled framework for capital allocation under probabilistic uncertainty (Brieman n.d.; Thorp 2008; MacLean et al. 2011).

For a binary outcome with success probability p and net odds b , the optimal Kelly fraction is given by

$$f^* = \frac{bp - (1 - p)}{b},$$

where:

- f^* denotes the optimal fraction of available capital to allocate to the risky position;
- p represents the estimated probability of a favourable outcome;
- $1 - p$ is the probability of an unfavourable outcome;
- b denotes the net payoff ratio (i.e., the profit per unit stake if the outcome is favourable).

obtained by maximising expected logarithmic utility. In financial applications, this formulation links predicted probabilities directly to position sizing, making the reliability of probability estimates critical for stable portfolio growth.

2.6.2 Mean-Variance Portfolio Theory (MPT)

Introduced by Markowitz (1952), MPT frames construction as an optimisation problem balancing expected return against risk (Markowitz 1952).

Formally, the mean-variance optimisation problem can be written as:

$$\max_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\mu} - \lambda \mathbf{w}^\top \Sigma \mathbf{w},$$

where:

- $\mathbf{w}$ denotes portfolio weights;

- μ expected returns;
- Σ the covariance matrix;
- λ a risk-aversion parameter.

MPT emphasises diversification and covariance structure across assets, stated by Markowitz (1959), contrasting with single-asset Kelly Sizing. While Kelly maximises growth under independent bets, MPT explicitly accounts for cross-asset risk interactions.

The integration of probability-based sizing with diversification principles motivates hybrid allocation approaches (Peterson 2018).

3 | Analysis and Research Questions

The literature on stock movement prediction reveals several consistent findings:

- Deep sequence models frequently outperform classical statistical baselines when evaluated using temporally ordered validation.
- Incorporating textual information alongside price-based features can provide incremental predictive gains.
- Architectural and training enhancements, such as adversarial learning and multi-scale modelling, improve statistical performance.

Despite these advances, important limitations remain:

- Evaluation is typically restricted to accuracy, correlation, or classification-based metrics, with limited consideration of economic outcomes.
- Filtering strategies, such as threshold-based label exclusion, are often applied, simplifying the prediction task and potentially inflating reported performance.
- Transaction costs, turnover, and capital constraints are frequently omitted from evaluation.
- Model reliability is generally assumed to be static, despite well-documented regime shifts and volatility clustering in financial markets.
- Methodological heterogeneity across studies makes cross-paper comparisons and ablation analysis difficult.

These observations motivate a more integrated formulation of the stock prediction problem that jointly considers predictive accuracy, probabilistic reliability, and capital allocation under realistic market constraints.

3.1 Problem Definition

The objective of this dissertation is to develop and evaluate an end-to-end probabilistic stock movement prediction and capital allocation framework, assessed under realistic walk-forward market conditions.

3.1.1 Prediction Task

The predictive component consists of two primary objectives.

1. **Directional Probability Estimation:** Estimate the probability that a given asset will experience a positive return over a fixed forecast horizon H .
2. **Directional Magnitude Estimation:** Model return magnitude using ordinal multi-class classification rather than continuous regression. Returns are discretised into ordered buckets, enabling estimation of expected return categories while preserving directional structure.

Formally, for asset i at time t , the model estimates:

$$\hat{p}_{i,t} = \mathbb{P}(R_{i,t+H} > 0 \mid \mathcal{F}_t),$$

where $R_{i,t+H}$ denotes the return over horizon H , and $\mathcal{F}_t$ represents the information set available at time t .

3.1.2 Decision Layer

Predicted probabilities are not the final objective. Instead, they are transformed into investment decisions through two sequential components:

1. **Probability Calibration:** Post-hoc calibration methods, including isotonic regression and temperature scaling, are applied to improve probabilistic reliability.
2. **Capital Allocation:** Calibrated probabilities are converted into position sizes using alternative allocation strategies:
 - Full Kelly
 - Fractional Kelly
 - Mean–Variance Portfolio Theory (MPT)
 - Hybrid fractional Kelly–MPT allocation

Early Exit Mechanism. For positions held over multi-day horizons ($H > 1$), the framework includes a dynamic early-exit rule: if the next-day predicted probability indicates a sufficiently high likelihood of negative return, the position is liquidated prior to maturity. This mechanism allows adaptive horizon adjustment and mitigates adverse drift under changing market conditions.

This structure enables the joint evaluation of predictive accuracy, probabilistic reliability, and economic performance.

3.2 Constraints and Assumptions

The proposed framework operates under realistic financial and statistical constraints, reflecting both empirical asset pricing practice and practical portfolio management considerations.

3.2.1 Temporal Integrity and Information Structure

Strict temporal integrity is enforced throughout the pipeline to eliminate lookahead bias. Targets are defined using forward returns, while all features are constructed using information available at or prior to time t . Model training, validation, and testing follow a walk-forward (expanding window) procedure to ensure that parameter estimation relies exclusively on historical data.

This design mirrors the predictive regression framework of Fama and French (1988), where returns over $(t, t + T)$ are regressed on state variables observed at time t , thereby preserving causal ordering and economic validity.

3.2.2 Market Non-Stationarity

Financial markets are inherently non-stationary. The data-generating process for returns, volatility, and cross-asset correlations evolves across macroeconomic regimes. Consequently:

- Model reliability is time-varying,
- Calibration may degrade across structural breaks,
- Estimated probabilities may drift out of distribution.

The allocation layer must therefore remain robust to prediction instability and regime shifts rather than assuming parameter constancy.

3.2.3 Capital, Risk, and Leverage Constraints

Capital is finite and position sizes must satisfy feasibility and risk constraints. In particular:

- Leverage limits restrict exposure,

- Portfolio weights must satisfy budget constraints,
- Estimation error can induce over-allocation.

Both Kelly sizing and Mean-Variance (MPT) optimisation are highly sensitive to input estimation error. Overconfident probability estimates or inflated Sharpe ratios can lead to excessive leverage and catastrophic drawdowns. As demonstrated in the literature on backtest overfitting and estimation error amplification (Bailey et al. 2013; de Prado 2018), robust capital allocation requires shrinkage, fractional Kelly scaling, or risk caps to mitigate tail risk.

3.3 Research Questions

This dissertation is guided by the following research questions, structured to evaluate the predictive, probabilistic, and economic components of the proposed framework.

RQ1. Predictive Performance

Does incorporating textual information alongside price-based features improve directional and magnitude prediction performance under strict temporal validation?

Specifically:

- Which transformer-based language model provides the strongest predictive signal?
- Which feature configurations perform best for:
 - Binary classification (directional prediction),
 - Ordinal multi-class classification (return magnitude),
 - Regression-based return estimation (exact price prediction)

RQ2. Probability Calibration

Does probability calibration improve the reliability and economic stability of probability-based capital allocation strategies?

In particular:

- Does calibration reduce model overconfidence?
- Does improved calibration lead to smoother capital growth under Kelly-based sizing?
- How does calibration influence drawdown behaviour and risk-adjusted performance?

RQ3. Meta-Features and Reliability Modelling

Does incorporating previous prediction outcomes and realised error statistics as meta-features improve model stability under non-stationary market conditions?

This question evaluates whether explicit modelling of time-varying predictive reliability enhances robustness and reduces performance degradation across regimes.

RQ4. Allocation Strategy Comparison

How do different capital allocation strategies compare in terms of economic performance when applied to identical predictive signals?

The strategies evaluated include:

- Full Kelly,
- Fractional Kelly,
- Mean-Variance Portfolio Theory (MPT),
- Hybrid fractional Kelly-MPT allocation.

RQ5. Interaction Effects

How do probability calibration and capital allocation method interact to influence long-term capital growth, volatility, and maximum drawdown?

This question investigates system-level effects beyond isolated component analysis.

4 | System Design

TODO: Application of the Kelly Criterion to Prediction Markets Bernhard K Meister, probability and

TODO: Walk-Forward/ expanding window training in simulation, White 2000, bailey et al. 2014, de prado 2018, gu, kelly and xiu 2010

This chapter describes the conceptual architecture of the proposed framework. The system is designed as an end-to-end probabilistic decision pipeline, transforming raw market and textual information into calibrated investment allocations evaluated under walk-forward simulation. The focus of this chapter is the structural design of the framework rather than implementation-specific details.

4.1 Overall Architecture

The proposed system follows a structured, layered architecture that transforms raw financial and textual data into capital allocation decisions through a strictly sequential pipeline.

The architecture consists of the following high-level components:

1. **Data Ingestion and Preprocessing.** Stock price data and associated textual data (e.g., social media posts) are collected and cleaned and joined. T
2. **Feature Engineering.** Tweets are passed through transformer-based language models to extract contextual representations and are then aggregated for each ticker. In parallel, technical indicators are computed from historical price series. Sector-level indicators are also incorporated. These feature sets are merged to form a temporally aligned initial master dataset.
3. **Initial Predictive Benchmarking.** The constructed dataset is used to benchmark predictive performance across multiple model families. At this stage, models are primarily evaluated using their raw outputs in order to assess statistical predictive capability independently of downstream decision mechanisms. Probability calibration is therefore not applied to the binary classification or regression benchmarks. However, for the ordinal multi-class formulation, temperature scaling is applied to stabilise the softmax probability distribution, as the predicted class probabilities are subsequently used to estimate expected return across ordered outcome buckets.
4. **Meta-Feature Construction.** A second-stage dataset is constructed by augmenting the master dataset with meta-features derived from historical predictions and realised errors. These features capture time-varying predictive reliability in a leak-safe manner.
5. **Enhanced Predictive Benchmarking.** Models are re-evaluated using the augmented dataset to assess whether reliability-aware signals improve predictive stability and accuracy.
6. **Investment Simulation and Allocation.** Within the simulation layer, models are trained on a per-asset basis under a walk-forward regime. Predicted probabilities are optionally calibrated before being mapped to position sizes using selected capital allocation strategies. Portfolio performance is then evaluated under realistic trading assumptions.

All components operate under strict temporal integrity constraints, ensuring that predictions and allocations are generated using only information available at the decision time.

The framework supports multiple predictive formulations in parallel, including binary classification, ordinal multi-class classification, and regression-based return estimation. These model outputs are integrated through stacking mechanisms, whereby higher-level predictors learn from base-model outputs and their historical performance characteristics.

The overall architecture therefore constitutes a probabilistic decision system rather than a standalone predictive model.

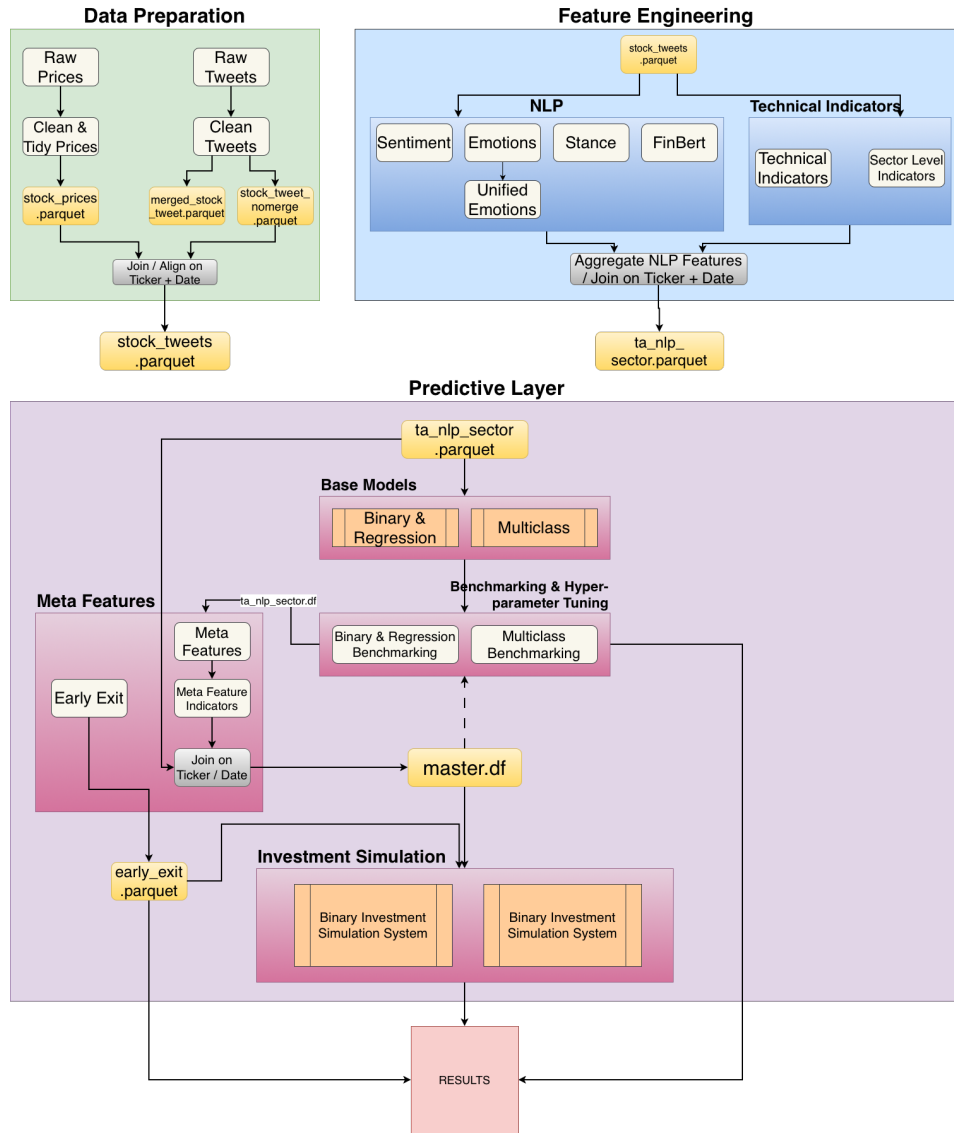


Figure 4.1: End-to-end architecture of the proposed prediction and investment framework.

4.2 Data Flow and Temporal Integrity

Maintaining temporal integrity is central to the system design. All components operate under strict chronological constraints to prevent look-ahead bias.

At time t , the information set $\mathcal{F}_t$ includes:

- Historical price data up to time t ,
- Derived technical indicators,
- Textual features extracted from documents published no later than time t ,
- Historical model predictions and lagged error-based reliability signals computed strictly from data available before time t .

Model training and evaluation follow a walk-forward structure. Parameters are estimated on historical windows and applied to future unseen data without reusing future information. Meta-features are constructed in a leak-safe manner using lagged reliability statistics.

This design ensures that all reported predictive and economic results reflect realistic deployment conditions.

4.3 Model Families and Outputs

The framework supports three predictive formulations.

4.3.1 Binary Classification

Binary models estimate the probability of a positive return over a variable forecast horizon H . The output is a probabilistic estimate of directional movement.

4.3.2 Ordinal Multi-Class Classification

Instead of modelling returns as continuous variables, ordinal classification discretises returns into ordered magnitude buckets. This design preserves directional information while allowing estimation of expected return categories. The ordered structure enables modelling of both direction and magnitude without assuming linear return dynamics.

4.3.3 Regression-Based Prediction

Regression models directly estimate expected return values. For evaluation consistency, regression outputs can be transformed into directional signals by considering the sign of the predicted return.

These three model families allow systematic comparison between probabilistic, ordinal, and continuous return formulations under identical evaluation conditions.

4.4 Calibration and Meta-Learning Design

Raw predictive probabilities are not assumed to represent well-calibrated estimates of true event likelihoods. However, probability calibration is not applied during the predictive benchmarking stage aside from ordinal multi-class formulation. Instead, calibration is introduced within the investment simulation layer, where predicted probabilities directly influence capital allocation decisions.

Post-Hoc Calibration in Simulation

During simulation, model outputs are transformed using post-hoc calibration methods (e.g., isotonic regression or temperature scaling) prior to capital allocation. This ensures that probability magnitudes used for staking reflect more reliable confidence estimates.

Importantly, calibration is applied only after walk-forward prediction generation. At decision time t , the model prediction is produced using information available up to $t - 1$, and calibration is fitted strictly on historical data within the walk-forward training window. This preserves temporal integrity and prevents lookahead bias.

Meta-Learning and Reliability Modelling

Beyond calibration, the framework incorporates model self-assessment through the construction of reliability-aware meta-features. These features are derived from historical model predictions and realised outcomes under a walk-forward regime.

Stacking and Reliability Integration

These reliability signals are incorporated into extended datasets that augment base predictive features. In this design, stacking mechanisms allow secondary models to learn how base-model outputs behave over time, effectively modelling time-varying predictive quality.

Rather than assuming static model reliability, the system explicitly accounts for performance drift and regime-dependent volatility. This layered design separates predictive modelling, probability correction, and reliability modelling, enabling controlled evaluation of their individual and combined economic effects.

TODO: stacking diagram

4.5 Benchmarking and Investment Simulation Design

The framework separates predictive benchmarking from full investment simulation. This separation enables rapid model experimentation while preserving a realistic economic evaluation stage.

Predictive Benchmarking Layer

The benchmarking layer is designed for efficient model comparison and hyperparameter tuning. Using walk-forward evaluation, models are assessed on statistical predictive metrics without probability calibration or capital allocation.

This stage serves two primary purposes:

- Rapid comparison of model families (binary, regression, ordinal multi-class),
- Systematic hyperparameter tuning and feature selection.

Raw model outputs are evaluated directly to measure predictive quality in isolation. By excluding calibration and allocation mechanisms at this stage, the framework ensures that statistical performance is not conflated with downstream economic effects.

Investment Simulation Layer

The investment simulation layer extends the predictive outputs into capital allocation decisions under realistic trading constraints.

At each decision point, model predictions—generated using only historical data—are optionally passed through a post-hoc calibration stage to improve probabilistic reliability. Calibration is applied exclusively within the simulation layer, as probability magnitudes directly influence position sizing.

Calibrated probabilities are then mapped to capital allocations according to a selected staking rule. The system supports:

- Growth-optimal allocation (Kelly-based),
- Fractional Kelly sizing,
- Mean-Variance portfolio optimisation,
- Hybrid allocation combining growth and diversification principles.

For multi-day positions, an adaptive early-exit mechanism allows dynamic liquidation if subsequent predictions indicate deteriorating conditions.

Portfolio value evolves multiplicatively through time, incorporating transaction costs and rebalancing effects. Performance is continuously tracked using growth and risk metrics, including volatility, drawdown, and turnover.

By holding predictive inputs constant while varying calibration and allocation logic, the simulation layer isolates the economic contribution of probability reliability and capital allocation design.

5 | Implementation

This chapter describes the practical implementation of the proposed framework. While the previous chapter outlined the conceptual architecture of the system, this chapter focuses on the technical realisation of each component, including data preparation, feature construction, model training procedures, and the investment simulation environment.

Particular attention is given to ensuring temporal integrity, reproducibility, and robustness in the experimental pipeline.

5.1 Data Acquisition and Pre-processing

The dataset used in this project is based on the StockNet benchmark introduced by Xu and Cohen (2018). It spans the period from 1 January 2014 to 1 January 2016 and contains 88 U.S. stocks across nine sectors. The selection focuses on highly capitalised firms, providing both sectoral diversity and broad market coverage.¹

Price data consists of daily OHLCV variables obtained from Yahoo Finance, while textual data consists of time-stamped Twitter posts associated with individual tickers. Pre-processing includes tweet cleaning and deduplication, ticker/date alignment, chronological sorting, and forward target construction. The resulting data sources are merged into a unified time-indexed master dataset. **TODO: dataset sizes**

5.2 Feature Engineering

Feature engineering combines structured market signals, tweet-derived signals, and reliability-aware meta-features. **TODO: Appendix showing master_df dataset columns**

5.2.1 Technical Indicators

Technical indicators are generated at both the stock and sector levels in order to capture trend, momentum, volatility, and volume dynamics. This allows the model to learn from both asset-specific behaviour and broader sector structure.

At the stock level, the implementation computes exponential moving averages (EMA), MACD, RSI, Stochastic RSI, ATR, Bollinger Bands, OBV, and rolling returns.

```
def apply_ta_indicators(df_group):
    df_group = df_group.copy()
    df_group.set_index(pd.DatetimeIndex(df_group['date']), inplace=True)

    df_group.ta.ema(length=12, append=True)
    df_group.ta.ema(length=26, append=True)
```

¹Add link to the StockNet repository here.

```

df_group.ta.ema(length=50, append=True)
df_group.ta.macd(fast=12, slow=26, signal=9, append=True)
df_group.ta.rsi(length=14, append=True)
df_group.ta.stochrsi(length=14, append=True)
df_group.ta.atr(length=14, append=True)
df_group.ta.obv(append=True)

bb = ta.bbands(df_group['close'], length=20, std=2)
df_group['BB_upper'] = bb['BBU_20_2.0']
df_group['BB_middle'] = bb['BBM_20_2.0']
df_group['BB_lower'] = bb['BBL_20_2.0']

return df_group.reset_index(drop=True)

```

Listing 5.1: Per-ticker technical indicator generation.

To incorporate sector context, average OHLCV values are first computed for each sector-date pair, and the same technical analysis pipeline is then applied to these aggregated sector series before merging them back to ticker-level rows.

```

for col in ['open', 'high', 'low', 'close', 'volume']:
    ta_nlp[f'sector_{col}_mean'] = (
        ta_nlp.groupby(['date', 'sector'])[col].transform('mean')
    )

```

Listing 5.2: Construction of sector-average features.

5.2.2 Textual Features

Textual information is converted into structured daily signals through a staged NLP pipeline. Signals are first computed at the tweet level and then aggregated by ticker and date before merging with stock and technical features.

Four main classes of textual features are extracted:

- **General sentiment**, using a BERT sentiment model;
- **Emotion scores**, using an emotion classification transformer;
- **Financial stance**, using a finance-tuned RoBERTa model;
- **FinBERT sentiment**, producing positive, negative, and neutral financial sentiment probabilities.

```

tokenizer = AutoTokenizer.from_pretrained(
    "nlptown/bert-base-multilingual-uncased-sentiment"
)
bert_model = AutoModelForSequenceClassification.from_pretrained(
    "nlptown/bert-base-multilingual-uncased-sentiment"
)

def cal_sentiment_score(text):
    inputs = tokenizer.encode(
        text, return_tensors="pt", truncation=True, max_length=512
    ).to(device)
    with torch.no_grad():

```

```

    outputs = bert_model(inputs)
    return int(torch.argmax(outputs.logits)) + 1

```

Listing 5.3: Per-tweet sentiment scoring.

```

classifier = pipeline(
    "text-classification",
    model="j-hartmann/emotion-english-distilroberta-base",
    top_k=None,
    device=device
)

def emotion_analysis(text):
    analysis = classifier(text, top_k=None)
    return [e["score"] for e in analysis]

```

Listing 5.4: Tweet-level emotion scoring.

```

def stance_score(text):
    result = nlp(text)
    return result[0]["label"], result[0]["score"]

```

Listing 5.5: Financial stance detection.

```

classifier = pipeline(
    "text-classification",
    model="ProsusAI/finbert",
    top_k=None,
    device=0 if torch.cuda.is_available() else -1
)

```

Listing 5.6: Batch FinBERT inference.

After tweet-level inference, the resulting signals are aggregated at the daily ticker level using summary statistics such as mean scores and class proportions. Missing tweet-derived features on non-tweet days are filled with neutral defaults so that all trading days remain represented.

5.2.3 Meta-Features

In addition to market and textual features, the framework incorporates meta-features derived from prior model predictions. These are intended to capture time-varying predictive reliability under non-stationary market conditions.

Meta-features are generated using a walk-forward out-of-sample procedure with periodic refitting. For each ticker, a model is trained on an initial historical window, used to predict a future block, then refit on an expanded window before the next block is predicted. This ensures that all meta-features are based on genuine historical forecasts rather than in-sample fits.


```

k = 0
while k < len(valid_idx):
    train_end = valid_idx[k]
    pred_start = train_end + 1
    pred_end = min(pred_start + refit_interval - 1, valid_idx[-1])

    train_df = df_tkr.iloc[:train_end + 1].copy()
    pred_df = df_tkr.iloc[pred_start:pred_end + 1].copy()

    scaler = StandardScaler()
    X_train = scaler.fit_transform(train_df[feature_cols])
    X_pred = scaler.transform(pred_df[feature_cols])

    model.fit(X_train, y_train)
    preds = model.predict(X_pred)

    k += refit_interval

```

Listing 5.7: Walk-forward out-of-sample meta-feature generation.

Two types of meta-features are constructed from the two models and their bi-directional variants:

- **Regression-based**, including predicted returns, absolute errors, squared errors, and directional correctness;
- **Classification-based**, including predicted up-probabilities, Brier scores, log-loss, and correctness indicators.

```

df["regression_abs_err_ret_1d"] = (
    df["ret_1d_meta"] - df["regression_pred_ret_1d"]
).abs()

df["brier"] = (df["prob_up_1d"] - df["up_1d_meta"]) ** 2
df["logloss"] = -(
    df["up_1d_meta"] * np.log(df["prob_up_1d"]) +
    (1 - df["up_1d_meta"]) * np.log(1 - df["prob_up_1d"])
)

```

Listing 5.8: Regression and classification reliability features.

To prevent leakage, all rolling reliability statistics are computed only after applying a one-step shift. Consequently, meta-features available at time t depend only on prediction outcomes observed up to $t - 1$.

```

df["regression_mae_20"] = (
    df["regression_abs_err_ret_1d"].shift(1).rolling(20).mean()
)
df["brier_20"] = df["brier"].shift(1).rolling(20).mean()
df["logloss_20"] = df["logloss"].shift(1).rolling(20).mean()
df["acc_20"] = df["correct"].shift(1).rolling(20).mean()

```

Listing 5.9: Leak-safe rolling reliability features.

5.3 Target Construction and Horizon Design

Targets are constructed for three predictive formulations: return regression, binary direction prediction, and ordinal multi-class return bucketing. In addition, variable-horizon targets are generated so that positions can be evaluated over different holding periods H .

5.3.1 Regression and Binary Targets

For a given asset at time t , the H -day forward return is defined as

$$r_t^{(H)} = \frac{P_{t+H}}{P_t} - 1,$$

where P_t denotes the closing price at time t . The corresponding binary direction target is

$$y_t^{(H)} = \begin{cases} 1 & \text{if } r_t^{(H)} > 0 \\ 0 & \text{otherwise.} \end{cases}$$

```
def add_horizon_targets(df, H, price_col='close'):
    df = df.sort_values(['ticker', 'date']).copy()
    df[f'ret_{H}d'] = df.groupby('ticker')[price_col].shift(-H)/df[price_col] - 1
    df[f'target_{H}d'] = (df[f'ret_{H}d'] > 0).astype(int)
    return df
```

Listing 5.10: Construction of H -day return and binary targets.

5.3.2 Ordinal Multi-Class Targets

For the ordinal formulation, returns are discretised into ordered buckets

$$\mathcal{B} = \{-\infty, -0.02, -0.005, 0, 0.005, 0.02, \infty\},$$

producing $K = 6$ return classes. The class index is

$$c_t = \text{bucket}(r_t^{(H)}).$$

```
BUCKET_BINS = [-np.inf, -0.02, -0.005, 0.0, 0.005, 0.02, np.inf]

def bucketize_with_edges(x, edges):
    return np.clip(
        np.digitize(x, edges[1:-1], right=True),
        0,
        len(edges)-2
    )
```

Listing 5.11: Ordinal return bucket construction.

5.3.3 Sequence Horizon Design

Models operate on rolling feature windows of length L :

$$X_t = [x_{t-L+1}, \dots, x_t],$$

paired with the future target $r_t^{(H)}$. Training sequences are created by sliding this window across contiguous historical observations.

```
for i in range(seq_len-1, usable_end+1):
    X.append(data[i-seq_len+1:i+1])
    y.append(targets[i+H])
```

Listing 5.12: Sequence construction for horizon prediction.

5.4 Model Selection, Training and Calibration

5.4.1 Hyperparameter Tuning and Feature Selection

Hyperparameter tuning and feature selection are performed using Optuna (Akiba et al. 2019). This is particularly suitable for the project because it supports efficient black-box optimisation while allowing search spaces to be defined dynamically.

Feature sets are organised hierarchically into three groups:

- **Base:** price-based and technical features together with meta-features and their associated error statistics;
- **NLP:** all Base features augmented with textual signals such as sentiment, emotion, stance, and FinBERT sentiment;
- **Sector:** all NLP features further augmented with sector-level indicators.

For binary and ordinal classification, the tuning objective is to maximise MCC. For regression, the objective is to minimise MAE. The search space includes model family, feature set, sequence length, hidden dimensions, dropout, batch size, learning rate, weight decay, and early stopping settings.

```
def objective(trial):
    params = {
        'feature_set': trial.suggest_categorical(
            'feature_set', list(group_sets)
        ),
        'model_type': trial.suggest_categorical(
            'model_type', ['LSTM', 'BiLSTM', 'GRU', 'BiGRU']
        ),
        'sequence_length': trial.suggest_int(
            'sequence_length', 5, 20, step=5
        ),
        'hidden1': trial.suggest_categorical('hidden1', [64, 128, 256]),
        'hidden2': trial.suggest_categorical('hidden2', [32, 64, 128]),
        'dropout': trial.suggest_float('dropout', 0.0, 0.8, step=0.1),
        'learning_rate': trial.suggest_float(
            'learning_rate', 1e-6, 1e-2, log=True
        )
    }
```

```
    ),
}
```

Listing 5.13: Optuna-based hyperparameter tuning.

The resulting Optuna configurations are stored in CSV files and reused during benchmarking and simulation, allowing expensive search procedures to be decoupled from final model evaluation.

5.4.2 Model Training and Calibration

Models used in the investment simulation are not tuned inside the simulation itself. Instead, the best-performing architectures and hyperparameter configurations identified during benchmarking are loaded from the stored Optuna results.

For each selected horizon H , a separate model is trained independently for each asset using the best configuration associated with that horizon. This allows the framework to specialise predictions to individual tickers while remaining consistent with benchmark-validated settings.

```
optuna_df = pd.read_csv(OPTUNA_CSV)
best_cfg = optuna_df.iloc[0]
```

Listing 5.14: Loading the best benchmarking configuration.

Before capital allocation, predicted probabilities are optionally calibrated to improve probabilistic reliability.

Temperature scaling. For ordinal multi-class prediction, temperature scaling is used to rescale logits while preserving class ranking.

```
def temperature_scale(logits, T):
    return torch.softmax(logits / T, dim=1)
```

Listing 5.15: Temperature scaling.

Isotonic regression. For binary simulation, isotonic regression is used as a non-parametric calibration method fitted on validation probabilities.

```
iso = IsotonicRegression(out_of_bounds="clip")
iso.fit(val_probs, val_labels)
calibrated_probs = iso.predict(test_probs)
```

Listing 5.16: Isotonic regression calibration.

These calibration methods ensure that probabilities passed to the capital allocation layer more closely reflect empirical outcome frequencies.

5.5 Investment Simulation Framework

The investment simulation layer converts predictive model outputs into sequential trading decisions. Predictions are generated using the model configurations identified during the bench-

marking stage and applied within a walk-forward framework that mimics realistic deployment conditions.

5.5.1 Per-Asset Training and Walk-forward Retraining

Models are trained independently for each asset to capture asset-specific dynamics for a prediction horizon H .

Training follows an expanding walk-forward procedure. At each decision point, the model is trained using historical observations up to time t , then used to generate predictions for future observations. The model is periodically refit as new data becomes available.

```
for t in range(train_start, len(data)-H):
    train_df = data.iloc[:t]
    model.fit(X_train, y_train)

    pred = model.predict(X_t)
```

This design ensures that all predictions are produced strictly out-of-sample.

5.5.2 Portfolio Construction

Before capital allocation, the simulation first determines which assets are eligible for investment on a given day. This portfolio construction stage acts as a filtering and diversification layer: it removes weak signals, reduces concentration, and ensures that subsequent capital allocation is performed only over a smaller set of higher-conviction candidates. The binary and ordinal multi-class simulations apply this filtering differently, but both ultimately use a softmax-based action selection rule within each sector before proceeding to position sizing.

Binary System In the binary simulation, each ticker receives a calibrated probability of an upward move, denoted p_i . The first portfolio gate is applied at the *sector level*. For each sector s , the simulation computes the mean calibrated probability across all stocks in that sector on that day:

$$\bar{p}_s = \frac{1}{|S_s|} \sum_{i \in S_s} p_i.$$

A sector is retained only if

$$\bar{p}_s \geq \tau_{\text{sector}},$$

where τ_{sector} is the sector threshold. This means the binary simulation does *not* filter stocks individually at the first stage. Instead, it asks whether the sector as a whole is sufficiently strong on average. If the answer is no, all stocks in that sector are discarded for that day.

```
df = daily_predictions_df.copy()
df = df.groupby('sector').filter(
    lambda g: g['calibrated_prediction'].mean() >= sector_threshold
)
```

Listing 5.17: Sector-level filtering in the binary simulation.

After the surviving sectors are identified, the simulation evaluates individual stocks within those sectors. For ticker i , the Kelly edge is computed as

$$f_i^* = p_i - \frac{1 - p_i}{b_i},$$

where b_i is the historical payout ratio. If average winning and losing returns are available, expected return may also be estimated as

$$\hat{r}_i = p_i \bar{r}_i^+ - (1 - p_i) \bar{r}_i^-.$$

Only stocks with positive edge are retained as candidates for selection.

Multi-Classification System In the ordinal multi-class simulation, each ticker receives a probability distribution over ordered return buckets. Let $P_i(c)$ denote the probability of bucket c , and let μ_c denote the historical mean return associated with that bucket. The expected return for ticker i is then computed as

$$\hat{r}_i = \sum_{c=0}^{K-1} P_i(c) \mu_c.$$

The probability of an upward move is obtained by summing the probabilities of all buckets corresponding to non-negative returns:

$$p_i^\uparrow = \sum_{c \in \mathcal{C}_+} P_i(c),$$

where $\mathcal{C}_+$ is the set of buckets whose bin edge is non-negative.

```
Pc = ordinal_to_class_probs(P_cum)
expected_rise = float((Pc[0] * mu_c).sum())

positive_buckets = [i for i in range(K_used) if BUCKET_BINS[i] >= 0]
prob_up = float(Pc[0, positive_buckets].sum())
```

Listing 5.18: Expected return and up-probability in the ordinal simulation.

Unlike the binary simulation, the ordinal multi-class simulation does *not* apply a sector-average confidence filter. Instead `ticker_threshold` is used at the *ticker level*. A ticker survives only if the expected return is above the threshold and it is positive.

$$p_i^\uparrow \geq \tau_{\text{ticker_threshold}} \quad \text{and} \quad \hat{r}_i > 0.$$

Thus, the filter is applied row-by-row rather than by first averaging over the sector.

```
df = df[(df['prob_up'] >= ticker_threshold) & (df['expected_rise'] > 0)]
```

Listing 5.19: Ticker-level filtering in the ordinal simulation.

After this filtering stage only valid candidates are retained.

Softmax action selection For both binary and ordinal multi-class simulations, the final stock choice is made using the same softmax-style action selection mechanism. Within each surviving sector, candidates are ranked, the top- K names are retained, and a softmax distribution is applied over their scores. In the binary case, the ranking score is the calibrated probability; in the ordinal case, it is the expected return. One stock is then sampled from this sector-specific top- K set.

Formally, for scores $z_1, \dots, z_K$ within a sector, the selection probability of candidate j is

$$\Pr(j) = \frac{\exp(\alpha z_j)}{\sum_{m=1}^K \exp(\alpha z_m)},$$

where α controls how sharply the selection favours the highest-ranked candidate.

```
top_group = group.sort_values(score_col, ascending=False).head(max(1, top_k))

logits = top_group[score_col].to_numpy() * 100.0 * alpha
logits = logits - logits.max()
weights = np.exp(logits)
probs = weights / weights.sum()

choice_idx = np.random.choice(len(top_group), p=probs)
choice = top_group.iloc[choice_idx]
```

Listing 5.20: Softmax-based action selection within each sector.

This means portfolio construction is not a simple ranking step. Instead, it combines filtering, diversification, and probabilistic action selection before capital allocation is applied. In the binary simulation, filtering begins with a sector-level average-confidence gate; in the ordinal simulation, filtering is applied directly to individual tickers using both upward probability and expected return. In both cases, softmax action selection is then used within each surviving sector to determine the final candidate trades.

5.5.3 Capital Allocation

Predicted probabilities or expected returns are converted into position sizes using risk-aware allocation rules. The primary allocation strategy is based on the Kelly criterion, which determines the optimal fraction of capital to wager given an edge in probabilistic forecasts.

```
kelly_fraction = (p * payoff - (1 - p)) / payoff
stake = capital * kelly_fraction * kelly_multiplier
```

Listing 5.21: Fractional Kelly position sizing.

While Kelly sizing determines the amount of capital allocated to individual positions, cross-asset portfolio allocation is further refined using Mean-Variance Portfolio Theory (MPT). Given expected returns μ and covariance matrix Σ . **TODO: fix formula**

```
def compute_mpt_weights(mu, Sigma, risk_lambda=1.0):
    import scipy.optimize as opt

    def obj(w):
```

```

return -(mu @ w - risk_lambda * (w @ Sigma @ w))

cons = ({'type': 'eq', 'fun': lambda w: np.sum(w) - 1.0},)
bounds = [(0.0, 1.0)] * len(mu)

x0 = np.ones(len(mu)) / len(mu)
res = opt.minimize(obj, x0, method='SLSQP',
                  bounds=bounds, constraints=cons)
return res.x

```

Listing 5.22: Mean–variance portfolio weight optimisation.

A hybrid Kelly–MPT strategy is also implemented. In this approach, Kelly sizing first determines the candidate trade budget based on predicted probabilities, after which MPT redistributes this capital across selected assets using their expected returns and covariance structure.

```

kelly_budget = capital * kelly_fraction

weights = compute_mpt_weights(expected_returns, covariance_matrix)

investment_amounts = kelly_budget * weights

```

Listing 5.23: Hybrid Kelly–MPT allocation.

This hybrid approach allows the system to incorporate both probabilistic forecast confidence (via Kelly sizing) and cross-asset diversification effects (via mean–variance optimisation).

5.5.4 Early Exit Mechanism

Positions may be closed prior to the full horizon if subsequent predictions indicate deteriorating expected returns. At each new time step $t + 1$, the model re-evaluates the position using updated inputs.

If the predicted return falls below a defined threshold, the position is liquidated early. **TODO: early exit df columns in appendix and graphs for early exit**

$$\text{Exit if } \hat{r}_{t+1,H} < \theta$$

```

if predicted_next_day_return < early_exit_threshold and regression_direction ==
    -1 and regression_confidence > 0.5:
    close_position()

```

This mechanism reduces exposure during adverse market movements and improves risk control. Auxiliary models (LSTM, GRU, BiLSTM, BiGRU), stored in `early_exit_df.parquet`, is trained using historical data up to the 100-day mark. The model then generates next-day directional predictions for each asset using both binary classification (up or down), regression-based return direction and its confidence in its prediction are then used to trigger an early exit.

5.5.5 Portfolio Tracking and Constraints

Portfolio value evolves multiplicatively as positions are opened and closed over time. If V_t denotes portfolio value at time t , then after applying position returns:

$$V_{t+1} = V_t \left(1 + \sum_i w_i r_{i,t} \right)$$

where w_i is the capital weight allocated to asset i .

```
portfolio_value *= (1 + np.sum(weights * realised_returns))
```

Transaction costs, capital limits, drawdown controls, and position size constraints are applied to ensure realistic trading behaviour. The specific parameter values used in the simulation (including transaction fee assumptions, capital caps, and risk limits) are provided in Appendix A.1.

5.6 Reproducibility and Robustness Measures

Several safeguards are implemented to ensure that results are reproducible and free from methodological bias.

5.6.1 Random Seed Control and Deterministic Execution

To improve reproducibility and ensure fair comparison across experiments, a fixed random seed of 42 is used throughout the notebooks. This seed is applied consistently to Python's random module, NumPy, and PyTorch. For GPU execution, deterministic cuDNN behaviour is enabled and benchmark mode is disabled to reduce nondeterministic variation between runs.

```
def set_global_seeds(seed: int = 42):
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)
    torch.backends.cudnn.deterministic = True
    torch.backends.cudnn.benchmark = False

set_global_seeds(42)
```

Listing 5.24: Global seed control for reproducible execution.

This does not guarantee bitwise-identical results in every environment, but it substantially improves experimental consistency and comparability across model configurations.

5.6.2 Data Handling

Due to the non-continuous nature of real-world financial data, the system partitions each ticker's history into contiguous trading periods before sequence generation. This ensures that recurrent models are trained only on temporally coherent windows and do not implicitly treat large gaps in observation as ordinary day-to-day transitions.

Within each contiguous period, rolling input sequences of length L are constructed and paired with their corresponding H -step-ahead targets. Periods that are too short to support both the input window and the forward horizon are discarded.

```
def create_contiguous_sequences(data: np.ndarray, targets: np.ndarray,
                               contiguous_periods: list,
                               sequence_length: int, H: int):

    X, y = [], []

    for start_idx, end_idx in contiguous_periods:
        usable_end = end_idx - H
        period_length = usable_end - start_idx + 1

        if period_length < sequence_length + 1:
            continue

        for i in range(start_idx + sequence_length - 1, usable_end + 1):
            X.append(data[i - sequence_length + 1 : i + 1])
            y.append(targets[i + H])

    return np.array(X) if X else np.array([]), np.array(y) if y else np.array([])
```

Listing 5.25: Construction of contiguous training sequences.

5.6.3 Chronological Splits and No-Shuffle Training

Financial time-series exhibit strong temporal dependence; therefore, datasets are split chronologically rather than randomly. Training always uses earlier observations, while validation and test sets contain later observations.

This approach follows established recommendations for time-series forecasting (Arratia et al. 2025).

```
X_tr, X_val, y_tr, y_val = train_test_split(X_seq, y_seq, test_size=0.2,
                                             shuffle=False)

train_loader = DataLoader(
    train_ds,
    batch_size=train_bs,
    shuffle=False,
    drop_last=False,
    num_workers=NUM_WORKERS_TRAIN,
    pin_memory=PIN_MEMORY,
    persistent_workers=PERSISTENT_WORKERS,
)

val_loader = DataLoader(
    val_ds,
    batch_size=val_bs,
    shuffle=False,
    drop_last=False,
    num_workers=NUM_WORKERS_EVAL,
    pin_memory=PIN_MEMORY,
```

```
    persistent_workers=PERSISTENT_WORKERS,  
)
```

Listing 5.26: Chronological splitting and no-shuffle data loading.

Shuffling is explicitly disabled to preserve temporal ordering.

Leakage Prevention Strict temporal integrity is enforced throughout the pipeline. Features are computed only from information available prior to the prediction time, and rolling statistics are constructed using lagged values.

For example, rolling reliability metrics use a one-step shift before aggregation:

```
df["mae_20"] = (  
    df["abs_error"]  
    .shift(1)  
    .rolling(20)  
    .mean()  
)
```

This ensures that features available at time t depend only on information observed up to $t - 1$, preventing lookahead bias and maintaining realistic evaluation conditions.

6 | Experiments & Results

6.1 Evaluation Metrics

Evaluation is conducted across three complementary layers: predictive performance, probabilistic reliability, and economic outcomes. This multi-layered framework ensures that statistical accuracy, probability validity, and financial viability are jointly assessed.

Classification and Ordinal Tasks. For directional and ordinal multi-class prediction, the primary metric is the Matthews Correlation Coefficient (MCC), which is particularly appropriate for imbalanced classification problems and was used in prior stock prediction literature Xu and Cohen 2018 allowing for direct comparison.

For binary classification, MCC is defined as:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}},$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives respectively.

Additional classification metrics include:

- F1-score,
- Precision,
- Recall,
- Directional Accuracy.

Regression Tasks. For return magnitude estimation, the primary metric is Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|.$$

MAE is preferred over Mean Squared Error (MSE) in financial return forecasting because financial returns exhibit heavy tails and outliers, with MAE being less sensitive to extreme deviations it therefore provides a more robust estimate of average prediction error (Brassington 2017).

To enable direct comparison with classification models, regression outputs are converted into directional signals based on the sign of the predicted return, and the MCC is computed on the resulting binary outcomes.

6.1.1 Calibration and Reliability Metrics

To assess probabilistic reliability, both classification-based and regression-based calibration metrics are used for meta features.

Classification Calibration.

- Brier Score
- Log Loss (Cross-Entropy)
- Rolling-window reliability measures (e.g., 20-day brier score, log loss and directional accuracy)

These metrics evaluate the consistency between predicted probabilities and realised outcomes.

Regression-Based Reliability Metrics. For meta-modelling of predictive reliability, additional diagnostics are computed:

- MAE and MSE on next-day realised returns,
- Directional accuracy derived from regression sign,
- Rolling-window reliability measures (e.g., 20-day MAE, RMSE, and directional accuracy).

Rolling reliability statistics are constructed in a leak-safe manner, ensuring that only historical prediction errors are used when generating meta-features.

6.1.2 Economic Metrics

The ultimate objective of the framework is economic viability rather than statistical accuracy alone. Allocation strategies are therefore evaluated using established portfolio performance metrics that capture both growth and risk characteristics.

- **Cumulative Return (Return on Investment (ROI)):**

Cumulative return measures total capital growth over the evaluation period and reflects the compounding effect of sequential returns. It is fundamental in assessing whether a strategy generates positive long-term wealth (Thorp n.d.; Elton et al. 2007).

$$CR = \prod_{t=1}^T (1 + r_t) - 1,$$

where r_t denotes the portfolio return at time t .

- **Volatility:**

Volatility represents the dispersion of portfolio returns and serves as a proxy for total risk. In mean-variance theory, variance (or standard deviation) is the primary measure of portfolio risk (Markowitz 1952).

$$\sigma = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2},$$

where $\bar{r}$ is the mean portfolio return.

- **Sharpe Ratio:**

Originally termed the “reward-to-variability ratio,” the Sharpe Ratio was introduced by Sharpe (1994) as a measure of risk-adjusted performance. It evaluates excess return relative to return volatility, allowing comparison of portfolios on a risk-normalised basis.

$$\text{Sharpe} = \frac{\bar{r} - r_f}{\sigma},$$

where r_f is the risk-free rate and σ is the standard deviation of returns.

- **Maximum Drawdown:**

Maximum drawdown quantifies the largest peak-to-trough decline in portfolio value over the evaluation horizon. It captures downside risk and capital erosion, which are particularly important under leverage-sensitive allocation methods such as Kelly sizing.

$$\text{MDD} = \max_{t \in [1, T]} \left(\frac{\max_{s \leq t} P_s - P_t}{\max_{s \leq t} P_s} \right),$$

where P_t denotes portfolio value at time t .

All economic metrics are computed under walk-forward simulation with transaction costs incorporated, ensuring realistic assessment of capital growth, risk exposure, and trading frictions.

6.2 Experimental Setup

The evaluation is divided into two stages. First, models are benchmarked to assess their predictive performance under fixed train, validation, and test splits. Second, the best-performing configurations are transferred into the investment simulation, where they are retrained in a walk-forward manner and evaluated in terms of economic performance.

6.2.1 Evaluation Splits

For the benchmarking stage, each ticker is divided into fixed date ranges for training, validation, and testing:

- Train: [2014-01-01, 2015-08-01),
- Validation: [2015-08-01, 2015-10-01),
- Test: [2015-10-01, 2016-01-01).

In addition to comparing model families and feature groups, the experiments also vary the prediction horizon H . Rather than restricting evaluation to next-day forecasting, the framework considers multiple holding periods so that the effect of short- and medium-horizon prediction can be examined. This is particularly important because the usefulness of technical, textual, and reliability-aware signals may differ depending on the forecast horizon.

For the investment simulation, the evaluation protocol is more realistic. An initial training window is used to fit the first per-asset models, after which predictions are generated sequentially through time over the last **100 days** of the dataset. Models are retrained periodically using only historical data available prior to the decision date. In this way, the simulation mimics live deployment rather than a single static train-test experiment.

6.2.2 Compared Models and Feature Sets

Model families evaluated:

- LSTM
- BiLSTM
- GRU
- BiGRU

These architectures are evaluated across three predictive formulations: binary classification, regression, and ordinal multi-class classification.

Feature sets are organised hierarchically in order to measure the contribution of different information sources:

- **Base:** price-based features, technical indicators, and meta-features derived from prior model predictions and errors;
- **NLP:** all Base features augmented with textual features, including sentiment, emotion, stance, and FinBERT representations;
- **Sector:** all NLP features further augmented with sector-level features and sector technical indicators.

Each model-feature combination is evaluated across multiple forecast horizons H , allowing the analysis to compare not only which architecture performs best, but also whether performance is sensitive to the length of the prediction window. This is particularly relevant for the subsequent investment simulation, where both model confidence and realised returns depend directly on the chosen holding horizon.

The full list of compared model families, feature subsets, horizon values, and tuning ranges is provided in Appendix A.2.

6.3 Predictive Performance

This section evaluates the statistical forecasting performance of the proposed models before introducing probability calibration, meta-features, or downstream capital allocation. The aim is to isolate pure predictive quality and identify the strongest model families, feature sets, and forecast horizons to carry forward into the investment simulation stage. Particular attention is given to direct comparison with prior work, especially Xu and Cohen (2018), in order to assess whether the proposed architectures and feature combinations provide measurable improvement over established baselines.

6.3.1 Binary Classification Results

For binary classification, models are first evaluated without post-hoc calibration and without the addition of reliability-aware meta-features. This provides the fairest comparison against the baseline literature, including Xu and Cohen (2018), whose setting also focuses primarily on directional prediction from price and text signals.

Table 6.1 summarises the strongest binary classification results for the 1-day prediction horizon. Across the tested configurations, the strongest performance is achieved by the BiGRU model using sentiment-enhanced features, with a mean directional accuracy of 0.525 and a mean MCC of 0.086. This is the highest MCC observed among the binary models reported, and slightly exceeds the best previously reported MCC of 0.080 from the HFA model of Xu and Cohen (2018). Although the absolute gain is modest, it is meaningful given the difficulty of the task and the class imbalance typical of financial movement predictions accompanied with the fact that no trades were filtered.

To place the results in context, Table 6.2 compares the best-performing model in this study against representative baselines from the literature. While the HFA model of Xu and Cohen (2018) reports a higher average directional accuracy, the BiGRU sentiment model achieves a slightly higher MCC, suggesting improved balance in directional prediction quality. **TODO: reference properly**

In directional accuracy terms, the best-performing configuration in this study also improves on older statistical baselines such as ARIMA, which reported an average accuracy of 0.514, and compares favourably with neural and text-aware baselines such as HAN and HFA. These results suggest that bidirectional recurrent architectures, when combined with sentiment features, can provide a small but consistent improvement in directional stock prediction over prior approaches.

Table 6.1: Top binary classification results for the 1-day prediction horizon.

Model	Feature Set	Dir Acc	MCC	F1	Precision	Recall
BiGRU	sentiment	0.525	0.086	0.331	0.476	0.360
BiLSTM	sentiment	0.525	0.077	0.363	0.488	0.404
BiGRU	sector	0.528	0.069	0.389	0.495	0.435
GRU	sentiment	0.521	0.069	0.379	0.528	0.423
LSTM	base	0.516	0.069	0.405	0.508	0.486
BiLSTM	sector_sentiment	0.519	0.064	0.307	0.505	0.325
GRU	base	0.514	0.063	0.358	0.499	0.415
LSTM	sector_unified_emotion	0.514	0.060	0.373	0.483	0.412
LSTM	sentiment	0.522	0.060	0.425	0.517	0.490
GRU	sector_sentiment	0.513	0.059	0.356	0.459	0.414
BiLSTM	base	0.518	0.057	0.357	0.490	0.408

Table 6.2: Comparison against reported binary prediction baselines.

Model	Avg. Accuracy	MCC
ARIMA (?)	0.514	-0.020
HAN (?)	0.576	0.051
HFA (Xu and Cohen 2018)	0.582	0.080
BiGRU + Sentiment (ours)	0.525	0.086

More broadly, the results indicate that no single feature group dominates across all architectures, but sentiment-based extensions appear among the strongest-performing configurations, particularly for BiGRU and BiLSTM. Sector-enhanced and base-only variants also remain competitive, implying that the predictive gains from text are present but not uniformly large across all models. This reinforces the importance of evaluating multiple feature combinations rather than assuming that additional input sources always improve performance.

Overall, for the 1-day binary classification task, the BiGRU sentiment model is selected as the strongest benchmarked candidate. It achieves the highest MCC in the comparison and marginally outperforms the HFA benchmark of Xu and Cohen (2018), making it a strong candidate for subsequent investment simulation.

6.3.2 Regression Results

For the regression task, performance is evaluated primarily using Mean Absolute Error (MAE), since the objective is to predict forward return magnitude rather than only directional movement. Mean Squared Error (MSE) is also reported as a secondary measure, while directional accuracy is included to show how often the sign of the predicted return matches the realised movement.

Table 6.3 presents the 1-day regression results ranked by MAE. The strongest configuration is the BiLSTM model with base features, achieving the lowest MAE of 0.01155, followed closely by BiLSTM with NLP features and LSTM with base features. Overall, the differences in MAE across models are relatively small, suggesting that all recurrent architectures perform similarly when predicting short-horizon return magnitude.

A notable pattern is that the best-performing configurations by MAE are not always the strongest in directional metrics. For example, while BiLSTM with base features achieves the lowest MAE, its MCC remains close to zero, indicating that accurate return magnitude estimation does not necessarily translate into strong directional discrimination. This highlights an important distinction between regression and classification performance in financial forecasting in that a model may minimise numerical prediction error while still offering limited value for directional

Table 6.3: Regression results for the 1-day prediction horizon, ordered by MAE (lower is better).

Model	Feature Set	Dir Acc	MAE	MSE	MCC	F1	Recall
BiLSTM	base	0.507	0.011549	0.000270	0.002	0.361	0.522
BiLSTM	emotion	0.510	0.011571	0.000270	0.002	0.314	0.433
LSTM	sector_sentiment	0.496	0.011589	0.000272	-0.002	0.344	0.488
BiLSTM	sector_sentiment	0.496	0.011590	0.000271	-0.012	0.386	0.548
BiGRU	base	0.494	0.011598	0.000272	-0.005	0.380	0.520
GRU	sector_sentiment	0.501	0.011609	0.000272	0.002	0.425	0.566
LSTM	emotion	0.507	0.011616	0.000273	0.002	0.412	0.581
GRU	base	0.502	0.011626	0.000272	0.002	0.364	0.496
BiGRU	all_nlp	0.501	0.011627	0.000273	-0.013	0.384	0.525
BiGRU	sector_sentiment	0.499	0.011636	0.000273	0.002	0.402	0.539
GRU	all_nlp	0.504	0.011664	0.000274	0.022	0.370	0.464
LSTM	base	0.505	0.011741	0.000280	0.022	0.408	0.550

decision-making.

More broadly, the results suggest that base and NLP-enhanced feature sets perform competitively, while sector-enhanced variants do not consistently outperform simpler inputs in the regression setting. This contrasts slightly with the classification results, where sentiment-enhanced models appeared more consistently beneficial. Consequently, although regression provides a useful estimate of return magnitude, its practical value for subsequent trading decisions must be interpreted alongside directional and economic performance rather than MAE alone.

Overall, for the 1-day regression benchmark, the BiLSTM base model is selected as the strongest candidate on the basis of MAE, while the narrow spread in results across models suggests that feature set choice has a more limited effect on regression error than on directional classification quality.

6.3.3 Ordinal Multi-Class Results

For the ordinal multi-class task, models are evaluated on their ability to predict ordered return buckets rather than only binary direction. This formulation preserves partial magnitude information while remaining more robust than direct regression. Performance is assessed using both ordinal classification metrics, namely micro-accuracy and macro-F1, and directional metrics derived by collapsing the ordered buckets into up/down movement.

Table 6.4 summarises the strongest ordinal multi-class results for the 1-day prediction horizon. The best MCC is achieved by the BiGRU model with base features, which attains a mean MCC of 0.078 together with a directional accuracy of 0.516. This makes it the strongest overall ordinal configuration in terms of balanced directional discrimination. In contrast, the highest micro-accuracy and macro-F1 are achieved by GRU and LSTM variants using sector-emotion features, indicating that the models which best recover the full ordered bucket structure are not necessarily the same as those that best capture directional movement.

This distinction is important as the ordinal formulation introduces a more difficult prediction problem than binary classification, as the model must distinguish between multiple ordered return ranges rather than simply predicting up or down **TODO: insert confusion matrix**. As expected, absolute micro-accuracy remains low, since correctly identifying the exact return bucket is considerably harder than recovering direction alone. However, the corresponding directional metrics remain competitive, suggesting that ordinal models can still provide useful directional information while also preserving magnitude structure. **TODO: insert confusion matrix** A notable pattern is that sector-emotion feature sets appear frequently among the

Model	Feature Set	Micro Acc	Macro F1	MCC	Dir Acc	F1	Recall
BiGRU	base	0.152	0.104	0.078	0.516	0.362	0.400
BiGRU	sector_emotion	0.178	0.119	0.075	0.519	0.370	0.389
LSTM	sector_emotion	0.186	0.120	0.071	0.522	0.441	0.496
BiGRU	sentiment	0.153	0.107	0.071	0.510	0.361	0.388
GRU	sector_emotion	0.191	0.124	0.070	0.522	0.424	0.462
BiLSTM	sector_unified_emotion	0.167	0.116	0.065	0.513	0.397	0.426
GRU	base	0.156	0.106	0.063	0.517	0.374	0.397
GRU	all_nlp	0.165	0.118	0.059	0.515	0.407	0.453
LSTM	sentiment	0.181	0.116	0.055	0.510	0.411	0.465
LSTM	base	0.169	0.117	0.054	0.515	0.369	0.386
BiLSTM	emotion	0.149	0.113	0.046	0.503	0.398	0.435
BiLSTM	base	0.154	0.104	0.044	0.501	0.350	0.393

Table 6.4: Ordinal multi-class results for the 1-day prediction horizon.

strongest ordinal configurations, particularly for LSTM and GRU models. This suggests that richer contextual features may be more useful when the prediction task requires finer-grained discrimination across return ranges. At the same time, the BiGRU base model achieves the best MCC overall, indicating that simpler feature sets can still remain highly effective when combined with a stronger recurrent architecture.

6.4 Calibration Results

6.4.1 Isotonic Regression Results

To assess the effect of post-hoc calibration on binary directional prediction, isotonic regression is applied to the predicted probabilities produced by the classification models. The aim is not to change the ranking of predictions, but to improve the reliability of the probability estimates used downstream in decision-making.

Table 6.5 compares the strongest 1-day binary classification result obtained without isotonic regression against the strongest result obtained after isotonic calibration. Without calibration, the best-performing configuration is the BiGRU model with sentiment features, which achieves an MCC of 0.086 and a directional accuracy of 0.525. After isotonic regression, the best directional accuracy increases slightly to 0.527, again for a BiGRU model with sentiment features, while the highest MCC after calibration is 0.082 from the BiLSTM sentiment model.

Table 6.5: Best 1-day binary classification result before and after isotonic regression.

Setting	Model	Feature Set	Dir Acc	MCC
Without isotonic regression	BiGRU	sentiment	0.525	0.086
With isotonic regression	BiLSTM	sentiment	0.527	0.082

These results suggest that isotonic regression provides only a modest change in pure predictive classification performance. Directional accuracy improves slightly, but the strongest MCC is marginally lower than the uncalibrated best result. This is not unexpected, since calibration is primarily intended to improve probabilistic reliability rather than maximise classification metrics directly. Its importance therefore lies less in improving headline predictive accuracy and more in making model probabilities more meaningful for downstream capital allocation.

6.4.2 Temperature Scaling Results

To assess the effect of post-hoc calibration in the ordinal multi-class setting, temperature scaling and threshold (τ) tuning are applied to the raw ordinal outputs. As with isotonic regression in the binary setting, the purpose of this procedure is not primarily to maximise classification accuracy, but to improve the reliability and interpretability of the class probabilities used downstream.

Table 6.6 compares the strongest 1-day ordinal multi-class result obtained without temperature scaling or τ -tuning against the strongest result obtained after calibration. Without calibration, the best-performing configuration is the BiGRU model with sector-emotion features, achieving an MCC of 0.084 and a directional accuracy of 0.524. After temperature scaling and τ -tuning, the strongest MCC is achieved by the BiGRU model with base features, with an MCC of 0.078 and a directional accuracy of 0.516.

Table 6.6: Best 1-day ordinal multi-class result before and after temperature scaling and τ -tuning.

Setting	Model	Feature Set	Dir Acc	MCC
Without temperature scaling/ τ -tuning	BiGRU	sector_emotion	0.524	0.084
With temperature scaling/ τ -tuning	BiGRU	base	0.516	0.078

These results indicate that temperature scaling and τ -tuning do not improve the headline predictive metrics in the ordinal setting. In fact, both MCC and directional accuracy are slightly lower after calibration. This is not unexpected, since temperature scaling is designed to improve probability calibration rather than to optimise directional classification metrics directly. Its value therefore lies in producing more reliable ordinal class probabilities and expected-return estimates for downstream portfolio construction and capital allocation, rather than in improving benchmark predictive scores alone.

Although the calibrated models do not outperform the uncalibrated models on MCC, they remain competitive, suggesting that the calibration stage preserves much of the underlying predictive signal while improving the interpretability of the resulting probabilities.

6.5 Meta-Feature and Ablation Results

6.5.1 Effect of Meta-Features

To evaluate the contribution of reliability-aware meta-features, we compare model behaviour with and without the inclusion of prior prediction outcomes and rolling error statistics. This ablation is performed across the binary classification, ordinal multi-class classification, and regression settings. In addition to predictive metrics, training dynamics are examined through the corresponding loss curves in order to assess whether meta-features improve optimisation stability or generalisation.

Effects on Training To evaluate the influence of reliability-aware meta-features on the training process, we compare the aggregate training and validation loss behaviour of the strongest-performing BiGRU models with and without the inclusion of meta-features. Rather than presenting individual loss curves in the main text, Table 6.7 summarises the area under the training and validation loss curves across the primary feature groups.

Across the binary and ordinal classification settings, the inclusion of meta-features generally improves training stability and reduces the magnitude of validation loss. In particular, the sector feature configuration exhibits the largest improvement, where the validation loss area decreases from 28.07 to 24.38 when meta-features are included. Similar behaviour is observed for the base feature set, where validation loss drops substantially from 75.99 to 27.66.

Table 6.7: Effect of meta-features on the training dynamics of the BiGRU models (1-day horizon). Loss behaviour is summarised using the area under the training and validation loss curves.

Feature Set	Without Meta-Features		With Meta-Features	
	Train Loss AUC	Validation Loss AUC	Train Loss AUC	Validation Loss AUC
Base	13.63	75.99	17.67	27.66
NLP	10.89	35.57	14.52	38.90
Sector	11.58	28.07	9.31	24.38

Although the NLP feature set exhibits a slightly higher validation loss when meta-features are added, the overall pattern suggests that reliability-aware inputs help the optimiser learn representations that generalise more consistently across epochs. This behaviour is consistent with the interpretation that meta-features provide additional information about the recent reliability of model predictions, allowing the training process to better adapt to periods of higher or lower predictive signal.

Overall, the results suggest that meta-features act not only as additional predictive inputs but also as a form of implicit regularisation in the presence of non-stationary financial data. By incorporating information about recent prediction performance, the model appears better able to stabilise training and reduce overfitting.

For completeness, the full set of training and validation loss curves for all architectures and feature-set combinations is provided in Appendix A.3.

Effects on Results We then compare the strongest-performing models from the baseline experiments against their corresponding meta-feature counterparts. The comparison is performed separately for the binary classification, regression, and ordinal multi-class settings. For clarity, only the best-performing configuration from each task is presented here, while the full set of meta-feature results is reported in Appendix A.4.

Binary Classification Table 6.8 compares the best binary classification model obtained without meta-features against the strongest model trained using meta-feature inputs.

Table 6.8: Best binary classification results with and without meta-features (1-day horizon).

Setting	Model	Feature Set	Dir Acc	MCC	F1	Recall
No Meta	BiGRU	sentiment	0.525	0.086	0.331	0.360
Meta	GRU	meta_sector_finbert	0.516	0.081	0.358	0.400

The results indicate that the strongest meta-feature configuration remains competitive with the baseline model, although the original BiGRU sentiment model retains the highest MCC. However, the meta-feature models demonstrate improved F1 and recall, suggesting that the additional reliability-aware inputs may help the model capture a broader range of positive movement signals.

Regression Table 6.9 presents the comparison between the best baseline regression model and the strongest regression model trained with meta-features.

In the regression setting, the meta-feature model produces a modest improvement in both MAE and MSE while maintaining similar directional accuracy. This suggests that incorporating reliability-aware signals can slightly improve the precision of return magnitude predictions.

Ordinal Multi-Class Classification Table 6.10 reports the comparison between the strongest ordinal multi-class models with and without meta-features.

Table 6.9: Best regression results with and without meta-features (1-day horizon).

Setting	Model	Feature Set	Dir Acc	MAE	MSE
No Meta	BiLSTM	base	0.507	0.01155	0.000270
Meta	BiGRU	meta_sector	0.509	0.01135	0.000242

Table 6.10: Best ordinal multi-class results with and without meta-features (1-day horizon).

Setting	Model	Feature Set	Micro Acc	Macro F1	MCC
No Meta	BiGRU	base	0.152	0.104	0.078
Meta	BiLSTM	meta_sector_sentiment	0.163	0.111	0.083

In contrast to the binary classification setting, the ordinal multi-class task benefits more clearly from meta-features. The best meta-feature model improves micro accuracy, macro F1, and MCC relative to the baseline model, indicating that reliability-aware signals may be particularly beneficial when predicting finer-grained return categories.

Summary Overall, the inclusion of meta-features produces mixed but encouraging results across tasks. While the strongest baseline binary classification model remains competitive, meta-feature models achieve improvements in several metrics, particularly in the regression and ordinal multi-class settings. These findings suggest that incorporating information about recent predictive reliability can provide additional context for the model, helping it adapt to the non-stationary behaviour typical of financial time series.

6.5.2 Effect of Variable Horizon Design

Varying the prediction horizon has a substantial effect on performance across all three tasks. In general, extending the forecast horizon from 1 day to 5 days produces noticeably stronger results for the classification and ordinal multi-class settings, but leads to worse regression error. This suggests that longer-horizon directional structure may be easier for the models to capture than short-horizon return magnitude.

Table 6.11 compares the strongest-performing configurations at the 1-day and 5-day horizons for the binary classification, regression, and ordinal multi-class tasks, both with and without meta-features. The full set of horizon-specific results is reported in Appendix A.5.

For binary classification, the benefit of the 5-day horizon is clear. Without meta-features, the best MCC increases from 0.086 at 1 day to 0.152 at 5 days, while directional accuracy also rises slightly. A similar improvement is observed when meta-features are included, with MCC increasing from 0.081 to 0.145 and directional accuracy rising from 0.516 to 0.531. These results indicate that directional prediction becomes easier when the forecast window is extended, likely because longer-horizon price movements contain less day-to-day noise than next-day fluctuations.

A similar pattern is visible in the ordinal multi-class setting. Without meta-features, MCC rises from 0.078 to 0.145 when moving from a 1-day to a 5-day horizon, accompanied by gains in both micro-accuracy and directional accuracy. With meta-features, the same qualitative effect remains, with MCC increasing from 0.083 to 0.126 and micro-accuracy improving from 0.163 to 0.213. This suggests that longer horizons are particularly beneficial when the task requires prediction of ordered return ranges rather than simple direction alone.

In contrast, the regression setting shows the opposite behaviour. While directional accuracy and MCC increase modestly at the 5-day horizon, the mean absolute error deteriorates sharply. Without meta-features, MAE rises from 0.0117 to 0.0260, and with meta-features it rises from

0.0114 to 0.0253. This reflects the fact that return magnitudes naturally become larger and more variable over longer horizons, making precise continuous prediction more difficult even when directional discrimination improves slightly.

Overall, these results suggest that variable-horizon design is an important modelling choice rather than a minor implementation detail. Longer horizons appear especially beneficial for classification-based formulations, where they improve balanced directional performance and bucket prediction quality. By contrast, regression becomes less attractive as the horizon increases, since the gain in directional information is outweighed by a substantial increase in magnitude error.

Table 6.11: Comparison of the best-performing 1-day and 5-day models across tasks.

Task	Setting	Horizon	Model / Feature Set	Dir Acc	MCC	MAE / Micro Acc
Binary	No Meta	1-day	BiGRU / sentiment	0.525	0.086	–
Binary	No Meta	5-day	GRU / sector_sentiment	0.529	0.152	–
Binary	Meta	1-day	GRU / meta_sector_finbert	0.516	0.081	–
Binary	Meta	5-day	BiGRU / meta	0.531	0.145	–
Regression	No Meta	1-day	LSTM / base	0.505	0.022	0.0117
Regression	No Meta	5-day	LSTM / sector_finbert	0.533	0.055	0.0260
Regression	Meta	1-day	BiGRU / meta_sector	0.509	0.028	0.0114
Regression	Meta	5-day	BiLSTM / meta_sector_emotion	0.526	0.042	0.0253
Ordinal	No Meta	1-day	BiGRU / base	0.516	0.078	0.152
Ordinal	No Meta	5-day	BiGRU / sector	0.533	0.145	0.208
Ordinal	Meta	1-day	BiLSTM / meta_sector_sentiment	0.525	0.083	0.163
Ordinal	Meta	5-day	GRU / meta_emotion	0.530	0.126	0.213

6.6 Investment Simulation Results

For the investment simulation experiments, the models with the highest Matthews Correlation Coefficient (MCC) from both the 1-day and 5-day horizons were selected for the binary and ordinal multi-class prediction tasks. The corresponding feature configurations were then used to generate trading signals over the last **100 days** within the investment simulation framework.

6.6.1 Binary Simulation Results

For the binary classification setting without meta features, the following models were selected based on their MCC performance:

- **1-day horizon:** BiGRU with sentiment features (Directional Accuracy = 0.525, MCC = 0.086)
- **5-day horizon:** GRU with sector_sentiment features (Directional Accuracy = 0.529, MCC = 0.152)

Figure 6.1 illustrates the cumulative portfolio returns obtained when using the selected binary models without meta features.

The **1-day horizon models generated significantly weaker results**, shown in the table above 6.12, with Kelly-only strategy achieved the best performance among the 1-day configurations with a final equity of \$100,967, corresponding to a return of only 0.97%. Both the hybrid and MPT-only strategies remained close to the initial capital level, with the hybrid approach

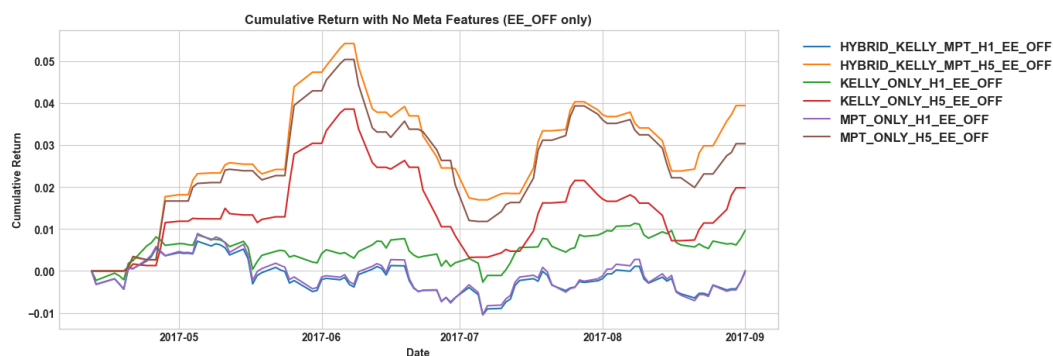


Figure 6.1: Cumulative returns for binary classification strategies without meta features.

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
HYBRID KELLY MPT	5	OFF	100000	103939	3.94	2.21	0.044
MPT	5	OFF	100000	103031	3.03	1.74	0.044
KELLY	5	OFF	100000	101982	1.98	1.36	0.037
KELLY	1	OFF	100000	100968	0.97	1.04	0.024
MPT	1	OFF	100000	100013	0.01	0.03	0.031
HYBRID KELLY MPT	1	OFF	100000	99982	-0.02	-0.00	0.027

Table 6.12: Binary investment simulation results without meta features.

producing a slight negative return. This suggests that the predictive signals generated at a 1-day horizon are substantially noisier and less capable of generating consistent trading profits.

This difference between horizons is clearly visible in Figure 6.1. While the 5-day strategies exhibit sustained upward growth throughout the simulation period, the 1-day strategies fluctuate around the initial capital level with limited cumulative gains.

Effects of Meta Features To evaluate the impact of incorporating meta features derived from previous predictions and model performance, the binary investment simulation was repeated using models trained with meta feature inputs. The selected models were:

- **1-day horizon:** GRU with meta_sector_finbert features (Directional Accuracy = 0.516, MCC = 0.081)
- **5-day horizon:** BiGRU with meta features (Directional Accuracy = 0.531, MCC = 0.145)

The resulting cumulative portfolio performance is shown in Figure 6.2.

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
MPT	5	OFF	100000	112465	12.47	4.34	0.069
HYBRID KELLY MPT	5	OFF	100000	108665	8.66	4.26	0.049
KELLY	5	OFF	100000	102763	2.76	1.79	0.039
KELLY	1	OFF	100000	101051	1.05	1.26	0.021
HYBRID KELLY MPT	1	OFF	100000	100249	0.25	0.29	0.023
MPT	1	OFF	100000	99981	-0.02	0.00	0.034

Table 6.13: Binary investment simulation results with meta features.

The inclusion of meta features (Table 6.13) leads to a substantial improvement in trading performance, particularly for the 5-day horizon models. The strongest performance was achieved by the MPT-only allocation strategy, which increased the initial capital to \$112,465, corresponding

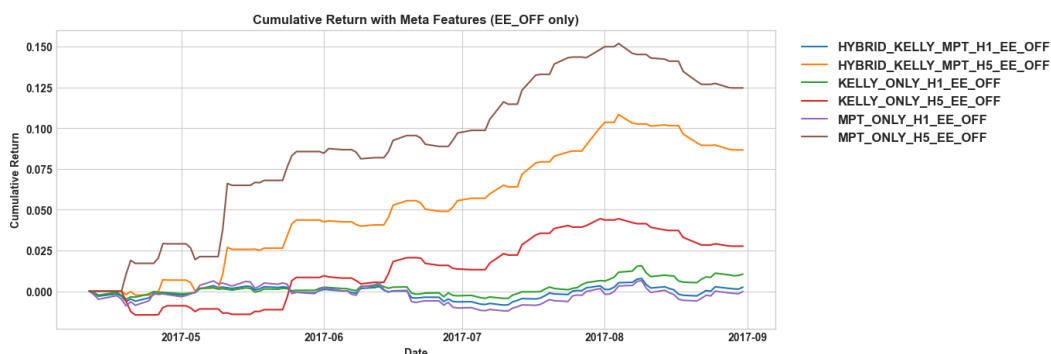


Figure 6.2: Cumulative returns for binary classification strategies incorporating meta features.

to a total return of 12.47%. This strategy also achieved strong risk-adjusted performance, with a Sharpe ratio of 4.34.

The hybrid Kelly–MPT strategy also performed strongly, achieving a final portfolio value of \$108,664 and a total return of 8.66%. These results represent a significant improvement over the corresponding non-meta strategies, suggesting that the meta features provide useful additional information regarding the reliability of model predictions.

Although the 1-day horizon strategies show modest improvements with the addition of meta features, their overall profitability remains limited compared to the 5-day horizon models. The best 1-day configuration achieved a return of approximately 1.05%, indicating that short-horizon predictions still struggle to generate strong trading signals even when additional predictive information is incorporated.

Overall, the results indicate that **meta features can significantly enhance portfolio performance**, particularly when combined with longer prediction horizons. The improvements are especially pronounced for the mean–variance allocation strategy, suggesting that the additional predictive signals may improve the estimation of expected returns used by the portfolio optimisation process.

6.6.2 Ordinal Multi-Class Simulation Results

For the ordinal multi-class prediction task, the best-performing models were selected based on their MCC scores during model benchmarking. The selected configurations were:

- **1-day horizon:** BiGRU with base features (Directional Accuracy = 0.516, MCC = 0.078, F1 = 0.152)
- **5-day horizon:** BiGRU with sector features (Directional Accuracy = 0.533, MCC = 0.145, F1 = 0.208)

Using these models, trading signals were generated for the investment simulation over a 100-day test period. The cumulative return curves for the ordinal strategies without meta features are provided in Appendix A.5.

The results in Table 6.14 demonstrate a clear difference between prediction horizons. Similar to the binary classification experiments, the **5-day horizon models produce substantially stronger trading performance**. The best-performing strategy was the MPT-only allocation, which increased the initial capital from \$100,000 to \$110,561, corresponding to a total return of 10.56% as shown in Table 6.15.

The hybrid Kelly–MPT approach also performed well, producing a 5.23% return with the highest Sharpe ratio among the strategies. In contrast, the 1-day horizon strategies resulted in negative

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
MPT	5	OFF	100000	110561	10.56	4.59	0.055
HYBRID KELLY MPT	5	OFF	100000	105233	5.23	5.06	0.025
KELLY	5	OFF	100000	103345	3.35	3.81	0.022
MPT	1	OFF	100000	99730	-0.27	-0.19	0.033
KELLY	1	OFF	100000	99054	-0.95	-0.67	0.035
HYBRID KELLY MPT	1	OFF	100000	99039	-0.96	-0.68	0.035

Table 6.14: Ordinal multi-class investment simulation results without meta features.

returns across all allocation methods, suggesting that short-horizon ordinal predictions remain too noisy to generate consistent profitable signals.

The cumulative return trajectories for these strategies are illustrated in Appendix A.5, where the 5-day strategies exhibit a clear upward trend, while the 1-day strategies fluctuate around the initial capital level.

Effects of Meta Features To evaluate whether incorporating meta features improves ordinal prediction performance, the simulation was repeated using models trained with meta features. The selected configurations were:

- **1-day horizon:** BiLSTM with meta_sector_sentiment features (Directional Accuracy = 0.525, MCC = 0.083, F1 = 0.163)
- **5-day horizon:** GRU with meta_emotion features (Directional Accuracy = 0.530, MCC = 0.126, F1 = 0.213)

The cumulative returns for these strategies are shown in Appendix A.10.

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
KELLY	5	OFF	100000	103609	3.61	2.22	0.041
HYBRID KELLY MPT	5	OFF	100000	102926	2.93	1.54	0.048
MPT	5	OFF	100000	102825	2.83	1.47	0.049
KELLY	1	OFF	100000	101719	1.72	1.22	0.036
HYBRID KELLY MPT	1	OFF	100000	101467	1.47	1.11	0.034
MPT	1	OFF	100000	100942	0.94	0.75	0.032

Table 6.15: Ordinal multi-class investment simulation results with meta features.

In contrast to the binary classification experiments, incorporating meta features does not improve overall trading performance for the ordinal multi-class setting. While the strategies remain profitable for the 5-day horizon, the best return achieved is 3.61%, substantially lower than the 10.56% return achieved without meta features.

However, the inclusion of meta features does appear to stabilise the short-horizon strategies. Unlike the non-meta models, the 1-day strategies with meta features produce positive returns across all allocation approaches. This suggests that meta features may improve the reliability of short-term predictions, even though they do not lead to superior long-horizon profitability.

The cumulative return curves illustrating these results are provided in Appendix A.10.

6.6.3 Effect of Early Exit

In addition to the baseline investment simulations, we evaluated the effect of an *early exit* mechanism. Early exit allows a position to be closed before the full prediction horizon if the predicted signal reverses or becomes unfavourable. This mechanism is intended to reduce downside risk and lock in profits earlier when market conditions change.

Early exit can only be applied to the **5-day prediction horizon**, since a 1-day horizon does not allow intermediate exits. Therefore, all results presented in this section correspond to the 5-day models previously identified as the best-performing configurations.

Binary Results without Meta Features Table 6.16 compares portfolio performance with early exit enabled and disabled for the binary classification models without meta features.

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
HYBRID KELLY MPT	5	ON	100000	106569	6.57	4.42	0.036
KELLY	5	ON	100000	105946	5.95	4.38	0.033
MPT	5	ON	100000	104813	4.81	3.39	0.035
HYBRID KELLY MPT	5	OFF	100000	103939	3.94	2.21	0.044
MPT	5	OFF	100000	103031	3.03	1.74	0.044
KELLY	5	OFF	100000	101982	1.98	1.36	0.037

Table 6.16: Binary investment simulation results with and without early exit (no meta features, 5-day horizon).

The introduction of early exit leads to a noticeable improvement in portfolio performance across all allocation strategies. The hybrid Kelly–MPT strategy achieves the highest return of 6.57%, compared to 3.94% without early exit. Similar improvements are observed for the Kelly-only and MPT-only strategies, indicating that the early exit mechanism is effective in reducing losses and capturing profits sooner.

The frequency of early exits is summarised below.

Mode	Total Early Exits	Avg Early Exits per Day	Total Trades
KELLY	341	3.41	424
MPT	337	3.37	401
HYBRID KELLY MPT	269	2.69	329

Table 6.17: Early exit statistics for binary models without meta features.

Binary Results with Meta Features Table 6.18 presents the corresponding results for the models trained with meta features.

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
MPT	5	OFF	100000	112465	12.47	4.34	0.069
HYBRID KELLY MPT	5	OFF	100000	108665	8.66	4.26	0.049
KELLY	5	OFF	100000	102763	2.76	1.79	0.039
MPT	5	ON	100000	103572	3.57	1.81	0.049
HYBRID KELLY MPT	5	ON	100000	102711	2.71	1.40	0.048
KELLY	5	ON	100000	101390	1.39	1.05	0.034

Table 6.18: Binary investment simulation results with and without early exit (meta features, 5-day horizon).

Unlike the non-meta models, enabling early exit does not improve performance when meta features are used. In fact, early exit reduces the overall returns across all allocation strategies. The best-performing configuration remains the MPT-only strategy without early exit, achieving a return of 12.47%.

Early exit activity for the meta models is summarised in Table 6.19.

Overall, the results suggest that early exit can improve trading performance when models rely solely on direct predictive signals. However, when meta features are incorporated, the models

Mode	Total Early Exits	Avg Early Exits per Day	Total Trades
KELLY	353	3.53	453
MPT	317	3.17	374
HYBRID KELLY MPT	309	3.09	366

Table 6.19: Early exit statistics for binary models with meta features.

already appear to produce more stable predictions, and early exit may prematurely close positions that would otherwise remain profitable.

Ordinal Results without Meta Features Table 6.20 presents the simulation results for the ordinal models trained without meta features.

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
MPT	5	OFF	100000	110561	10.56	4.59	0.055
HYBRID KELLY MPT	5	OFF	100000	105233	5.23	5.06	0.025
KELLY	5	OFF	100000	103345	3.35	3.81	0.022
HYBRID KELLY MPT	5	ON	100000	102695	2.70	2.80	0.024
MPT	5	ON	100000	102615	2.62	1.43	0.046
KELLY	5	ON	100000	101469	1.47	1.77	0.021

Table 6.20: Ordinal multi-class investment simulation results without meta features.

The best performing strategy in this setting was the MPT-only allocation without early exit, which increased the initial capital to \$110,561, corresponding to a return of 10.56%. The hybrid Kelly–MPT strategy also produced strong performance with a return of 5.23% and the highest Sharpe ratio among the strategies.

Introducing the early exit mechanism reduced the overall returns for all strategies in the ordinal setting. Although early exit slightly reduced volatility and drawdowns, the earlier closing of positions appears to have limited the ability of profitable trades to realise their full gains.

The frequency of early exits during the simulation is summarised in Table 6.21.

Mode	Total Early Exits	Avg Early Exits per Day	Total Trades
KELLY	561	5.61	712
MPT	313	3.13	404
HYBRID KELLY MPT	307	3.07	405

Table 6.21: Early exit statistics for ordinal models without meta features.

Ordinal Results with Meta Features Table 6.22 summarises the results obtained when meta features were incorporated into the ordinal models.

When meta features are introduced, the ordinal strategies remain profitable but overall returns are lower compared to the non-meta models. The best-performing configuration is the MPT-only strategy with early exit enabled, producing a return of 4.86%. The hybrid Kelly–MPT approach achieves a similar performance with a return of 4.55%.

Overall, the ordinal results indicate that incorporating additional prediction classes allows the allocation strategies to capture larger market movements, particularly in the 5-day horizon setting. However, the impact of meta features appears mixed, improving stability in some cases but not consistently increasing total returns.

Mode	Horizon	Early Exit	Start Equity	End Equity	ROI (%)	Sharpe	Volatility
MPT	5	ON	100000	104860	4.86	2.59	0.047
HYBRID KELLY MPT	5	ON	100000	104553	4.55	2.46	0.046
KELLY	5	OFF	100000	103609	3.61	2.22	0.041
HYBRID KELLY MPT	5	OFF	100000	102926	2.93	1.54	0.048
MPT	5	OFF	100000	102825	2.83	1.47	0.049
KELLY	5	ON	100000	102814	2.81	2.06	0.034

Table 6.22: Ordinal multi-class investment simulation results with meta features.

Mode	Total Early Exits	Avg Early Exits per Day	Total Trades
KELLY	342	3.42	473
MPT	324	3.24	423
HYBRID KELLY MPT	267	2.67	357

Table 6.23: Early exit statistics for ordinal models with meta features.

6.6.4 Drawdown and Risk

Binary System without Meta Features The drawdown profiles of the binary system without meta-features are illustrated in Figure A.11. A pronounced and sustained market downturn is observed between June and July 2017, during which all strategies experience their largest drawdowns.

To facilitate comparison, Table 6.26 presents a subset of key drawdown statistics across representative configurations. Full results are provided in Appendix A.7.1.

Strategy	Max Drawdown
MPT_ONLY_H5_EE_OFF	-3.67%
HYBRID_KELLY_MPT_H5_EE_OFF	-3.53%
KELLY_ONLY_H5_EE_OFF	-3.40%
MPT_ONLY_H5_EE_ON	-1.12%
HYBRID_KELLY_MPT_H5_EE_ON	-1.13%

Table 6.24: Summary of maximum drawdown for selected binary strategies without meta-features. Full table in Appendix.

At the 5-day horizon without early exit, all allocation strategies exhibit substantial drawdowns, exceeding -3.4%. These drawdowns occur over an extended period (early June to early July 2017), indicating persistent exposure to adverse market conditions rather than isolated volatility spikes.

Mechanistically, this behaviour arises from the *fixed-horizon commitment* of the system. Without early exit, positions are held until maturity regardless of evolving market conditions, allowing losses to compound over multiple time steps. This is further reflected in the rolling Sharpe ratios (Figure 6.3), where extreme negative values are observed, with minimum Sharpe reaching as low as -10.34. Such values indicate significant downside volatility relative to realised returns, consistent with prolonged drawdown periods.

Enabling early exit leads to a marked improvement in risk characteristics. As shown in Table 6.26, maximum drawdown is reduced by approximately a factor of three, from around -3.5% to approximately -1.1%. Additionally, drawdowns become shorter in duration and exhibit faster recovery, as also evident in Figure A.11.

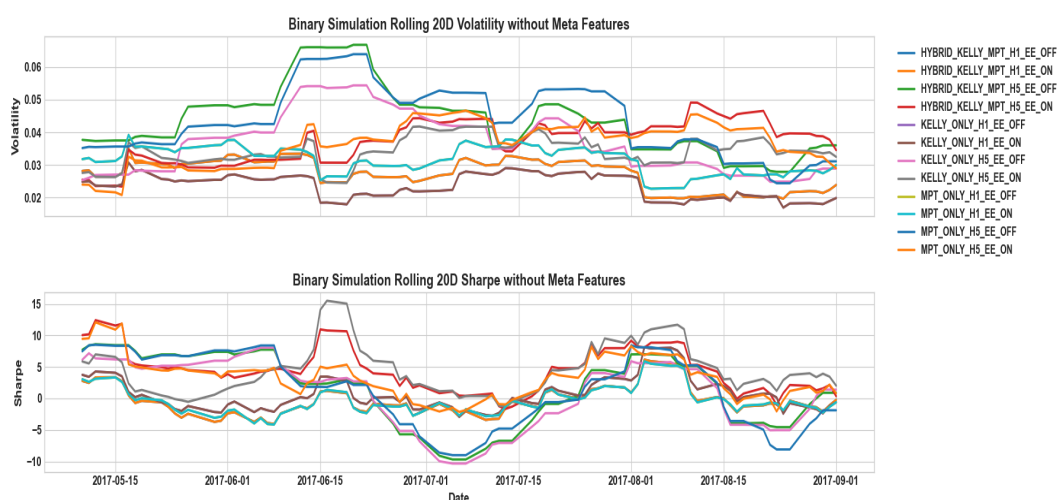


Figure 6.3: Sharpe ratio and volatility rolling 20-day over the binary investment simulation period without meta features.

This improvement can be attributed to the role of early exit as a form of *adaptive horizon control*. By allowing positions to be liquidated when predicted downside risk increases, the system truncates loss trajectories and prevents sustained capital erosion. This is corroborated by improvements in rolling risk-adjusted performance: mean rolling Sharpe increases significantly (e.g., from 1.51 to 4.51 for hybrid strategies), while minimum Sharpe values become substantially less negative, indicating reduced tail risk.

At the 1-day horizon, drawdowns are consistently smaller across all strategies, typically below -2% . For example, HYBRID KELLY MPT H1 records a maximum drawdown of -1.74% . This reflects the reduced exposure window, which limits the accumulation of adverse movements. However, this reduction in risk comes at the cost of weaker performance, as shown in Figure 6.3, where mean rolling Sharpe is near zero or negative for several strategies (e.g., -0.44 for hybrid H1).

To further quantify this trade-off, Table 6.27 summarises key rolling risk metrics. Full results are again deferred to Appendix A.7.1.

Strategy	Mean Volatility	Mean Sharpe
HYBRID_KELLY_MPT_H5_EE_OFF	0.0434	1.51
HYBRID_KELLY_MPT_H5_EE_ON	0.0374	4.51
KELLY_ONLY_H5_EE_OFF	0.0365	0.89
KELLY_ONLY_H5_EE_ON	0.0342	4.47
HYBRID_KELLY_MPT_H1_EE_OFF	0.0272	-0.44

Table 6.25: Rolling risk statistics (20-day window) for selected binary strategies without meta-features.

These results highlight a fundamental trade-off between *exposure and signal quality*. Shorter horizons reduce volatility and drawdown magnitude but produce weaker predictive signals, leading to poorer risk-adjusted returns. Conversely, longer horizons enable stronger signal extraction but introduce significant downside risk when not paired with adaptive control mechanisms.

Across allocation strategies, no single method consistently minimises drawdown in the absence of early exit. MPT and hybrid strategies exhibit similar drawdown behaviour under these

conditions, suggesting that diversification alone is insufficient to mitigate risk when predictive signals are weak or noisy. Instead, effective risk management is primarily driven by *dynamic decision mechanisms*, particularly early exit.

Overall, these findings demonstrate that, in the binary system without meta-features, **risk is primarily governed by horizon length and the presence of adaptive exit mechanisms**. Without such controls, longer horizons lead to substantial and prolonged drawdowns, whereas early exit significantly stabilises performance by limiting downside exposure.

Binary System without Meta Features The risk profile of the binary system without meta-features is summarised visually in Figure A.11 and Figure A.12. The drawdown curves in Figure A.11 show that the most severe losses are concentrated between June and July 2017, particularly for the 5-day strategies without early exit. In contrast, the rolling 20-day volatility and Sharpe profiles in Figure A.12 show that these same configurations also experience the weakest risk-adjusted behaviour over the same period.

To keep the main text focused, only a compact comparison of representative drawdown outcomes is shown in Table 6.26, while the full drawdown and rolling-risk tables are reported in Appendix A.7.1 and Appendix A.7.1.

Mode	Horizon	Early Exit	Max Drawdown	End Drawdown
MPT	5	OFF	-3.67%	-1.91%
HYBRID	5	OFF	-3.53%	-1.40%
KELLY	5	OFF	-3.40%	-1.80%
MPT	5	ON	-1.12%	-0.19%
HYBRID	5	ON	-1.14%	-0.19%

Table 6.26: Selected drawdown outcomes for the binary system without meta-features. Full results are reported in Appendix A.7.1.

At the 5-day horizon without early exit, all three allocation rules experience large and persistent peak-to-trough losses, with maximum drawdowns above -3.4% . Figure A.11 makes clear that these losses are not brief shocks but prolonged deterioration phases, indicating that fixed-horizon holding leaves the system exposed when conditions deteriorate.

This is also reflected in the rolling risk behaviour in Figure A.12. The 5-day strategies without early exit exhibit both elevated volatility and strongly negative rolling Sharpe episodes, confirming that losses are accompanied by poor short-window risk-adjusted performance rather than simply temporary mark-to-market fluctuations.

Enabling early exit substantially improves downside control. As seen in Table 6.26, maximum drawdown falls from roughly -3.5% to around -1.1% for the 5-day MPT and Hybrid strategies. Figure A.11 also shows that the EE_ON variants recover much more quickly and avoid the long, deep troughs visible in the EE_OFF runs. This supports the interpretation of early exit as a form of *adaptive horizon control*: rather than holding a position until maturity regardless of changing evidence, the strategy truncates exposure when the updated signal becomes unfavourable.

The 1-day strategies show a different pattern. Their drawdowns are consistently smaller, generally remaining below -2% , which is expected given the shorter holding exposure. However, Figure A.12 shows that lower drawdown does not necessarily translate into stronger risk-adjusted performance. In particular, several 1-day strategies exhibit near-zero or negative rolling Sharpe over extended periods, indicating that reduced exposure comes at the cost of weaker signal extraction.

For a compact numerical comparison of rolling risk, Table 6.27 reports selected 20-day statistics, while the full tables are deferred to Appendix A.7.1.

Mode	Horizon	Early Exit	Mean Volatility	Mean Sharpe	Min Sharpe
HYBRID	5	OFF	0.0434	1.51	-9.69
HYBRID	5	ON	0.0374	4.51	-1.66
KELLY	5	OFF	0.0365	0.89	-10.34
KELLY	5	ON	0.0342	4.47	-1.22
HYBRID	1	OFF	0.0272	-0.44	-4.05

Table 6.27: Selected rolling 20-day risk statistics for the binary system without meta-features. Full results are reported in Appendix A.7.1.

Taken together, the binary system without meta-features exhibits a clear trade-off between exposure and reliability. Longer horizons provide stronger return opportunities, but without adaptive controls they also generate larger and more persistent drawdowns. Early exit therefore appears to be the main mechanism stabilising downside risk in the non-meta configuration.

Binary System with Meta Features The corresponding risk profile for the binary system with meta-features is shown in Figure 6.4 and Figure 6.5. Compared with the non-meta case, the most important visual difference is that drawdown periods are redistributed later in the test window, with the most severe stress now concentrated in late July and August rather than June and July. This suggests that meta-features materially change how the system responds to changing market conditions.

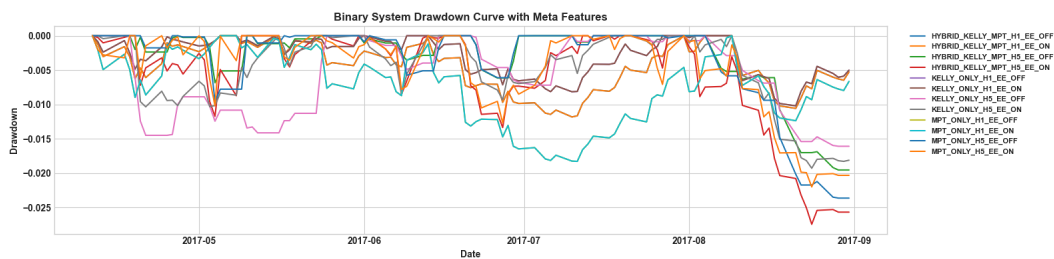


Figure 6.4: Binary system drawdown curves with meta-features.

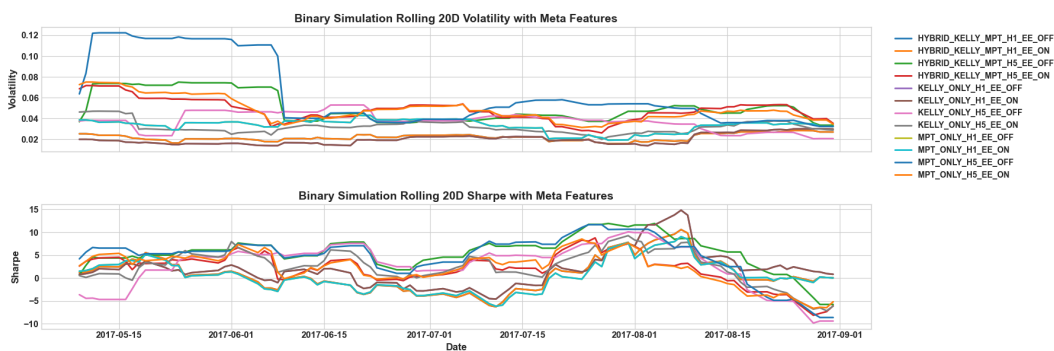


Figure 6.5: Binary system rolling 20-day volatility and Sharpe ratio with meta-features.

A compact comparison of key drawdown outcomes is provided in Table 6.28, while full drawdown tables are reported in Appendix A.7.2.

The impact of meta-features on drawdown is mixed. On the one hand, the meta-feature system achieves substantially stronger average performance, as established earlier in the return analysis.

Mode	Horizon	Early Exit	Max Drawdown	End Drawdown
HYBRID	5	ON	-2.75%	-2.57%
MPT	5	OFF	-2.36%	-2.36%
MPT	5	ON	-2.20%	-2.03%
HYBRID	5	OFF	-1.96%	-1.96%
KELLY	5	ON	-1.93%	-1.81%

Table 6.28: Selected drawdown outcomes for the binary system with meta-features. Full results are reported in Appendix A.7.2.

On the other hand, Figure 6.4 shows that this improvement does not uniformly translate into smaller drawdowns. In particular, HYBRID H5 ON reaches a drawdown of -2.75% , materially larger than its non-meta counterpart.

This suggests that meta-features act less as a direct drawdown suppressor and more as a form of *adaptive confidence modulation*. By conditioning on prior predictive errors and reliability signals, the system becomes more selective and, in favourable regimes, more aggressive. That can improve average return and mean Sharpe, but it can also increase exposure to adverse episodes when the inferred reliability signal is temporarily misaligned.

That trade-off is evident in Figure 6.5. Relative to Figure A.12, the meta-feature system achieves noticeably stronger rolling Sharpe on average for several 5-day configurations, but volatility is also higher, especially for MPT-based strategies. This is consistent with stronger conviction in the allocation layer: capital is being deployed more effectively on average, but with greater regime sensitivity.

For a compact numerical comparison, Table 6.29 reports selected rolling-risk statistics. Full results are reported in Appendix A.7.2.

Mode	Horizon	Early Exit	Mean Volatility	Mean Sharpe	Min Sharpe
HYBRID	5	OFF	0.0499	5.54	-5.76
MPT	5	OFF	0.0630	4.78	-8.61
KELLY	5	OFF	0.0375	2.76	-9.82
HYBRID	5	ON	0.0478	2.01	-8.13
KELLY	1	OFF	0.0201	1.93	-4.58

Table 6.29: Selected rolling 20-day risk statistics for the binary system with meta-features. Full results are reported in Appendix A.7.2.

A key result is that average risk-adjusted performance improves markedly in the meta-feature system. For example, the Hybrid 5-day OFF configuration increases its mean rolling Sharpe from 1.51 to 5.54. However, minimum Sharpe values remain strongly negative in several cases, so tail risk is reduced only partially. In other words, meta-features improve *average* behaviour more than they eliminate downside excursions.

At the 1-day horizon, meta-features appear more stable. Drawdowns remain modest and mean Sharpe improves relative to the non-meta case, suggesting that rapid feedback helps the model estimate its own reliability more effectively over short horizons.

Overall, the binary meta-feature system demonstrates a different risk-return balance from the non-meta baseline. Meta-features improve average risk-adjusted performance and alter the timing of drawdown episodes, but they also introduce greater regime dependence. The result is a system that is stronger on average, yet not uniformly safer, reinforcing the need to evaluate

meta-features jointly with allocation and exit logic rather than in isolation.

Ordinal Multi-Class System without Meta Features The risk behaviour of the ordinal multi-class system without meta-features is illustrated by the drawdown curves in Figure A.13 and summarised numerically in Table 6.30. Compared with the binary system, the ordinal setting exhibits a more differentiated risk structure across strategies, reflecting the richer signal provided by ordered return classes rather than a simple up/down decision.

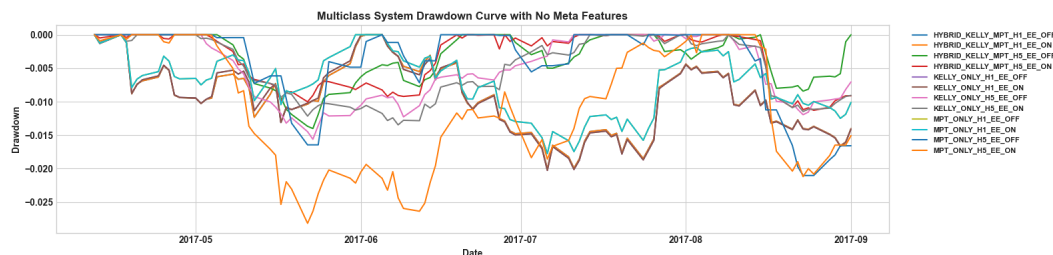


Figure 6.6: Ordinal multi-class system drawdown curves without meta-features.

Table 6.30: Selected rolling 20-day risk statistics for the ordinal multi-class system without meta-features. Full rolling-risk tables are reported in Appendix A.7.3.

Mode	Horizon	Early Exit	Mean Volatility	Mean Sharpe
KELLY	5	OFF	0.0194	5.03
MPT	5	OFF	0.0529	4.84
HYBRID	5	OFF	0.0232	4.32
HYBRID	5	ON	0.0226	3.17
KELLY	5	ON	0.0195	1.88
MPT	5	ON	0.0430	1.68
MPT	1	OFF	0.0327	-0.01
KELLY	1	OFF	0.0344	-0.52
HYBRID	1	OFF	0.0344	-0.54

The drawdown curves in Figure A.13 show that the most severe losses are not uniformly concentrated in a single period. Instead, different strategies experience their major drawdowns at different points in the test window, indicating that the ordinal signal interacts more strongly with the allocation rule. Full drawdown statistics are reported in Appendix A.7.3.

A key observation is that the strongest-performing 5-day strategies without meta-features generally maintain relatively controlled downside given their return profile. In particular, the Hybrid and Kelly 5-day OFF configurations achieve comparatively modest maximum drawdowns of -1.40% and -1.56% respectively, while the MPT 5-day OFF strategy reaches -2.10%. By contrast, the 1-day ordinal strategies exhibit weaker overall behaviour, with drawdowns of approximately -1.78% to -2.03% despite much lower return generation.

This trade-off is reinforced by Table 6.30. The 5-day OFF configurations sustain the strongest average rolling Sharpe over the test period, especially KELLY H5 OFF, which combines the lowest mean rolling volatility with the highest mean rolling Sharpe among the non-meta ordinal configurations shown. In contrast, the 1-day strategies display weak or negative average rolling Sharpe, implying that their smaller or comparable drawdowns do not translate into attractive risk-adjusted performance.

This indicates that, in the ordinal non-meta setting, the additional information carried by ordered class predictions is most effectively exploited at the 5-day horizon. The system appears capable of

extracting more persistent signal over longer holding periods, while the 1-day horizon remains too noisy to support consistently favourable risk-adjusted outcomes.

The role of early exit is more nuanced here than in the binary case. Figure A.13 suggests that early exit does not uniformly reduce downside risk in the ordinal system. For example, MPT H5 ON exhibits the largest maximum drawdown in this setting at -2.82% , exceeding its corresponding EE_OFF variant. Likewise, Table 6.30 shows that several EE_ON configurations achieve weaker average Sharpe than their fixed-horizon counterparts. This suggests that, when ordinal predictions already provide magnitude-sensitive information, premature liquidation may interfere with the ability of profitable positions to realise their full expected move.

Overall, the ordinal multi-class system without meta-features appears strongest when paired with longer horizons and fixed holding periods. Relative to the binary case, the richer ordinal signal supports better average risk-adjusted performance, but the benefits depend materially on the interaction between signal structure and allocation logic rather than on early exit alone.

Ordinal Multi-Class System with Meta Features The corresponding ordinal system with meta-features is shown in Figure 6.7 and summarised in Table 6.31. The most immediate visual difference is that drawdown periods become more synchronised across strategies, with the most severe losses concentrated primarily between June and July 2017. This suggests that meta-features alter not only the strength of the predictive signal but also the timing and concentration of risk exposure across strategies.

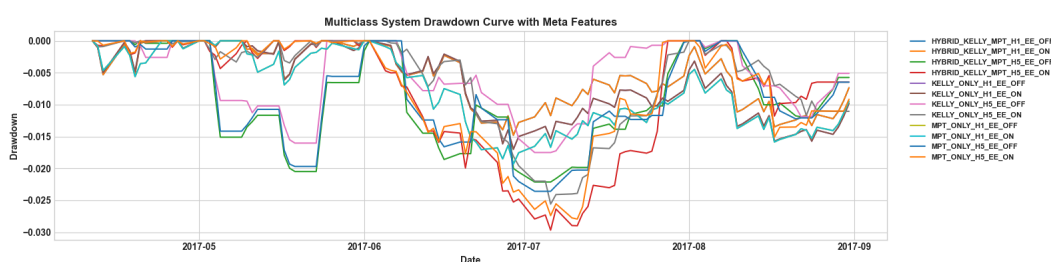


Figure 6.7: Ordinal multi-class system drawdown curves with meta-features.

Table 6.31: Selected rolling 20-day risk statistics for the ordinal multi-class system with meta-features. Full results are provided in Appendix A.7.4.

Mode	Horizon	Early Exit	Mean Volatility	Mean Sharpe
MPT	5	ON	0.0455	2.58
KELLY	5	ON	0.0342	2.27
HYBRID	5	ON	0.0455	2.13
KELLY	5	OFF	0.0410	1.87
HYBRID	5	OFF	0.0496	1.81
MPT	5	OFF	0.0509	1.60
HYBRID	1	OFF	0.0319	0.70
KELLY	1	OFF	0.0327	0.48
MPT	1	OFF	0.0311	0.18

Full drawdown and rolling-risk tables for the meta-feature ordinal system are reported in Appendix A.7.4 and Appendix A.7.4.

Compared with the non-meta ordinal system, maximum drawdowns are generally larger for the 5-day strategies. The worst case is HYBRID H5 ON at -2.97% , followed closely by MPT H5 ON at -2.79% and KELLY H5 ON at -2.56% . Even the 5-day OFF variants show deeper

drawdowns than several of their non-meta counterparts. This indicates that the inclusion of meta-features does not directly reduce downside excursion in the ordinal setting.

However, Table 6.31 shows that the meta-feature system remains competitive in terms of average rolling Sharpe, particularly for the 5-day strategies with early exit enabled. In contrast to the non-meta ordinal case, the strongest average Sharpe values now occur for the EE_ON configurations, especially MPT H5 ON. This suggests that, once meta-features are incorporated, early exit becomes more compatible with the signal structure, potentially because the meta-information improves short-term reliability estimates used to trigger liquidation.

From a volatility perspective, the meta-feature ordinal system is generally more variable at the 5-day horizon. As shown in Table 6.31, several 5-day strategies maintain elevated mean rolling volatility relative to their non-meta counterparts, particularly the MPT and Hybrid configurations. This highlights a trade-off: meta-features improve responsiveness and risk-adjusted returns in some configurations, but at the cost of increased variability.

At the 1-day horizon, meta-features provide a clearer improvement over the non-meta system. Mean rolling Sharpe becomes positive across all strategies, in contrast to the near-zero or negative values observed previously. This suggests that meta-features are particularly beneficial when the base predictive signal is weak, enabling the system to condition on recent performance and mitigate short-horizon noise.

Taken together, the ordinal multi-class system with meta-features exhibits a different balance from the non-meta version. While drawdowns are not reduced, and may increase at longer horizons, the consistency of risk-adjusted performance improves in several key configurations, particularly when combined with early exit. This supports the interpretation of meta-features as a *dynamic reliability signal*, allowing the system to adapt its behaviour based on recent predictive performance under changing market conditions.

6.6.5 Allocation Strategy Comparison

The relative strength of the allocation strategies depends mainly on prediction horizon and signal quality, as shown in Tables 6.12, 6.13, 6.14, and 6.15.

For the **1-day horizon**, KELLY is the strongest strategy overall. In the binary system without meta-features (Table 6.12), KELLY H1 achieves the highest ROI (0.97%) and the highest Sharpe ratio (1.04). The same pattern holds in the binary system with meta-features (Table 6.13), where KELLY H1 again produces the best 1-day ROI (1.05%) and Sharpe (1.26). In the ordinal system with meta-features (Table 6.15), KELLY H1 remains the best 1-day configuration, with the highest ROI (1.72%) and highest Sharpe (1.22). The only exception is the ordinal system without meta-features (Table 6.14), where all 1-day strategies are unprofitable and MPT H1 is simply the least poor performer.

For the **5-day horizon**, the best strategy depends on whether the objective is **maximum return** or **better risk-adjusted performance**. In terms of ROI, MPT performs best in the strongest-return settings: MPT H5 OFF achieves the highest return in the binary meta case (12.47%, Table 6.13) and in the ordinal no-meta case (10.56%, Table 6.14). However, in terms of Sharpe ratio, HYBRID is often more attractive. In the binary no-meta case, HYBRID H5 ON is best on both ROI and Sharpe (6.57%, Sharpe 4.42, Table 6.16). In the ordinal no-meta case, HYBRID H5 OFF has the highest Sharpe (5.06, Table 6.20) despite MPT H5 OFF having the larger ROI.

These patterns suggest that KELLY is best suited to **short-horizon and noisier environments**, where the signal is weaker and concentrated position sizing is more effective than diversification. MPT performs best when the signal is **stronger and more persistent**, particularly at the 5-day horizon, where diversification across opportunities translates into higher absolute returns. HYBRID performs best when a **balance between return and risk** is required, especially in the

stronger 5-day settings where Kelly-style sizing is useful but unconstrained aggressiveness would increase downside risk.

6.7 Summary of Findings

The empirical results show that predictive performance and economic performance do not always align, but several consistent patterns emerge. In the benchmark experiments, the strongest 1-day binary result was achieved by a BiGRU with sentiment features, reaching an MCC of 0.086 and slightly outperforming the Xu and Cohen benchmark on this metric. For regression, the best 1-day model was a BiLSTM with base features, although the gains were mainly in return magnitude error rather than directional quality. In the ordinal setting, the strongest 1-day result was obtained by a BiGRU with base features, with an MCC of 0.078.

Across tasks, extending the prediction horizon from 1 day to 5 days consistently improved directional performance for binary and ordinal classification. This translated into materially stronger investment performance in the simulation stage, where 5-day strategies clearly dominated 1-day strategies. The best binary simulation result was achieved by the meta-feature 5-day MPT strategy, which returned 12.47%, while the best ordinal simulation result was the 5-day non-meta MPT strategy, which returned 10.56%.

Meta-features produced mixed but important effects. In the benchmark stage, they improved regression and ordinal multi-class results more clearly than binary classification, and they also improved training stability. In the simulation stage, meta-features substantially improved the binary system, especially at the 5-day horizon, but did not improve long-horizon ordinal returns to the same extent. Early exit was also horizon- and model-dependent: it improved the binary non-meta system by reducing drawdowns and increasing Sharpe, but often reduced returns in the stronger meta-feature and ordinal settings.

Overall, the results suggest that the most effective configurations are classification-based, horizon-aware, and tightly linked to downstream allocation. Longer-horizon models produce more reliable directional signals, meta-features can enhance robustness when matched to the right formulation, and the final portfolio outcomes depend as much on calibration and allocation design as on raw predictive accuracy.

7 | Discussion

7.1 Overview

This dissertation set out to evaluate whether stock prediction models become more practically useful when they are embedded within an end-to-end decision framework that combines multi-modal forecasting, calibration, reliability-aware meta-features, and risk-aware capital allocation. This framing was central to the project from the outset: the objective was not simply to maximise predictive accuracy in isolation, but to assess whether predictive signals could be translated into economically meaningful trading performance under realistic walk-forward conditions. As defined in Chapter 1 and Chapter 3, the system was explicitly designed to bridge prediction and investment decision-making through a sequential pipeline of forecasting, calibration, and allocation. :contentReference[oaicite:0]index=0 :contentReference[oaicite:1]index=1

Overall, the results support this framing. The strongest findings did not arise from any single modelling component alone, but from the interaction between prediction horizon, signal formulation, reliability modelling, and portfolio construction. In particular, the experiments show that longer-horizon predictions were generally more useful than next-day predictions, that meta-features improved performance in several important settings, and that the allocation strategy played a decisive role in determining whether predictive gains translated into realised returns. These findings are consistent with the dissertation's broader claim that effective financial prediction systems must be evaluated as decision systems rather than as stand-alone classifiers. :contentReference[oaicite:2]index=2

7.2 Interpretation by Research Question

7.2.1 RQ1: Predictive Performance

The first research question asked whether incorporating textual information alongside price-based features improves directional and magnitude prediction under temporally valid evaluation. The results suggest that the answer is mixed but encouraging. Across the benchmarking experiments, sentiment- and sector-enhanced feature sets appeared repeatedly among the strongest-performing binary and ordinal configurations, indicating that text-derived signals can provide useful incremental information. At the same time, these gains were not universal across all architectures and formulations, suggesting that the value of textual information depends strongly on how it is combined with time-series structure. This is consistent with the dissertation's design choice to evaluate multiple model families and feature groups rather than assuming that additional modalities always improve performance. :contentReference[oaicite:3]index=3

A clearer result emerged with respect to prediction horizon. The 5-day models consistently outperformed the 1-day models in the binary and ordinal settings, both statistically and economically. This suggests that slightly longer horizons are easier for the models to learn, likely because next-day price movement is dominated by noise and microstructure effects, whereas 5-day movement contains more persistent directional structure. This interpretation is consistent

with the discussion in the literature review and research analysis that financial prediction quality is highly sensitive to horizon choice and non-stationarity. :contentReference[oaicite:4]index=4

By contrast, the regression results were less compelling. While regression models could achieve competitive MAE, their directional and investment value remained weaker than the classification-based approaches. This reinforces an important theme in the project: minimising numerical prediction error does not necessarily produce the most useful trading signal. In practical terms, ordinal and binary formulations were more effective because they aligned more naturally with the decision layer used in the simulation framework. :contentReference[oaicite:5]index=5

7.2.2 RQ2: Probability Calibration

The second research question concerned whether calibration improves reliability and downstream economic stability. The results indicate that calibration had only modest effects on headline predictive metrics, but this is not a contradiction. Calibration was never intended primarily to improve MCC or directional accuracy; rather, its purpose was to improve the quality of the probabilities that feed the capital allocation layer. This is especially important in a framework that uses Kelly-based sizing and portfolio optimisation, since both are highly sensitive to miscalibrated confidence estimates. The dissertation explicitly motivated calibration on these grounds, noting that raw neural probabilities are model confidence scores rather than true probabilities in the formal sense. :contentReference[oaicite:6]index=6

The practical implication is that calibration should be judged less by its effect on benchmark metrics and more by whether it stabilises allocation behaviour. In this dissertation, the investment simulation results suggest that calibration contributed most when paired with stronger predictive signals and well-matched allocation logic. This supports the claim that probability correction is a system-level improvement rather than a stand-alone predictive improvement. Put differently, calibration matters because the system consumes probabilities as decision inputs, not because it necessarily changes classification rankings. :contentReference[oaicite:7]index=7

7.2.3 RQ3: Meta-Features and Reliability Modelling

The third research question asked whether incorporating prior predictions and realised errors as meta-features improves stability under non-stationary market conditions. This was one of the most important and distinctive parts of the dissertation, and the results provide meaningful support for the approach.

Conceptually, the meta-features function as a form of reliability-aware adaptation. Rather than treating predictive confidence as static, the system conditions on recent prediction outcomes and rolling error behaviour, allowing it to respond implicitly to performance drift. This is closely aligned with the system design described in Chapters 4 and 5, where meta-features were introduced specifically to model time-varying predictive reliability in a leak-safe manner. :contentReference[oaicite:8]index=8

Empirically, the strongest gains from meta-features appeared in the binary investment simulations, where they substantially improved the 5-day results, especially under MPT allocation. In the ordinal setting, the effects were more mixed. Meta-features improved the weaker 1-day ordinal configurations more clearly, but did not consistently outperform the strongest non-meta 5-day ordinal strategies in absolute return. This suggests that reliability-aware signals are particularly valuable when the underlying predictive environment is noisy, unstable, or otherwise difficult. Where the base predictive task is already relatively strong, meta-features may improve stability or selectivity without necessarily increasing maximum return.

This interpretation also helps explain why the effect of meta-features differed by task. In binary classification, the signal is coarse and noisy, so reliability information can significantly improve

decision quality. In ordinal classification, the base model already carries richer directional and magnitude structure, so the marginal value of reliability features is smaller and may instead show up in improved consistency rather than higher peak return. This supports the view that meta-features are best understood not as a universal performance booster, but as an adaptive mechanism whose usefulness depends on the strength of the base predictive signal.

7.2.4 RQ4: Allocation Strategy Comparison

The fourth research question compared Kelly, MPT, and hybrid allocation. The results show clearly that the choice of allocation strategy is not secondary; it is central to realised performance.

At the 1-day horizon, Kelly-based allocation was generally the best performer. This likely reflects the fact that short-horizon signals are weaker and more sporadic, making concentrated sizing more effective than diversification. When the opportunity set is noisy, spreading capital too thinly can dilute what little signal exists. In this sense, Kelly acts as a focused bet-sizing mechanism that can still extract value when only a small number of signals are worth trading.

At the 5-day horizon, however, MPT often produced the highest absolute returns, particularly in the strongest binary-meta and ordinal-no-meta settings. This suggests that when the predictive signal is more persistent and cross-sectional opportunities are stronger, diversification becomes beneficial. In these cases, MPT is able to translate signal quality into higher aggregate return by allocating across multiple assets while controlling covariance structure. This outcome is consistent with the dissertation's motivation for including both Kelly and MPT, since they embody different assumptions about how predictive edge should be converted into position size. :contentReference[oaicite:9]index=9

The hybrid strategy often provided the best balance between return and risk. In several settings it achieved the strongest or near-strongest Sharpe ratios while avoiding the most extreme downside associated with pure Kelly sizing. This is arguably one of the most practically relevant findings of the dissertation: in realistic trading systems, the best approach may not be maximising growth or minimising variance in isolation, but combining both perspectives in a constrained way.

7.2.5 RQ5: Interaction Effects

The fifth research question concerned interaction effects between calibration, reliability modelling, and allocation. This is where the dissertation's integrated design matters most.

The results show that system components cannot be evaluated independently without losing important information. Early exit, for example, improved performance in some binary settings without meta-features, but reduced performance in some meta-feature settings. This implies that once the model already carries stronger reliability information, the additional intervention of early liquidation can become redundant or even harmful. Similarly, the best allocation strategy changed depending on horizon and feature configuration, demonstrating that there is no globally optimal decision rule independent of signal type.

These interaction effects strongly support the central argument of the dissertation: predictive modelling and capital allocation should not be treated as separate problems. The practical value of a forecast depends not only on whether it is statistically correct, but on how confidence is estimated, how reliability is modelled, and how the resulting signal is translated into capital deployment. The dissertation's framework was explicitly built around this idea, and the results validate that design choice. :contentReference[oaicite:10]index=10

7.3 Limitations

Several limitations should be acknowledged.

First, the underlying dataset imposes important constraints. The project is built on the StockNet-style benchmark setting, which covers a finite time period, a fixed set of U.S. stocks, and sparse tweet activity. As noted in the implementation chapter, tweet-derived features are not available uniformly across all days, requiring neutral filling and aggregation decisions that may weaken the textual signal. This limits external validity and may partly explain why some textual and meta-feature gains are inconsistent across tasks.

Second, although the simulation uses a realistic walk-forward design, the final investment evaluation still spans only the last 100 trading days. This is enough to compare strategies under controlled conditions, but it is not sufficient to claim robustness across many market regimes. A longer out-of-sample period, or evaluation across multiple market episodes, would strengthen the conclusions about stability and deployability.

Third, the decision layer remains sensitive to estimation error. This is especially true for Kelly and MPT, both of which rely on estimated probabilities, returns, or covariance structures. Even when calibration and meta-features help, errors in these inputs can still produce over-allocation, unstable exposure, or drawdown concentration. This is a known issue in portfolio construction and is already recognised in the dissertation’s discussion of capital, leverage, and non-stationarity constraints.

Fourth, the meta-feature search space was substantially larger than the baseline feature space. Although the tuning process still used 100 Optuna trials, the inclusion of many additional feature-set combinations means that the effective search coverage per configuration was lower. In other words, the meta-feature experiments were conducted under the same trial budget despite a significantly expanded design space. This means that some meta-feature configurations may not have been tuned as thoroughly as the baseline ones, and stronger results may have been possible with a larger search budget.

Fifth, transaction costs, capital caps, and portfolio controls were included, but the simulation does not capture all practical frictions of live trading. There is no explicit modelling of market impact, slippage beyond chosen assumptions, execution latency, or changing liquidity. As a result, the reported results should be interpreted as realistic simulations rather than evidence of immediate deployability.

Sixth, the study does not include live deployment or paper trading. This means the framework is still being evaluated in a historically replayed environment, not in a live adaptive market setting where behaviour could change in response to execution.

Finally, computational limitations constrained the breadth of experimentation. The project already evaluated many architectures, feature groups, horizons, and allocation rules, but further exploration of larger hyperparameter budgets, broader transformer comparisons, or alternative calibration and reliability formulations would likely reveal additional structure.

7.4 Practical Implications

Despite these limitations, the dissertation suggests several practical implications.

First, slightly longer-horizon signals appear more useful than next-day predictions for this dataset and framework. In practical terms, this implies that trading systems based on daily data may benefit more from modestly extended holding periods than from attempting to exploit highly noisy next-day movement.

Second, reliability-aware features appear particularly useful in unstable or noisy environments. This suggests that predictive systems in finance may benefit from explicitly modelling their own recent performance rather than assuming static confidence quality.

Third, allocation strategy should be matched to signal characteristics. Kelly-style sizing is well suited to short-horizon or sparse high-conviction opportunities, while MPT benefits more from stronger, more persistent cross-sectional signals. Hybrid approaches may offer the most robust practical compromise when both growth and risk control matter.

Most importantly, the results reinforce that high predictive accuracy alone is not enough. A forecasting model becomes useful only when its outputs can be translated into stable, well-sized decisions under realistic constraints.

7.5 Concluding Discussion

The central conclusion of the discussion is that the best-performing systems were those in which the predictive layer, reliability layer, and allocation layer were well aligned. Longer horizons improved extractable signal, meta-features improved adaptation in several settings, and allocation strategy determined whether those signals produced return, stability, or both.

This supports the dissertation's main claim: financial prediction should be treated as an integrated decision problem rather than a stand-alone forecasting task. The most useful models are not simply the most accurate, but those whose outputs are reliable enough, informative enough, and well enough matched to downstream allocation rules to produce meaningful economic value under non-stationary conditions.

8 | Conclusion

This dissertation investigated how stock trend prediction can be made more practically useful by moving beyond accuracy-driven benchmarking and evaluating models within an end-to-end investment decision framework. The project addressed the general problem introduced in Chapter 1: financial prediction is not valuable simply because a model achieves reasonable classification performance, but because its outputs can be converted into reliable and profitable decisions under realistic market conditions. The dissertation was explicitly designed to build on Xu and Cohen (2018) while extending the task beyond fixed next-day prediction toward multi-horizon forecasting, reliability-aware modelling, and downstream capital allocation.

To tackle this, the project developed and evaluated a multimodal pipeline combining historical price data, textual features, probability calibration, meta-features based on prior prediction behaviour, and multiple capital allocation strategies. Rather than focusing on a single formulation, the system was tested across binary classification, ordinal multi-class classification, and investment simulation under 1-day and 5-day horizons. This made it possible to assess not only predictive quality, but also whether model improvements translated into better financial outcomes.

Several clear findings emerged. First, the 5-day horizon consistently outperformed the 1-day horizon across both predictive and economic evaluation. In the classification experiments, longer horizons generally achieved stronger MCC values and produced substantially better simulation outcomes. In particular, the strongest classification configurations exceeded the Xu and Cohen (2018) baseline on MCC, which is especially important given the dissertation's emphasis on robust class-sensitive evaluation rather than accuracy alone. Xu and Cohen's reported benchmark achieved an MCC of approximately 0.080, whereas this project reached a higher MCC of approximately 0.086 for a 1-day horizon.

Second, these predictive gains were matched by strong economic results. In the binary investment setting without meta-features, the strongest configuration was the 5-day Hybrid Kelly-MPT strategy with early exit enabled, achieving a return of 6.57% and a Sharpe ratio of 4.42. With meta-features, the best binary result was achieved by the 5-day MPT strategy without early exit, which returned 12.47% with a Sharpe ratio of 4.34. In the ordinal multi-class setting without meta-features, the best ROI was produced by the 5-day MPT strategy at 10.56%, while the strongest risk-adjusted performance was achieved by the 5-day Hybrid strategy with a Sharpe ratio of 5.06. With meta-features, the best ordinal result was the 5-day MPT strategy with early exit enabled, achieving a return of 4.86% and a Sharpe ratio of 2.59.

Third, the results showed that meta-features were useful as a form of regime-aware reliability, but not uniformly so. In the binary system they produced a substantial improvement, especially in the stronger 5-day setting, suggesting that reliability-aware information can materially improve decision quality when the base task is noisy. In the ordinal system, meta-features improved the weaker 1-day case more clearly than the strongest 5-day case, indicating that their value is greatest when the underlying predictive signal is unstable or difficult. This supports the central idea that historical prediction behaviour can act as a useful proxy for time-varying model reliability.

Fourth, the project showed that allocation strategy matters as much as prediction quality. Kelly-based allocation was generally strongest at the 1-day horizon, where signals were weaker and more concentrated position sizing was beneficial. MPT performed best in the strongest 5-day

ROI settings, where diversification across stronger cross-sectional opportunities translated into larger absolute returns. The hybrid Kelly–MPT strategy often provided the best balance between return and risk, particularly in settings where the signal was strong enough to justify Kelly-style sizing but noisy enough that portfolio constraints remained valuable.

Overall, the dissertation demonstrates that the most effective stock prediction systems are not those that simply maximise benchmark metrics, but those that combine useful predictive structure with reliability-aware modelling and appropriate capital allocation. The strongest contributions of this project therefore lie not in proposing a new deep learning architecture, but in integrating multimodal forecasting, calibration, meta-feature reliability modelling, and portfolio construction into a unified evaluation framework grounded in economic outcomes.

8.1 Future work

There are several directions in which this work could be taken further.

The first is to extend the dataset and evaluation period. The present system was tested on a historically replayed benchmark over a limited out-of-sample window. A longer evaluation period covering a wider range of market regimes would strengthen claims about robustness, especially for meta-features and allocation behaviour under changing volatility conditions.

The second is to improve the feature space, particularly for text. While textual and meta-features did provide gains in several settings, the signal remains constrained by sparsity, aggregation quality, and the limited availability of text on some trading days. A natural next step would be to incorporate richer financial text sources, such as headlines, earnings call transcripts, or filtered social media content weighted by source reliability and engagement.

The third is to expand the reliability-modelling component. In this dissertation, meta-features were constructed from prior predictions and realised errors, which already showed promising results. Future work could explore more explicit regime-detection mechanisms, confidence forecasting, or uncertainty-aware architectures that directly model predictive reliability rather than only learning it through auxiliary features.

The fourth is to deepen the optimisation and tuning process. Although the meta-feature experiments still used 100 Optuna trials, the search space became substantially larger once many additional feature combinations were introduced. Increasing the tuning budget, applying more structured search strategies, or using staged tuning could improve model selection and potentially reveal stronger meta-feature configurations.

The fifth is to strengthen the decision layer. Future work could compare additional allocation schemes such as fractional Kelly, robust portfolio optimisation, downside-risk-aware objectives, or dynamic leverage constraints. More realistic execution assumptions could also be added, including transaction cost sensitivity analysis, market impact approximations, and live paper-trading evaluation.

Finally, a longer-term research direction would be to treat the system as a continuously adaptive decision process rather than a fixed forecasting pipeline. This could involve reinforcement learning for allocation, online calibration, or dynamic model selection across regimes.

In summary, this dissertation has shown that combining multimodal deep learning with reliability-aware modelling and principled allocation can produce economically meaningful improvements over accuracy-only evaluation. It also shows that, in key predictive terms such as MCC, the framework can surpass the Xu and Cohen benchmark while delivering much stronger downstream economic performance. The strongest future opportunities lie in improving signal quality, expanding robustness across regimes, and deepening the interaction between predictive modelling and downstream financial decision-making.

A | Appendices

A.1 Investment Simulation Parameters

This appendix summarises the parameters and constraints used in the investment simulation. These controls ensure realistic trading behaviour and limit excessive leverage or capital deployment during the simulation.

Table A.1: Key parameters used in the investment simulation framework.

Parameter	Description	Value
Initial Capital	Starting portfolio capital	\$100,000
Transaction Cost	Proportional cost applied per trade	0.1%
Kelly Fraction	Fractional Kelly scaling factor	0.25
Kelly Trade Cap	Maximum raw Kelly allocation per trade	0.50
MPT Window	Lookback window for covariance estimation	60 days
MPT Risk Aversion (λ)	Mean–variance risk penalty parameter	1.0
MPT Ridge Regularisation	Stabilisation term for covariance matrix	10^{-4}
MPT Max Weight	Maximum asset weight under MPT allocation	30%
MPT Long Only	Restrict portfolio weights to non-negative values	True
MPT Budget Fraction	Fraction of capital allocated through MPT	30%
Retraining Interval	Walk-forward model refitting frequency	200 days
Maximum Day Gap	Maximum allowed trading-day gap in data	5 days
Max Total Utilisation	Maximum deployed portfolio capital	70%
Max New Daily Utilisation	Maximum capital deployed per day	30%
Max Ticker Utilisation	Maximum exposure to a single ticker	50%
Max Sector Utilisation	Maximum exposure to a single sector	50%
Max Position Size	Maximum capital allocated per trade	10%
Max Positions per Ticker	Maximum simultaneous trades per ticker	3
Minimum Trade Size	Minimum trade value threshold	\$100
Drawdown Throttle	Utilisation reduction during drawdowns	(8%, 0.60), (15%, 0.35)

A.2 Compared Models and Feature Groups

The model families evaluated are:

- LSTM
- BiLSTM
- GRU
- BiGRU

The feature-group structure used during benchmarking is:

```

model_types = ["LSTM", "BiLSTM", "GRU", "BiGRU"]

horizon_steps = [1, 5]

feature_columns = [
    'open', 'high', 'low', 'close', 'volume',
    'roll_ret_1d', 'roll_ret_5d',
    'roll_ret_20d',

    'ema_12', 'ema_26', 'ema_50', 'macd_12_26_9', 'macdh_12_26_9',
    'macds_12_26_9', 'rsi_14', 'stochrsik_14_14_3_3', 'stochrsid_14_14_3_3',
    'atrr_14', 'bb_upper', 'bb_middle', 'bb_lower', 'obv',
]

sentiment_columns = [
    'sentiment',
]

emotion_columns = [
    'emotion_anger', 'emotion_disgust', 'emotion_fear', 'emotion_joy',
    'emotion_neutral', 'emotion_sadness', 'emotion_surprise',
    'emotion_anger_pct', 'emotion_disgust_pct', 'emotion_fear_pct',
    'emotion_joy_pct', 'emotion_neutral_pct', 'emotion_sadness_pct',
    'emotion_surprise_pct',
]

unified_emotion_columns = [
    'positive_emotion', 'negative_emotion', 'uncertainty_emotion',
    'positive_emotion_pct', 'negative_emotion_pct', 'uncertainty_emotion_pct',
]

stance_columns = [
    'stance_label', 'stance_score',
]

finbert_columns = [
    'finbert_label', 'finbert_score', 'finbert_up', 'finbert_down',
    'finbert_neutral',
]

sector_columns = ['sector_open_mean', 'sector_high_mean', 'sector_low_mean',
    'sector_close_mean',
    'sector_volume_mean', 'sector_ret_1d', 'sector_ret_5d',
    'sector_ret_20d', 'sector_range', 'sector_vol_20d',

    'ema_12_sector', 'ema_26_sector', 'ema_50_sector',
    'macd_12_26_9_sector',
    'macdh_12_26_9_sector', 'macds_12_26_9_sector', 'rsi_14_sector',
    'sector_bb_upper', 'sector_bb_middle', 'sector_bb_lower',
    'market_close', 'sector_rel_strength', 'sector_dispersion_1d']

meta_regression_columns = [
    'reg_pred_ret_1d_LSTM', 'reg_pred_ret_1d_BiLSTM', 'reg_pred_ret_1d_GRU',
    'reg_pred_ret_1d_BiGRU',
]

```

```

meta_classification_columns = [
    'cls_prob_up_1d_LSTM', 'cls_prob_up_1d_BiLSTM', 'cls_prob_up_1d_GRU',
    'cls_prob_up_1d_BiGRU',
]

meta_regression_err_columns = [
    'reg_abs_err_lag1_LSTM', 'reg_mae_20_LSTM',
    'reg_rmse_20_LSTM', 'reg_dir_acc_20_LSTM',

    'reg_abs_err_lag1_BiLSTM',
    'reg_mae_20_BiLSTM', 'reg_rmse_20_BiLSTM', 'reg_dir_acc_20_BiLSTM',

    'reg_abs_err_lag1_GRU',
    'reg_mae_20_GRU', 'reg_rmse_20_GRU', 'reg_dir_acc_20_GRU',

    'reg_abs_err_lag1_BiGRU',
    'reg_mae_20_BiGRU', 'reg_rmse_20_BiGRU', 'reg_dir_acc_20_BiGRU',]

meta_classification_err_columns = [
    'cls_brier_20_LSTM', 'cls_logloss_20_LSTM', 'cls_acc_20_LSTM',

    'cls_brier_20_BiLSTM', 'cls_logloss_20_BiLSTM', 'cls_acc_20_BiLSTM',

    'cls_brier_20_GRU', 'cls_logloss_20_GRU', 'cls_acc_20_GRU',

    'cls_brier_20_BiGRU', 'cls_logloss_20_BiGRU', 'cls_acc_20_BiGRU',
]

#Feature sets
feature_sets = {
    'base': feature_columns,
    'sentiment': feature_columns + sentiment_columns,,
    'emotion': feature_columns + emotion_columns,
    'unified_emotion': feature_columns + unified_emotion_columns,
    'finbert': feature_columns + finbert_columns,
    'all_nlp': feature_columns + sentiment_columns + emotion_columns +
        unified_emotion_columns + stance_columns + finbert_columns,
    'sector': feature_columns + sector_columns,
    'sector_sentiment': feature_columns + sector_columns + sentiment_columns,
    'sector_emotion': feature_columns + sector_columns + emotion_columns,
    'sector_unified_emotion': feature_columns + sector_columns +
        unified_emotion_columns,
    'sector_finbert': feature_columns + sector_columns + finbert_columns,
    'sector_all_nlp': feature_columns + sector_columns + sentiment_columns +
        emotion_columns + unified_emotion_columns + stance_columns +
        finbert_columns,

    'meta': feature_columns + meta_regression_columns +
        meta_classification_columns,

    'meta_err': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns,

    'meta_sentiment': feature_columns + meta_regression_columns +
        meta_classification_columns + sentiment_columns,
    'meta_stance': feature_columns + meta_regression_columns +

```

```

        meta_classification_columns + stance_columns,
'meta_emotion': feature_columns + meta_regression_columns +
        meta_classification_columns + emotion_columns,
'meta_unified_emotion': feature_columns + meta_regression_columns +
        meta_classification_columns + unified_emotion_columns,
'meta_finbert': feature_columns + meta_regression_columns +
        meta_classification_columns + finbert_columns,
'meta_all_nlp': feature_columns + meta_regression_columns +
        meta_classification_columns + sentiment_columns + emotion_columns +
        unified_emotion_columns + stance_columns + finbert_columns,
'meta_sector': feature_columns + meta_regression_columns +
        meta_classification_columns + sector_columns,
'meta_sector_sentiment': feature_columns + meta_regression_columns +
        meta_classification_columns + sector_columns + sentiment_columns,
'meta_sector_stance': feature_columns + meta_regression_columns +
        meta_classification_columns + sector_columns + stance_columns,
'meta_sector_emotion': feature_columns + meta_regression_columns +
        meta_classification_columns + sector_columns + emotion_columns,
'meta_sector_unified_emotion': feature_columns + meta_regression_columns +
        meta_classification_columns + sector_columns + unified_emotion_columns,
'meta_sector_finbert': feature_columns + meta_regression_columns +
        meta_classification_columns + sector_columns + finbert_columns,
'meta_sector_all_nlp': feature_columns + meta_regression_columns +
        meta_classification_columns + sector_columns + sentiment_columns +
        emotion_columns + unified_emotion_columns + stance_columns +
        finbert_columns,

'meta_err_sentiment': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + sentiment_columns,
'meta_err_stance': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + stance_columns,
'meta_err_emotion': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + emotion_columns,
'meta_err_unified_emotion': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + unified_emotion_columns,
'meta_err_finbert': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + finbert_columns,
'meta_err_all_nlp': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + sentiment_columns + emotion_columns +
        unified_emotion_columns + stance_columns + finbert_columns,
'meta_err_sector': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + sector_columns,
'meta_err_sector_sentiment': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + sector_columns + sentiment_columns,
'meta_err_sector_stance': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +
        meta_classification_err_columns + sector_columns + stance_columns,
'meta_err_sector_emotion': feature_columns + meta_regression_columns +
        meta_classification_columns + meta_regression_err_columns +

```

```

meta_classification_err_columns + sector_columns + emotion_columns,
'meta_err_sector_unified_emotion': feature_columns + meta_regression_columns +
meta_classification_columns + meta_regression_err_columns +
meta_classification_err_columns + sector_columns + unified_emotion_columns,
'meta_err_sector_finbert': feature_columns + meta_regression_columns +
meta_classification_columns + meta_regression_err_columns +
meta_classification_err_columns + sector_columns + finbert_columns,
'meta_err_sector_all_nlp': feature_columns + meta_regression_columns +
meta_classification_columns + meta_regression_err_columns +
meta_classification_err_columns + sector_columns + sentiment_columns +
emotion_columns + unified_emotion_columns + stance_columns +
finbert_columns,
}}

```

Listing A.1: Model families, horizon and feature-group structure used in benchmarking.

A.3 Loss Curves

. This appendix presents the full set of training and validation loss curves for all models and feature-set combinations evaluated in the study. The main body of the paper summarises the training behaviour using aggregated loss statistics, while the figures below provide the complete loss curves for all model and feature-set combinations.

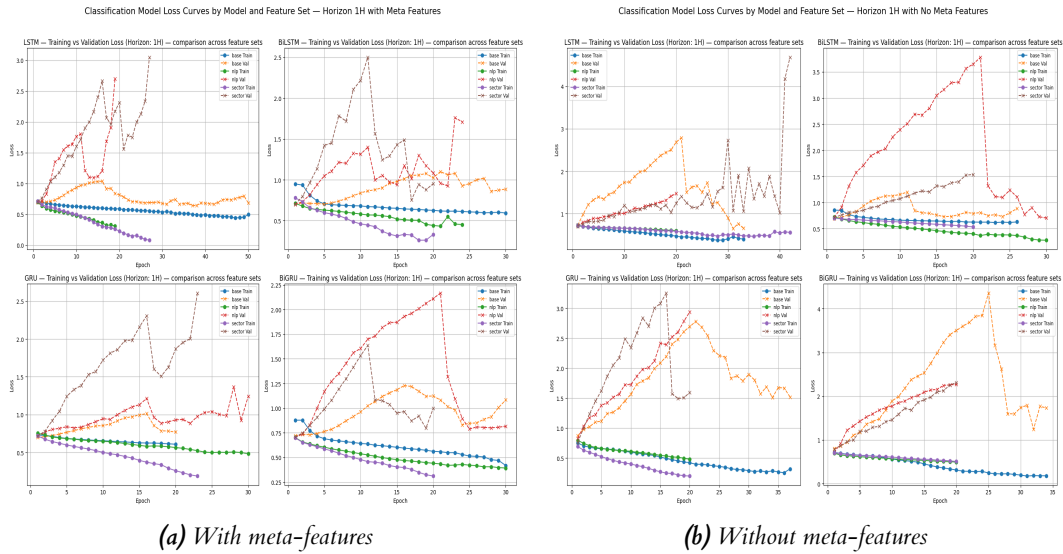


Figure A.1: Binary classification training and validation loss curves for all model and feature-set combinations.

TODO: Normalise pics

A.4 Meta-Feature Experimental Results

This appendix presents the complete set of experimental results obtained using reliability-aware meta-features. The tables report aggregated performance across all evaluated companies for the

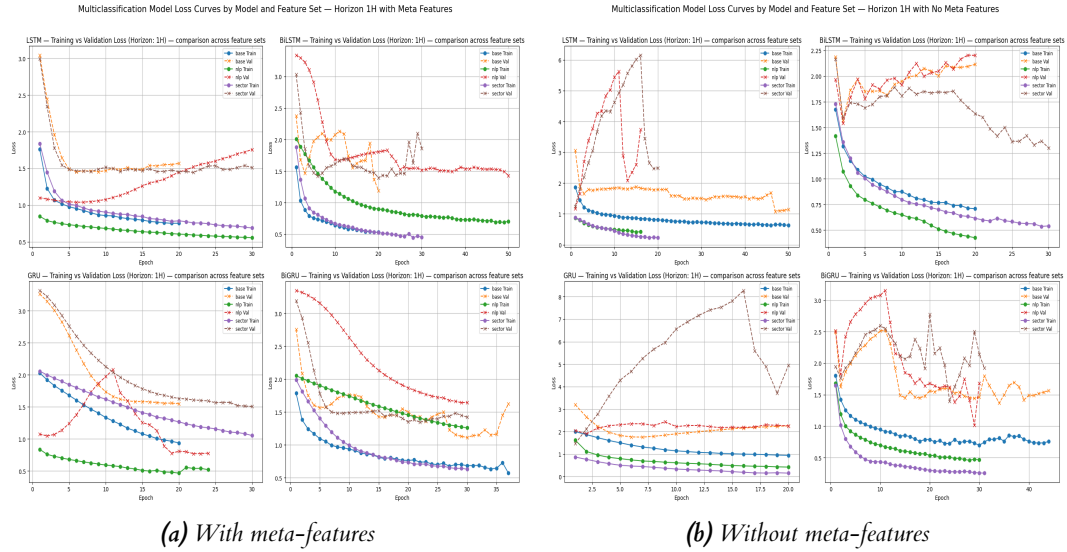


Figure A.2: Ordinal multi-class training and validation loss curves for all model and feature-set combinations.

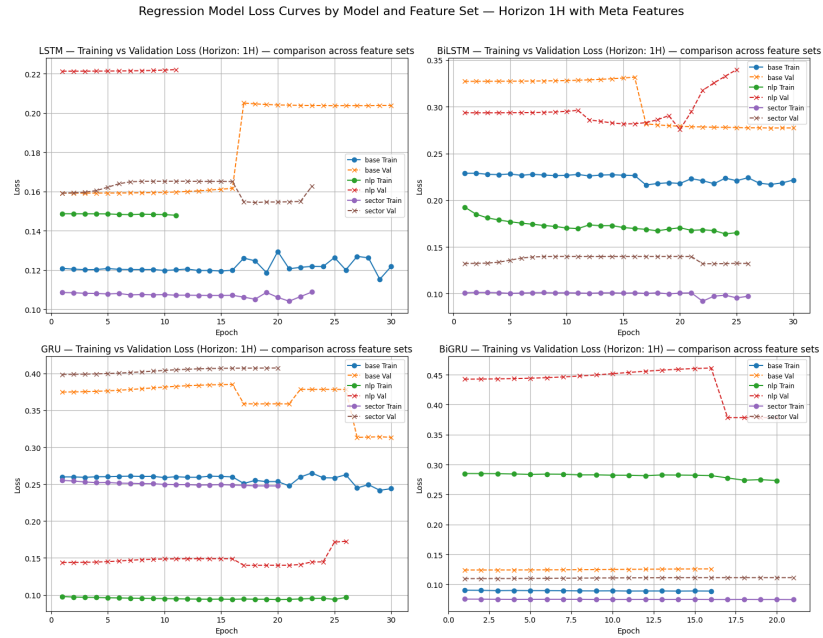


Figure A.3: Regression training and validation loss curves with meta-features.

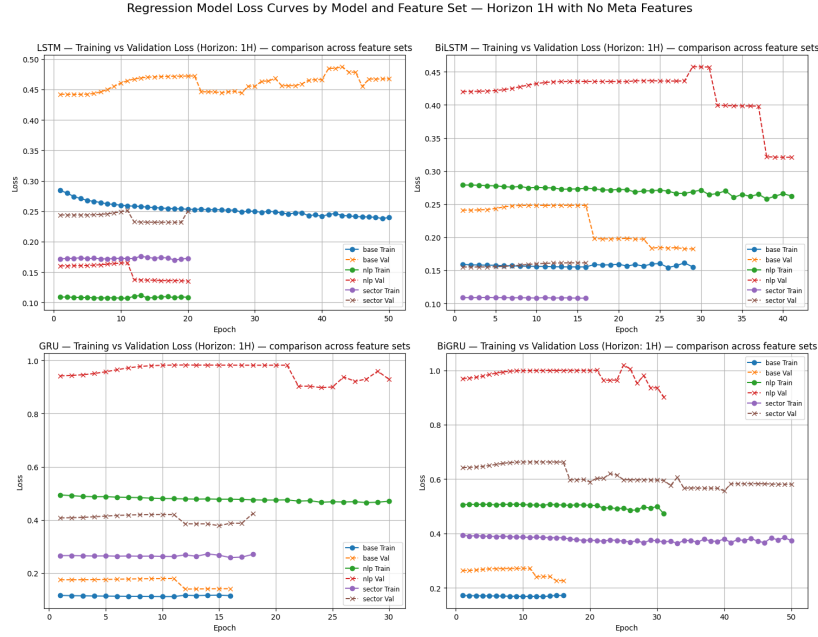


Figure A.4: Regression training and validation loss curves without meta-features.

1-day prediction horizon. Metrics are reported as mean values with the corresponding standard deviations across companies.

For clarity, the main body of the paper reports only the strongest-performing configurations, while the full experimental comparison is provided here. **TODO: fix table structure**

A.4.1 Binary Classification Results

Table A.2: Binary classification performance with meta-features (1-day horizon).

Model	Feature Set	Dir Acc	MCC	F1	Recall
GRU	meta_sector_fnbert	0.516 ± 0.063	0.081 ± 0.128	0.358 ± 0.242	0.400 ± 0.360
BiGRU	meta	0.526 ± 0.058	0.076 ± 0.116	0.403 ± 0.237	0.458 ± 0.367
LSTM	meta	0.513 ± 0.060	0.075 ± 0.119	0.341 ± 0.245	0.387 ± 0.374
GRU	meta	0.512 ± 0.062	0.072 ± 0.114	0.399 ± 0.246	0.478 ± 0.383
BiLSTM	meta_sector_fnbert	0.509 ± 0.055	0.067 ± 0.103	0.360 ± 0.238	0.409 ± 0.364
BiLSTM	meta_stance	0.520 ± 0.053	0.065 ± 0.103	0.422 ± 0.218	0.480 ± 0.354
BiGRU	meta_sector_unified_emotion	0.513 ± 0.060	0.061 ± 0.112	0.294 ± 0.254	0.330 ± 0.371
BiGRU	meta_fnbert	0.510 ± 0.060	0.055 ± 0.115	0.341 ± 0.257	0.398 ± 0.384
BiLSTM	meta	0.506 ± 0.053	0.054 ± 0.111	0.372 ± 0.257	0.449 ± 0.391
LSTM	meta_stance	0.510 ± 0.058	0.053 ± 0.114	0.369 ± 0.249	0.434 ± 0.385
GRU	meta_emotion	0.514 ± 0.058	0.051 ± 0.112	0.399 ± 0.233	0.454 ± 0.355
LSTM	meta_err_sector_stance	0.507 ± 0.052	0.028 ± 0.100	0.391 ± 0.253	0.464 ± 0.375

A.4.2 Regression Results

Table A.3: Regression performance with meta-features (1-day horizon).

Model	Feature Set	Dir Acc	MAE	MSE
BiGRU	meta_sector	0.509 ± 0.054	0.01135	0.000242
BiGRU	meta_err	0.511 ± 0.052	0.01132	0.000239
GRU	meta_err_sector_stance	0.504 ± 0.061	0.01132	0.000240
GRU	meta_unified_emotion	0.502 ± 0.053	0.01134	0.000242
GRU	meta	0.501 ± 0.060	0.01132	0.000240
BiLSTM	meta_err_sector_sentiment	0.505 ± 0.049	0.01130	0.000239
BiLSTM	meta	0.501 ± 0.053	0.01134	0.000241
LSTM	meta_sector	0.499 ± 0.051	0.01134	0.000240
LSTM	meta_err_emotion	0.500 ± 0.051	0.01132	0.000240
BiLSTM	meta_err_unified_emotion	0.507 ± 0.050	0.01130	0.000240
BiGRU	meta_emotion	0.499 ± 0.056	0.01134	0.000241
LSTM	meta	0.501 ± 0.048	0.01130	0.000239

A.4.3 Ordinal Multi-Class Results

Table A.4: Ordinal multi-class performance with meta-features (1-day horizon).

Model	Feature Set	Micro Acc	Macro F1	MCC
BiLSTM	meta_sector_sentiment	0.163 ± 0.069	0.111 ± 0.053	0.083 ± 0.100
BiGRU	meta_unified_emotion	0.159 ± 0.076	0.101 ± 0.051	0.081 ± 0.108
BiGRU	meta_sector_sentiment	0.165 ± 0.069	0.110 ± 0.050	0.071 ± 0.115
GRU	meta_sector_sentiment	0.151 ± 0.075	0.097 ± 0.050	0.070 ± 0.101
LSTM	meta_sector_sentiment	0.174 ± 0.068	0.117 ± 0.047	0.069 ± 0.113
BiLSTM	meta	0.157 ± 0.069	0.106 ± 0.055	0.064 ± 0.119
LSTM	meta	0.183 ± 0.065	0.123 ± 0.051	0.062 ± 0.108
BiGRU	meta	0.156 ± 0.074	0.107 ± 0.054	0.060 ± 0.139
BiLSTM	meta_unified_emotion	0.159 ± 0.069	0.112 ± 0.055	0.056 ± 0.115
GRU	meta	0.179 ± 0.069	0.122 ± 0.053	0.034 ± 0.113
GRU	meta_stance	0.165 ± 0.083	0.089 ± 0.053	0.024 ± 0.102
LSTM	meta_stance	0.172 ± 0.072	0.098 ± 0.054	0.022 ± 0.088

A.5 Variable-Horizon Results

This appendix reports the full 5-day horizon results for the binary classification, regression, and ordinal multi-class tasks, both with and without meta-features. These tables complement the summary comparisons presented in Section ?? by providing the full set of evaluated model and feature-set combinations.

A.5.1 Binary Classification Results: No Meta-Features

Table A.5: Binary classification results at the 5-day horizon without meta-features.

Model	Feature Set	Dir Acc	MCC	F1	Precision	Recall
GRU	sector_sentiment	0.529	0.152	0.388	0.608	0.427
BiLSTM	base	0.531	0.149	0.412	0.599	0.471
BiGRU	sentiment	0.538	0.144	0.406	0.591	0.440
BiLSTM	sector_finbert	0.524	0.133	0.351	0.616	0.365
BiGRU	base	0.518	0.125	0.373	0.574	0.428
BiLSTM	finbert	0.521	0.125	0.349	0.600	0.369
GRU	base	0.526	0.124	0.411	0.545	0.465
LSTM	finbert	0.515	0.124	0.359	0.552	0.419
LSTM	sector_sentiment	0.544	0.120	0.371	0.593	0.372
BiGRU	sector_finbert	0.523	0.119	0.422	0.579	0.472
GRU	emotion	0.519	0.116	0.364	0.527	0.410
LSTM	base	0.526	0.096	0.392	0.557	0.415

A.5.2 Binary Classification Results: With Meta-Features

Table A.6: Binary classification results at the 5-day horizon with meta-features.

Model	Feature Set	Dir Acc	MCC	F1	Precision	Recall
BiGRU	meta	0.531	0.145	0.349	0.624	0.367
BiLSTM	meta	0.530	0.143	0.365	0.603	0.390
GRU	meta	0.534	0.132	0.398	0.595	0.456
LSTM	meta_sentiment	0.521	0.121	0.369	0.585	0.415
BiGRU	meta_sentiment	0.515	0.116	0.372	0.630	0.406
LSTM	meta	0.524	0.115	0.361	0.580	0.386
GRU	meta_err_finbert	0.527	0.115	0.357	0.585	0.374
GRU	meta_sector	0.522	0.113	0.424	0.577	0.490
LSTM	meta_sector_finbert	0.529	0.113	0.401	0.601	0.425
BiLSTM	meta_sector	0.508	0.109	0.311	0.604	0.332
BiLSTM	meta_err_unified_emotion	0.528	0.097	0.330	0.553	0.348
BiGRU	meta_sector_sentiment	0.516	0.097	0.336	0.550	0.367

A.5.3 Regression Results: No Meta-Features

A.5.4 Regression Results: With Meta-Features

A.5.5 Ordinal Multi-Class Results: No Meta-Features

A.5.6 Ordinal Multi-Class Results: With Meta-Features

Table A.7: Regression results at the 5-day horizon without meta-features.

Model	Feature Set	Dir Acc	MCC	MAE	F1
LSTM	sector_finbert	0.533	0.055	0.025984	0.415
BiGRU	sentiment	0.526	0.035	0.026335	0.484
BiLSTM	sector_all_nlp	0.517	0.028	0.026248	0.383
GRU	all_nlp	0.515	0.027	0.026699	0.465
LSTM	base	0.526	0.025	0.026202	0.434
BiGRU	sector_sentiment	0.521	0.023	0.026253	0.419
LSTM	finbert	0.525	0.017	0.026291	0.406
GRU	base	0.529	0.012	0.026280	0.441
BiLSTM	emotion	0.511	0.006	0.026486	0.393
BiLSTM	base	0.534	0.005	0.026103	0.465
BiGRU	base	0.522	0.002	0.026219	0.463
GRU	sector_emotion	0.507	0.001	0.026552	0.391

Table A.8: Regression results at the 5-day horizon with meta-features.

Model	Feature Set	Dir Acc	MCC	MAE	F1
BiLSTM	meta_sector_emotion	0.526	0.042	0.025322	0.387
BiLSTM	meta_err_unified_emotion	0.534	0.039	0.025395	0.474
LSTM	meta_err_sector_emotion	0.533	0.038	0.025301	0.474
BiGRU	meta	0.526	0.029	0.025447	0.409
BiGRU	meta_sector_emotion	0.522	0.028	0.025453	0.492
GRU	meta_all_nlp	0.532	0.026	0.025237	0.415
LSTM	meta_err	0.526	0.016	0.025225	0.429
LSTM	meta_stance	0.517	0.008	0.025442	0.424
GRU	meta	0.500	0.005	0.025604	0.388
BiGRU	meta_all_nlp	0.514	0.003	0.025291	0.417
BiLSTM	meta_err	0.521	-0.005	0.025303	0.386
GRU	meta_err_sector_unified_emotion	0.523	-0.005	0.025585	0.462

Table A.9: Ordinal multi-class results at the 5-day horizon without meta-features.

Model	Feature Set	Micro Acc	Macro F1	MCC	Dir Acc	F1
BiGRU	sector	0.208	0.127	0.145	0.533	0.400
BiLSTM	sector_finbert	0.217	0.134	0.137	0.541	0.459
BiLSTM	base	0.214	0.129	0.128	0.520	0.428
LSTM	base	0.209	0.124	0.120	0.513	0.379
GRU	base	0.197	0.116	0.119	0.508	0.423
BiGRU	emotion	0.208	0.129	0.115	0.527	0.413
LSTM	sector_finbert	0.189	0.116	0.115	0.520	0.412
BiGRU	base	0.202	0.120	0.113	0.511	0.401
GRU	sector_all_nlp	0.222	0.122	0.110	0.511	0.536
LSTM	sentiment	0.206	0.123	0.109	0.522	0.443
BiLSTM	emotion	0.212	0.122	0.107	0.513	0.436
GRU	emotion	0.214	0.125	0.097	0.512	0.480

Table A.10: Ordinal multi-class results at the 5-day horizon with meta-features.

Model	Feature Set	Micro Acc	Macro F1	MCC	Dir Acc	F1
GRU	meta_emotion	0.213	0.128	0.126	0.530	0.377
BiLSTM	meta	0.216	0.118	0.109	0.512	0.546
BiGRU	meta	0.213	0.112	0.108	0.517	0.536
LSTM	meta_all_nlp	0.224	0.112	0.105	0.525	0.545
BiGRU	meta_emotion	0.211	0.114	0.102	0.514	0.324
LSTM	meta	0.221	0.114	0.099	0.516	0.536
BiLSTM	meta_sector	0.216	0.117	0.097	0.521	0.577
GRU	meta	0.216	0.117	0.092	0.526	0.435
GRU	meta_sector_stance	0.224	0.115	0.089	0.516	0.565
BiGRU	meta_sector	0.216	0.111	0.085	0.517	0.570
BiLSTM	meta_emotion	0.214	0.099	0.071	0.510	0.574
LSTM	meta_sector	0.224	0.116	0.067	0.509	0.571

A.6 Investment Simulation Cumulative Returns

TODO: Fix captions

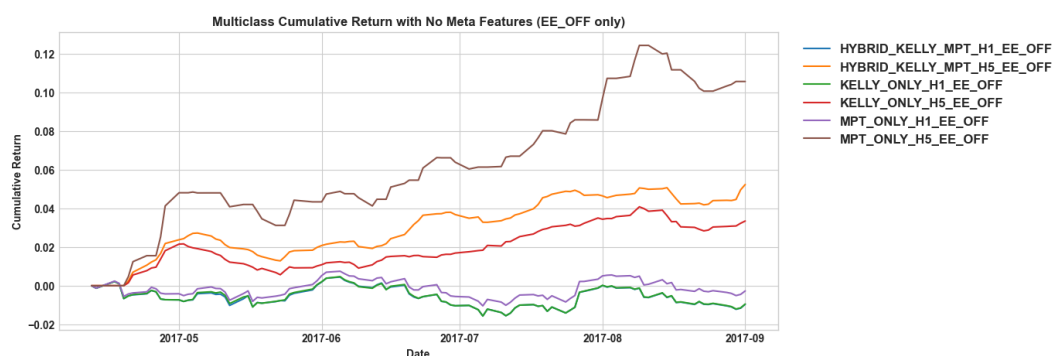


Figure A.5: Cumulative portfolio returns for ordinal multi-class strategies without meta features.

A.7 Appendix: Drawdown and Risk Tables and Graphs

A.7.1 Binary System without Meta Features

Full Drawdown Statistics

Rolling Risk Metrics (20-Day Window)

A.7.2 Binary System with Meta Features

Full Drawdown Statistics

Rolling Risk Metrics (20-Day Window)

A.7.3 Ordinal Multi-Class System without Meta Features

Full Drawdown Statistics

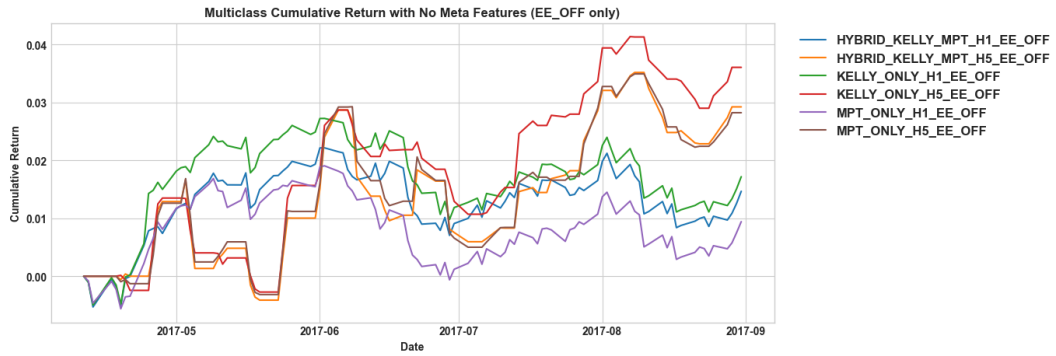


Figure A.6: Cumulative portfolio returns for ordinal multi-class strategies incorporating meta features.

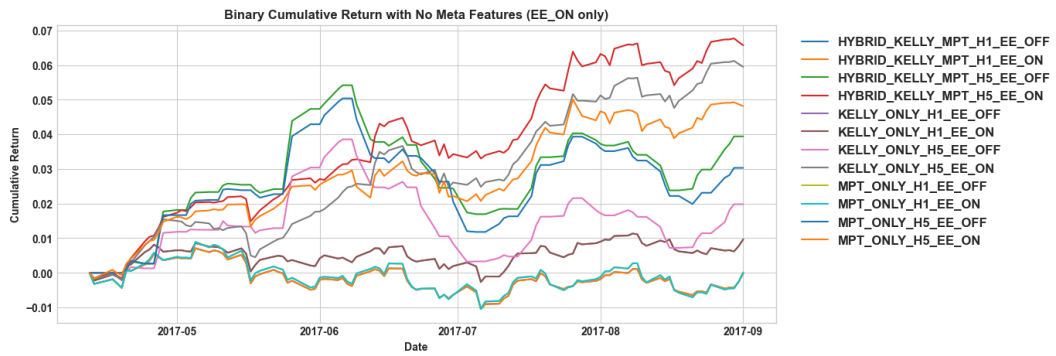


Figure A.7: Cumulative portfolio returns for ordinal multi-class strategies incorporating meta features.

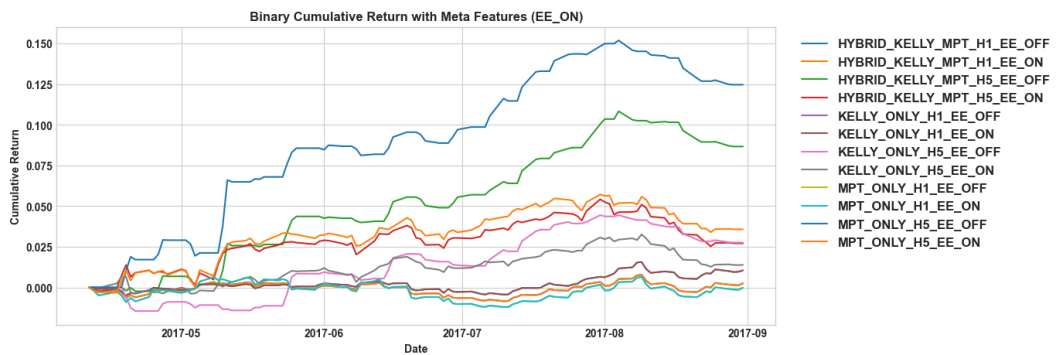


Figure A.8: Cumulative portfolio returns for ordinal multi-class strategies incorporating meta features.

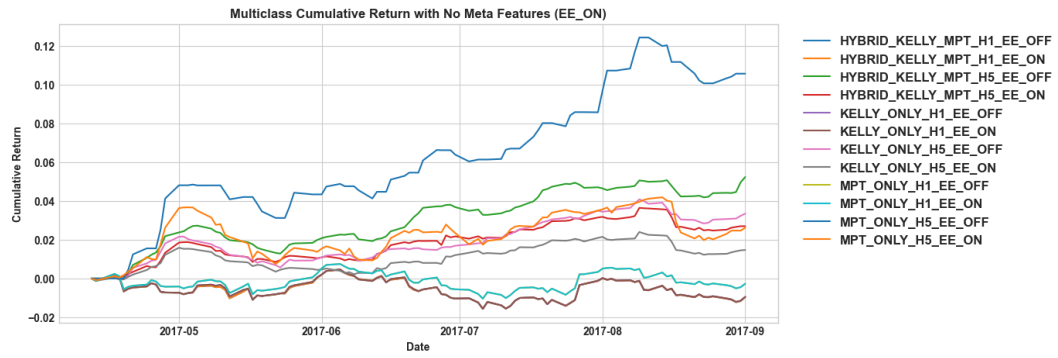


Figure A.9: Cumulative portfolio returns for ordinal multi-class strategies incorporating meta features.

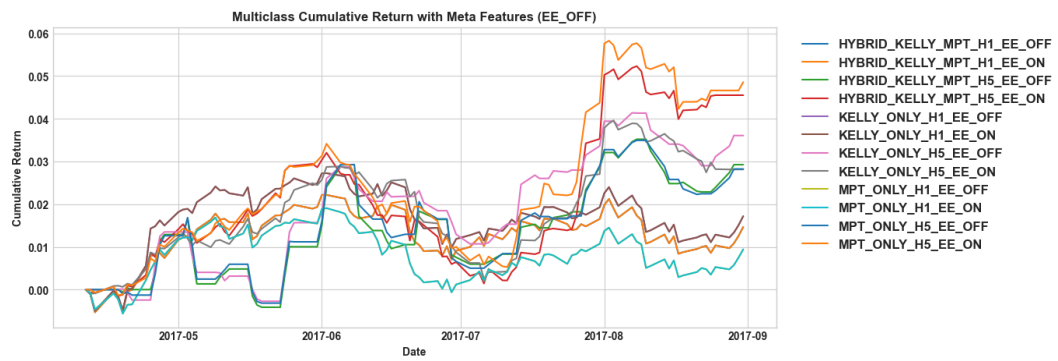


Figure A.10: Cumulative portfolio returns for ordinal multi-class strategies incorporating meta features.

Table A.11: Full drawdown statistics for the binary system without meta-features.

Mode	Horizon	Early Exit	Max DD	Peak Date	Trough Date
MPT	5	OFF	-3.6713%	2017-06-06	2017-07-05
HYBRID	5	OFF	-3.5307%	2017-06-06	2017-07-05
KELLY	5	OFF	-3.4001%	2017-06-06	2017-07-03
MPT	1	OFF	-1.9150%	2017-05-05	2017-07-06
MPT	1	ON	-1.9150%	2017-05-05	2017-07-06
HYBRID	1	OFF	-1.7415%	2017-05-05	2017-07-06
HYBRID	1	ON	-1.7415%	2017-05-05	2017-07-06
KELLY	5	ON	-1.1399%	2017-06-19	2017-06-29
HYBRID	5	ON	-1.1350%	2017-08-09	2017-08-17
MPT	5	ON	-1.1174%	2017-06-19	2017-07-03
KELLY	1	OFF	-1.1143%	2017-05-05	2017-07-06
KELLY	1	ON	-1.1143%	2017-05-05	2017-07-06

Table A.12: End-of-drawdown recovery values for the binary system without meta-features.

Mode	Horizon	Early Exit	End DD
MPT	5	OFF	-1.9096%
HYBRID	5	OFF	-1.4013%
KELLY	5	OFF	-1.8030%
MPT	1	OFF	-0.8726%
HYBRID	1	OFF	-0.7219%
KELLY	5	ON	-0.1640%
HYBRID	5	ON	-0.1869%
MPT	5	ON	-0.1857%
KELLY	1	OFF	-0.1652%

Rolling Risk Metrics (20-Day Window)

A.7.4 Ordinal Multi-Class System with Meta Features

Full Drawdown Statistics

Table A.13: Rolling volatility statistics (20-day window) for the binary system without meta-features.

Mode	Horizon	Early Exit	Mean Vol	Max Vol
HYBRID	5	ON	0.0374	0.0491
KELLY	5	ON	0.0342	0.0418
MPT	5	ON	0.0365	0.0467
HYBRID	5	OFF	0.0434	0.0668
MPT	5	OFF	0.0433	0.0639
KELLY	5	OFF	0.0365	0.0544
KELLY	1	OFF	0.0236	0.0307
MPT	1	OFF	0.0314	0.0392
HYBRID	1	OFF	0.0272	0.0350

Table A.14: Rolling Sharpe statistics (20-day window) for the binary system without meta-features.

Mode	Horizon	Early Exit	Mean Sharpe	Max Sharpe	Min Sharpe
HYBRID	5	ON	4.5074	12.4359	-1.6577
KELLY	5	ON	4.4724	15.5176	-1.2235
MPT	5	ON	3.3261	12.0925	-2.1337
HYBRID	5	OFF	1.5083	8.6150	-9.6907
MPT	5	OFF	1.4112	8.5256	-9.0081
KELLY	5	OFF	0.8861	8.0608	-10.3427
KELLY	1	OFF	0.6285	8.1392	-2.9141
MPT	1	OFF	-0.3875	5.9018	-4.1504
HYBRID	1	OFF	-0.4389	6.1727	-4.0468

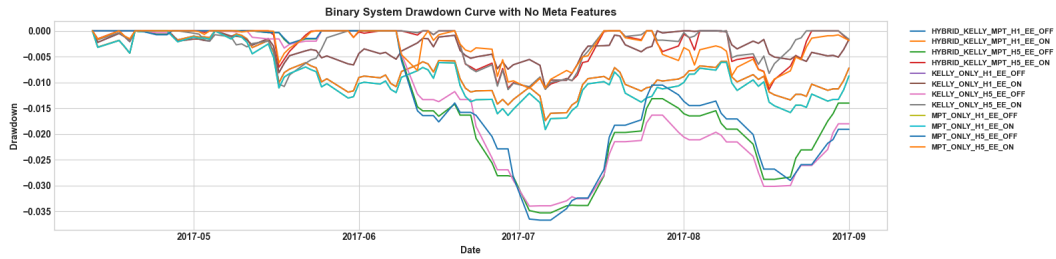
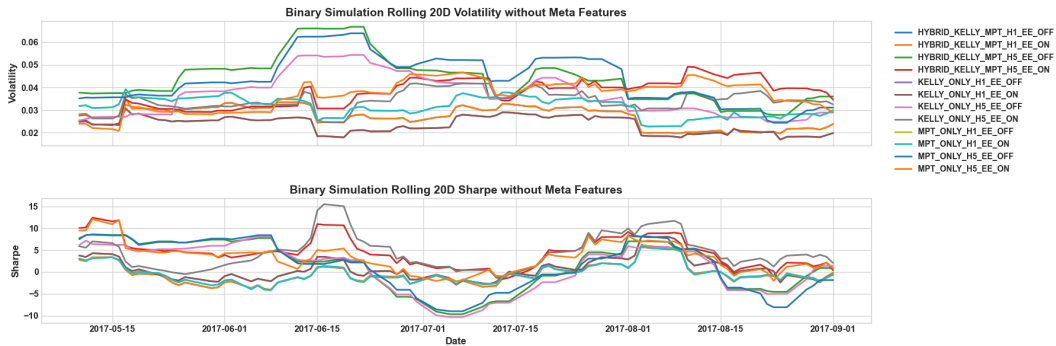
**Figure A.11:** Binary system drawdown curves without meta-features.**Figure A.12:** Binary system rolling 20-day volatility and Sharpe ratio without meta-features.

Table A.15: Full drawdown statistics for the binary system with meta-features.

Mode	Horizon	Early Exit	Max DD	Peak Date	Trough Date
HYBRID	5	ON	-2.7465%	2017-07-31	2017-08-24
MPT	5	OFF	-2.3648%	2017-08-04	2017-08-29
MPT	5	ON	-2.2019%	2017-07-31	2017-08-24
HYBRID	5	OFF	-1.9556%	2017-08-04	2017-08-29
KELLY	5	ON	-1.9302%	2017-08-09	2017-08-24
MPT	1	OFF	-1.8300%	2017-05-08	2017-07-11
MPT	1	ON	-1.8300%	2017-05-08	2017-07-11
KELLY	5	OFF	-1.6096%	2017-07-31	2017-08-29
HYBRID	1	OFF	-1.1828%	2017-05-08	2017-07-10
HYBRID	1	ON	-1.1828%	2017-05-08	2017-07-10
KELLY	1	OFF	-1.0215%	2017-08-09	2017-08-21
KELLY	1	ON	-1.0215%	2017-08-09	2017-08-21

Table A.16: End-of-drawdown recovery values for the binary system with meta-features.

Mode	Horizon	Early Exit	End DD
HYBRID	5	ON	-2.5691%
MPT	5	OFF	-2.3648%
MPT	5	ON	-2.0337%
HYBRID	5	OFF	-1.9556%
KELLY	5	ON	-1.8127%
MPT	1	OFF	-0.6648%
HYBRID	1	OFF	-0.5273%
KELLY	1	OFF	-0.5013%

Table A.17: Rolling volatility statistics (20-day window) for the binary system with meta-features.

Mode	Horizon	Early Exit	Mean Vol	Max Vol
HYBRID	5	OFF	0.0499	0.0750
MPT	5	OFF	0.0630	0.1223
KELLY	5	OFF	0.0375	0.0530
KELLY	5	ON	0.0318	0.0471
MPT	5	ON	0.0482	0.0751
HYBRID	5	ON	0.0478	0.0717
KELLY	1	OFF	0.0201	0.0302
HYBRID	1	OFF	0.0220	0.0285
MPT	1	OFF	0.0331	0.0432

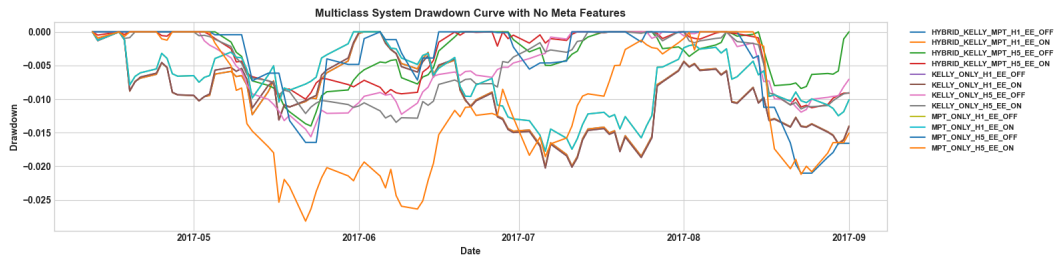
*Figure A.13: Ordinal multi-class system drawdown curves without meta-features.*

Table A.18: Rolling Sharpe statistics (20-day window) for the binary system with meta-features.

Mode	Horizon	Early Exit	Mean Sharpe	Max Sharpe	Min Sharpe
HYBRID	5	OFF	5.5423	11.9235	-5.7633
MPT	5	OFF	4.7818	11.6608	-8.6113
KELLY	5	OFF	2.7579	10.0893	-9.8231
KELLY	5	ON	2.4492	9.2632	-7.1356
MPT	5	ON	2.2816	8.5262	-6.6045
HYBRID	5	ON	2.0146	8.7736	-8.1283
KELLY	1	OFF	1.9283	14.8209	-4.5821
HYBRID	1	OFF	0.7741	10.5883	-6.2758
MPT	1	OFF	0.5262	9.0511	-6.1493

Table A.19: Full drawdown statistics for the ordinal multi-class system without meta-features.

Mode	Horizon	Early Exit	Max DD	Peak Date	Trough Date	End DD
MPT	5	ON	-2.8165%	2017-05-03	2017-05-22	-1.5070%
MPT	5	OFF	-2.1024%	2017-08-09	2017-08-23	-1.6585%
KELLY	1	OFF	-2.0267%	2017-06-05	2017-07-06	-1.4094%
KELLY	1	ON	-2.0267%	2017-06-05	2017-07-06	-1.4094%
HYBRID	1	OFF	-2.0061%	2017-06-05	2017-07-06	-1.4018%
HYBRID	1	ON	-2.0061%	2017-06-05	2017-07-06	-1.4018%
MPT	1	OFF	-1.7767%	2017-06-05	2017-07-06	-1.0110%
MPT	1	ON	-1.7767%	2017-06-05	2017-07-06	-1.0110%
KELLY	5	OFF	-1.5615%	2017-05-02	2017-05-23	-0.7049%
HYBRID	5	OFF	-1.4012%	2017-05-05	2017-05-23	0.0000%
KELLY	5	ON	-1.3464%	2017-05-01	2017-06-08	-0.9057%
HYBRID	5	ON	-1.1294%	2017-08-09	2017-08-25	-0.9121%

Table A.20: Rolling risk statistics for the ordinal multi-class system without meta-features.

Mode	Horizon	Early Exit	Mean Vol	Max Vol	Mean Sharpe	Min Sharpe
KELLY	5	OFF	0.0194	0.0335	5.0339	-8.4097
MPT	5	OFF	0.0529	0.0754	4.8418	-1.2871
HYBRID	5	OFF	0.0232	0.0343	4.3161	-4.3363
HYBRID	5	ON	0.0226	0.0328	3.1742	-6.0076
KELLY	5	ON	0.0195	0.0281	1.8827	-8.5642
MPT	5	ON	0.0430	0.0728	1.6803	-6.1705
MPT	1	OFF	0.0327	0.0403	-0.0050	-6.8335
MPT	1	ON	0.0327	0.0403	-0.0050	-6.8335
KELLY	1	OFF	0.0344	0.0427	-0.5186	-7.7029
KELLY	1	ON	0.0344	0.0427	-0.5186	-7.7029
HYBRID	1	OFF	0.0344	0.0427	-0.5414	-7.8300
HYBRID	1	ON	0.0344	0.0427	-0.5414	-7.8300

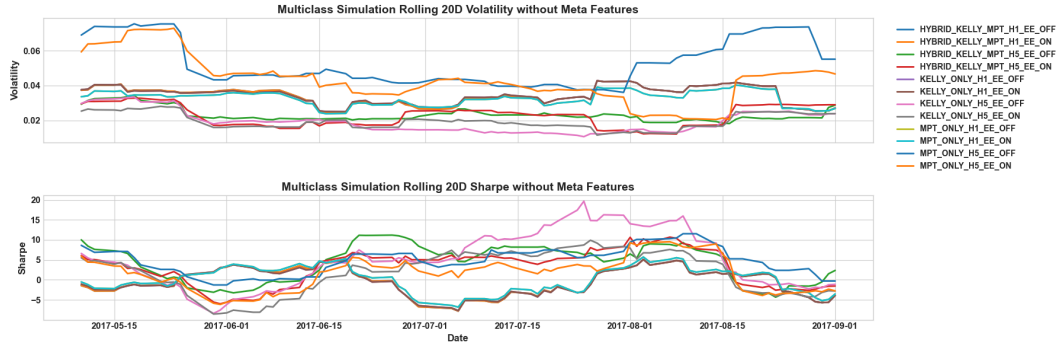


Figure A.14: Ordinal multi-class system rolling 20-day volatility and Sharpe ratio without meta-features.

Table A.21: Full drawdown statistics for the ordinal multi-class system with meta-features.

Mode	Horizon	Early Exit	Max DD	Peak Date	Trough Date	End DD
HYBRID	5	ON	-2.9671%	2017-06-02	2017-07-06	-0.6461%
MPT	5	ON	-2.7934%	2017-06-02	2017-07-11	-0.9154%
KELLY	5	ON	-2.5586%	2017-06-05	2017-07-06	-1.1046%
MPT	5	OFF	-2.3583%	2017-06-08	2017-07-03	-0.6476%
HYBRID	5	OFF	-2.2126%	2017-06-05	2017-07-03	-0.5747%
MPT	1	OFF	-1.9330%	2017-06-02	2017-06-29	-0.9468%
MPT	1	ON	-1.9330%	2017-06-02	2017-06-29	-0.9468%
KELLY	5	OFF	-1.7505%	2017-06-05	2017-07-03	-0.5098%
KELLY	1	OFF	-1.6994%	2017-06-02	2017-06-29	-0.9801%
KELLY	1	ON	-1.6994%	2017-06-02	2017-06-29	-0.9801%
HYBRID	1	OFF	-1.4798%	2017-06-02	2017-06-29	-0.7342%
HYBRID	1	ON	-1.4798%	2017-06-02	2017-06-29	-0.7342%

Table A.22: Rolling risk statistics for the ordinal multi-class system with meta-features.

Mode	Horizon	Early Exit	Mean Vol	Max Vol	Mean Sharpe	Min Sharpe
MPT	5	ON	0.0455	0.0766	2.5841	-8.0653
KELLY	5	ON	0.0342	0.0477	2.2652	-9.8862
HYBRID	5	ON	0.0455	0.0734	2.1259	-8.3034
KELLY	5	OFF	0.0410	0.0554	1.8737	-9.3084
HYBRID	5	OFF	0.0496	0.0699	1.8143	-5.1992
MPT	5	OFF	0.0509	0.0686	1.6046	-5.4583
HYBRID	1	OFF	0.0319	0.0424	0.7041	-6.2006
HYBRID	1	ON	0.0319	0.0424	0.7041	-6.2006
KELLY	1	OFF	0.0327	0.0488	0.4767	-6.8043
KELLY	1	ON	0.0327	0.0488	0.4767	-6.8043
MPT	1	OFF	0.0311	0.0391	0.1752	-8.9869
MPT	1	ON	0.0311	0.0391	0.1752	-8.9869

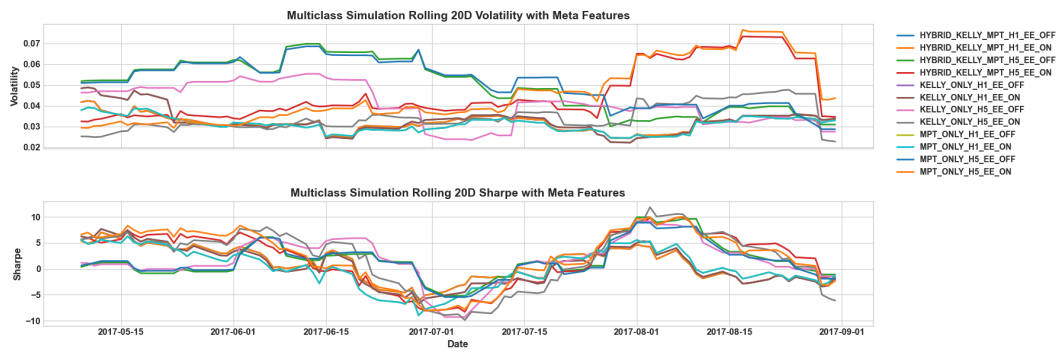


Figure A.15: Ordinal multi-class system rolling 20-day volatility and Sharpe ratio with meta-features.

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